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Biology+
Physics+
Chemistry+
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YEARS PAST PAPERS

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-Duru  

-Duryodhan_Chanpani  

-NMDCAT Aspirant 2022

-Tharparker Mithi  

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17 YRS PAST PAPERS

DURU
I.B.A.

PAST PAPER - 2000

SECTION-I Biology - 2000

- 1) Regarding bacteria:
- a) Cocci are rod - shaped
 - b) Bacilli are spiral shaped
 - c) All contain RNA
 - d) None of the these
- 2) Chlamydomonas *→ Unicellular*
- a) Is a multicultural alga
 - b) Has a single flagellum
 - c) Has no system of water regulation
 - d) Has a light sensitive structure
- 3) Cryptogams *→*
- a) Are non - flowering plants
 - b) From seeds + fruits
 - c) Include conifers + cycads
 - d) Include algae
- 4) Molluscs have all of the following except:
- a) A well developed nervous system
 - b) Kidneys
 - c) Nostrils for respiration
 - d) Circulatory system with a heart
- 5) Crustacea includes all of the following except:
- a) Crabs
 - b) Lobsters
 - c) Oysters
 - d) Crayfish
- 6) The dark reaction of photosynthesis:
- a) Takes place in the stroma of the chloroplast
 - b) Involves the fixing of water
 - c) Causes formation of lactic acid
 - d) Takes place in the cytoplasm
- 7) The pituitary gland secretes of the following except:
- a) Growth hormone
 - b) Calcitonin
 - c) Prolactin
 - d) Oxytocin
- 8) Thyroid stimulating hormone levels:
- a) Would be raised in a person with hyperthyroidism
 - b) Would be low in patients with hypothyroidism
 - c) Would be low in patient with hypothyroidism
 - d) Would be unaffected in hypothyroidism
- 9) Gastrulation in a frog's embryo involves all of the following except:
- a) Epiboly
 - b) Involution
 - c) Rotation
 - d) Invagination
- 10) Mesoderm gives rise to:
- a) Lens of the eye
 - b) Teeth enamel
 - c) Liver
 - d) Heart

11) Gaseous exchange in Planaria takes place by means of:

- a) Osmosis b) Diffusion
- c) Breathing d) Active transport

12) Retina contains light sensitive receptor cells called:

- a) Rods b) Cones
- c) Both rods and cones d) Optic nerves

13) Excretory system of grasshopper consists of:

- a) Nephridia b) Flame cells
- c) Malpighian tubules d) Kidneys

14) Which of the following Cnidarian lives in colonies?

- a) Tapeworm b) Obelia
- c) Jelly fish d) Sea anemone

15) The causative agent of malaria is:

- a) Plasmodium b) Trypanosoma
- c) Volvox d) Entamoeba histolytica

16) Earthworms feed on:

- a) Decaying vegetation b) Organic matter
- c) Soil d) All of the above.

17) Planaria feeds through the:

- a) Mouth b) Pharynx
- c) Esophagus d) Oral groove

18) Sugar is stored in the liver as:

- a) Starch b) Glycogen c) Maltose d) Cellulose

19) Function of the "tube feet" in Phylum Echinodermata is:

- a) Locomotion b) Anchoring to hard surface
- c) Grabbing the prey d) All of the above

20) What is the main role of vitamin K in the diet of humans?

- a) To participate in the formation of visual purple
- b) To promote the production of blood clotting factors
- c) To prevent skin disorders
- d) Deposition of Calcium in the bones
- e. All of the above

- 32) How many grams of NaCl are required to completely neutralize 100 ml of 1 M of H_2SO_4 .
a) 80 grams b) 40 grams c) 24 grams d) 8 grams
- 33) Cyclo Hexane has the formula:
a) C_6H_{14} b) C_6H_{12} c) C_6H_{16} d) C_6H_6
- 34) In $\text{H}_2\text{C} \equiv \text{CH}_2$ the molecule contains:
a) 4 sigma - 2 pi Bonds b) 2 sigma - 4 pi Bonds
c) 5 sigma - 1 pi Bonds b) 6 sigma Bonds
- 35) What is the general formula of Ester?
a) $\text{R}-\text{COO}-\text{R}$ b) $\text{R}-\text{CHO}$ c) $\text{R}-\text{OH}$ d) RX
- 36) The element is families of the periodic table classified as:
a) S- block element b) P - block element
c) d- block element d) d and f block elements
- 37) $\text{Al}_2\text{O}_3 \cdot 2\text{H}_2\text{O}$ is called)
a) Bauxite b) Gyolele c) Alum stone d) Alonite
- 38) What is the boiling point of HNO_3 ?
a) 86°C b) 40°C c) 76°C d) 66°C
- 39) The mole of Ethanol and one mole of Ethane have an equal:
a) Mass b) Number of atoms
c) Number of electrons d) Number of molecule
- 40) Alkenes undergo,
a) Addition reactions b) Substitution reactions
c) Both addition and substitution d) None of the above
- 41) During the electrolysis of H_2O , H_2 is liberated at:
a) Anode b) Cathode c) Diode d) None of the above
- 42) An example of covalent hydride is:
a) NaH b) CaH_2 c) SrH_2 d) AsH_3
- 43) In an equilibrium $K_{s.p}$ is expressed as:
$$\text{BaSO}_4 \rightleftharpoons \text{Ba}^{+2} + \text{SO}_4^{-2}$$

a) $K_{s.p} = [\text{BaSO}_4]$ b) $K_{s.p} = \frac{[\text{Ba}^{+2}][\text{SO}_4^{-2}]}{[\text{BaSO}_4]}$
c) $K_{s.p} = [\text{Ba}^{+2}][\text{SO}_4^{-2}]$ d) None of the above.

- 44) The pH of a solution contains 10^{-4} M $[H^+]$.
a) 4 b) 3 c) -3 d) 0
- 45) Energy can neither be created nor destroyed but can be transferred and transformed)
This is known as:
a) Law of conservation of energy b) Law of conservation of mass
c) Law of conservation of Momentum d) Hess's Law
- 46) At what temperature standard enthalpy changes are measured?
a) 298K b) 273 K c) 293°C d) 298°F
- 47) Rb is an element similar to potassium which can show the decomposition of Rubidium nitride.
a) $4\text{RbNO}_3 \longrightarrow 2\text{Rb}_2\text{O} + 4\text{NO}_2 + \text{O}_2$
b) $2\text{RbNO}_3 \longrightarrow 2\text{Rb} + 2\text{NO}_2 + \text{O}_2$
c) $2\text{RbNO}_3 \longrightarrow 2\text{RbNO}_2 + \text{O}_2$
d) $\text{Rb}(\text{NO}_3)_2 \longrightarrow \text{Rb}(\text{NO}_2)_2 + \text{O}_2$
- 48) The wavelength of radiation emits in Balmer series is:
a) b/w $4000 \text{ \AA}$ to $8000 \text{ \AA}$ b) $10,000 < d < 18000 \text{ \AA}$
c) $d < 8000 \text{ \AA}$ d) None of the above
- 49) Fedrle craft reaction classified as:
a) Halogenations and Hydrogenation b) Alkylation and Accylation
c) Hydrogenation and Halogenations d) Nitration
- 50) The formula for Sandhur is:
a) $\text{Pb}_3(\text{OH}_2) \cdot (\text{CO}_3)_2$ b) PbO_4 c) PbO d) Pb_3O_4
- 51) The atoms of the same elements, which have the same atomic number but different mass number are called)
a) Isobars b) Actanids
c) Allotropes d) Isotopes
- 52) In haber process
 $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$; increasing the pressure favors,
a) The forward reaction b. Reverse reaction
c) Neither reaction d) All of the above
- 53) $K_{s.p}$ is called,
a) Solubility product b) Concentration
c) Equilibrium d) Constant concentration

54) Thermodynamics is the study of heat energy and _____ energy relationship.

- a) Energy of activation
- b) Kinetic energy
- c) Chemical
- d) Ionization

55) The law of conservation of energy is also called)

- a) Second law of Thermodynamics
- b) First law of Thermodynamics
- c) III rd law of Thermodynamics
- d) None of the above

56) Rb is an element in the same group of Li and Na) Which one is true about Rb)

- a) Reacts slowly with H_2O
- b) Forms an insoluble hydroxide
- c) It is liberated at Cathode during the electrolysis of its salt in aqueous.
- d) Forms a sulphate Rb_2SO_4

57) Which statement concerning energy change is false?

- a) Electrical energy is produced by the reaction in a battery
- b) Light energy is absorbed during photosynthesis
- c) Light energy is produced when Methane burns in Oxygen
- d) Heat energy is absorbed when Sodium reacts with water.

58) Electrolysis of dil. H_2SO_4 gives:

- | Cathode | Anode |
|-----------|--------|
| a) H_2 | SO_2 |
| b) H_2 | O_2 |
| c) O_2 | H_2 |
| d) SO_2 | H_2 |

59) Carbohydrates contains;

- a) C and H only
- b) C, H and N only
- c) C, H, O only
- d) All of the above.

60) An oxide of titanium contains 60% Ti. What is its empirical formula for this oxide?

- a) TiO
- b) Ti_2O_3
- c) TiO_2
- d) Ti_2O_6

SECTION-III Physics

61) If the distance between two charged bodies is doubled and the charge is also doubled the interaction force will be:

- a) Half
- b) Four times
- c) Unchanged
- d) Doubled

62) Capacitance of a capacitor depends upon;

- a) Distance between the plates
- b) Area of the plates
- c) Dielectric used between the plates
- d) All of the above.

- 63) If a radio and a bulb each of resistance 3Ω are connected in series to a 12 volts battery, the potential difference across each will be;
a) 3V b) 6V c) 12V d) 9V
- 64) Choose the relation which is not correct:
a) Ampere = coulomb per second
b) Volt = coulomb per joule
c) Ohm = Volt per ampere
d) 1 coulomb = charge of 6.24×10^{18} electrons
- 65) A particle of charge 'q' and mass 'm' moving with a velocity 'v' enters perpendicularly in a uniform magnetic field 'B' describes a circular path. What will be the radius of that path?
a) mv/Bq b) mqv/B c) mvB/q d) mqB/v
- 66) Light of wavelength 15×10^{-7} m consists of photons whose energy is;
a) 1.7×10^{-15} J b) 4×10^{-19} J c) 1.1×10^{-27} J d) 1.1×10^{-48} J
- 67) The electrons structures of atoms are not involved in the emission of;
a) Spectral lines b) Photoelectrons
c) Gamma rays d) X-rays
- 68) When electron in hydrogen atom jumps from a higher orbit to the fifth orbit, the set of lines emitted are called)
a) Bracket series b) Paschen series
c) Lyman series d) Pfund series
- 69) Why gamma radiations are used to kill bacteria to sterilize surgical equipment etc)
a) Because they are chargeless
b) Because they have no mass
c) Because of its penetration
d) All of the above
- 70) The name of the most useful tracer is
a) Na-34 b) I-31
c) C-14 d) C-12
- 71) A wire of uniform cross-sectional area has a length of 10m, a resistance of 2Ω and resistivity $2 \times 10^{-7} \Omega \text{ cm}$. The cross-sectional area in m^2 is;
a) 2×10^{-4} b) 1×10^{-5} c) 0.5×10^{-5} d) 1×10^{-4}

- 72) The power of a load is one watt if;
- 1 ampere current flows through a conductor when a potential difference of 1 volt is applied
 - 0.1 ampere current flows when a potential difference of 0.1 volts is applied
 - 1 ampere current flows when a potential difference of 10 volts is applied
 - 10 amperes current flows when a potential difference of 10 volts is applied
- 73) If the rate of cost is Rs. 2 per kWh then a 500 W kettle operating for 30 minutes will cost;
- Rs. 0.50
 - Rs. 0.075
 - Rs. 0.1
 - 1.50
- 74) An A/C voltage measures V_0 volts in a circuit. Its peak value is given by;
- V_0
 - $V_0/\sqrt{2}$
 - $\sqrt{2}V_0$
 - $\sqrt{2}$
- 75) A stationary photon has mass;
- $1.67252 \times 10^{-27} \text{ kg}$
 - $9.1 \times 10^{-31} \text{ kg}$
 - Infinite
 - Zero
- 76) A battery of 6 volts is connected to two resistors of 30Ω and 20Ω joined together in series. Find the current through the circuit?
- 12A
 - 1.2A
 - 36A
 - 150A
- 77) An electric kettle of 1000 watts boils a certain quantity of water in 8 minutes. How much heat has been generated for boiling?
- 4000J
 - 480000J
 - 4800J
 - 5000J
- 78) The current passes through all the resistors combined in series should be ____.
- Same
 - the sum of the current passes from each resistor
 - Potential difference over the value of equivalent resistance
 - Both A and C
- 79) The red line of the Balmer series originates in a transition between levels having quantum numbers;
- 2 and 1
 - 3 and 1
 - 3 and 2
 - ∞ and 1
- 80) Solid state detector basically is;
- a PNP transistor
 - a NPN transistor
 - a PN-Junction diode
 - a light emitting diode
- 81) Michelson's interferometer is usually used to determine the;
- Wavelength of light
 - Velocity of light
 - Frequency of light
 - Dispersion of light

- 82) How is the image formed by a convex lens affected if the upper half of the lens is covered with a paper?
- a) The brightness of the image is reduced
 - b) The upper half of the image is cut off.
 - c) The size of the image becomes one half.
 - d) No effect at all
- 83) A convex lens gives a virtual image only when the object is placed;
- a) Between focus and centre of curvature
 - b) Between focus and optical center
 - c) Beyond $2f$
 - d) Between f and $2f$
- 84) If the pressure is increased, the melting point of ice;
- a) Decreases
 - b) Increases
 - c) Remains unchanged
 - d) None of the above
- 85) The process in which the pressure of system remains constant is called;
- a) Isothermal process
 - b) Isochoric process
 - c) Isobaric process
 - d) Adiabatic process
- 86) Under adiabatic compression the increase in temperature of a gas is proportional to;
- a) The pressure to which the gas is compressed
 - b) Ratio of the two specific heats
 - c) Universal gas constant
 - d) Work done on the gas
- 87) A 50 mh coil carries a current of 2 amp. The energy stored in its magnetic field is;
- a) 0.05J
 - b) 0.1 J
 - c) 10J
 - d) 100J
- 88) In amplitude modulation the waves that superimpose on electromagnetic waves are:
- a) Water waves
 - b) Sound waves
 - c) Light waves
 - d) Material waves
- 89) Quality of X-rays depends upon;
- a) Current of filament
 - b) Accelerating voltage
 - c) Material of the target
 - d) Both accelerating voltage and material of the target

90) According to Bohr's theory of hydrogen atom, the angular momentum of electron in an allowed orbit is;

- | | |
|---------------|---------------|
| a) nh/π | b) $nh/2\pi$ |
| c) $n/2\pi h$ | d) $2\pi h/n$ |

SECTION-IV ENGLISH

Structure and Written Expression

Structure

Directions: Questions 1-3 are incomplete sentences. Beneath each sentence you will see four words or phrases marked (A), (B), (C), and (D). Choose the one word or phrase that best completes the sentence. Then, on your answer sheet, find the number of the question and fill in the space that corresponds to the letter of the answer you have chosen. Fill in the space so that the letter inside the oval cannot be seen.

91) In order for people who spoke different languages to engage in trade-----, they often developed a simplified language called pidgin.

- | | |
|------------------------|---------------------------|
| a) With each the other | b) With each to the other |
| c) With each another | d) With each other |

92) The two main-----are permanent magnets and electromagnets.

- | | |
|---------------------|--------------------|
| a) Kinds of magnets | b) Kind of magnets |
| c) Kind magnets | d) Kinds magnets |

93) The Supreme Court does not hear a case unless-----, except those involving foreign ambassador.

- | | |
|----------------------|-------------------------------|
| a) A trial | b) Already tried |
| c) It already trying | d) It has already been tried) |

ANALOGIES

94) HOOF : HORSE

- | | |
|------------------|--------------------|
| a) Lion : Mane | b) Paw : Cat |
| c) Pace : Foot | d) Bicycle : Rider |
| e. Dog : Prairie | |

95) SHIP : DOC

- | | |
|------------------|------------------|
| a) Car : Park | b) Wheel : Steer |
| c) Horse : Ride | d) Raft : Float |
| e. Truck : Drive | |

SYNONYMS

96) EXPEDIENT

- | | |
|--------------|------------------|
| a) Scapegoat | b) Heterogeneous |
| c) Suitable | d) Apathetic |

97) VORACIOUS

- a) Reject
- c) Honest
- e. Inclusive

- b) Excitable
- d) Greedy

ANTONYMS

98) HAMPER:

- a) Ogle
- c) Assist
- e. Ventilate

- b) Clarify
- d) Uncover

99) GAUNT;

- a) Rotund
- c) Nervous
- e. Valiant

- b) Garish
- d) Stylish

100) UNWARY;

- a) Alert
- c) Angry
- e. Dutiful

- b) Sad
- d) Eager

Duru

**MODEL ENTRY TEST 2000
ANSWERY KEY**

SECTION I BIOLOGY		SECTION II CHEMISTRY		SECTION III PHYSICS		SECTION IV ENGLISH	
01	D	31	C	61	C	91	D
02	D	32	D	62	D	92	A
03	A	33	B	63	B	93	D
04	C	34	C	64	B	94	B
05	C	35	A	65	A	95	A
06	A	36	D	66	B	96	C
07	B	37	A	67	C	97	C
08	C	38	A	68	D	98	C
09	C	39	D	69	D	99	C
10	D	40	A	70	C	100	D
11	B	41	B	71	D		
12	C	42	D	72	A		
13	C	43	C	73	A		
14	C	44	A	74	B		
15	A	45	A	75	D		
16	D	46	A	76	B		
17	B	47	C	77	B		
18	B	48	A	78	D		
19	D	49	B	79	A		
20	B	50	D	80	C		
21	D	51	D	81	A		
22	h B C	52	A	82	A		
23	D	53	A	83	B		
24	B	54	C	84	C		
25	B	55	B	85	C		
26	D	56	D	86	D		
27	B	57	A	87	B		
28	A	58	B	88	B		
29	D	59	C	89	D		
30	D	60	C	90	B		

10 YRS PAST PAPERS

I.B.A.
PAST PAPER - 2001

- 1) ✓ Lacteals of Villi absorb:
- a) Amino Acids
 - b) Glucose
 - c) Hydrochloric Acid
 - d) Fatty Acids
- 2) Microtubules perform all of the following except:
- a) Transport of molecule
 - b) Movement in cilia and flagella
 - c) Movement of chromosomes
 - d) Protein synthesis
- 3) Which of the following is correct :
- a) Glucose + fructose = maltose
 - b) Glucose + galactose = sucrose
 - c) ~~Glucose~~ + glucose = maltose
 - d) Glucose + ~~glucose~~ = sucrose
- 4) Pepsin and trypsin:
- a) Are both secreted with acid digestive juices
 - b) Are both secreted by the pancreas
 - c) Breakdown protein to aminoacids
 - d) Are specific enzymes releasing peptide and polypeptide chains
- 5) Vitamins K
- a) is mostly found in green vegetables
 - b) deficiency causes Pellagra
 - c) promotes production of blood clotting factors
 - d) prevent night blindness
- 6) Regarding the respiratory system in humans:
- a) The trachea is completely surrounded by rings of cartilage
 - b) The trachea divides into 3 bronchi
 - c) The vocal cords lie in the larynx
 - d) The lungs are rigid structures with little elasticity
- 7) The function of oxygen in cellular respiration is:
- a) To combine with hydrogen from organic molecules to form carbon dioxide
 - b) To combine with Carbon from organic molecules to form carbon dioxide
 - c) To oxidize ADP to ATP
 - d) To combine with glucose to form carbon dioxide

- 8) Which of the following increase the rate of respiration most efficiently?
- a) Lack of oxygen in the alveoli
 - b) Excess of carbon dioxide in the blood
 - c) Excess of carbon dioxide in the lungs
 - d) Lack of oxygen in the tissues
- 9) The heart
- a) Pumps blood at a constant rate
 - b) Oxygenate blood
 - c) Is under some amount of voluntary control
 - d) Is stimulated by SA node
- 10) All the following are true about the skin except,
- a) It is the largest organ in the body
 - b) Vitamin E is synthesized here
 - c) It performs both secretory and excretory functions
 - d) Melanin is synthesized here
- 11) The motor nerve cell transmits impulses from
- a) The effector organ to the spinal cord
 - b) Receptor cells to the spinal cord
 - c) Receptor cells to the effector organ
 - d) Spinal cord to the effector organ
- 12) Removal of the thyroid gland in an adult would cause an increase in,
- a) Basal metabolic rate
 - b) Conversion of glycogen to glucose
 - c) Excretion of Na^+ from the kidney
 - d) Secretion of TSH
- 13) Insulin causes;
- a) Increased blood sugar
 - b) Increased calcium deposition in bones
 - c) Increased permeability of cells to glucose
 - d) Increased cholesterol in blood
- 14) Myofibrils:
- a) Are found in smooth muscles
 - b) Are the smallest fibres that make up a muscle?
 - c) Are crossed by transverse Y-Bands
 - d) Cannot contract

15) The mammalian placenta has all of the following functions except

- a) Excretion
- b) Nutrition
- c) Hormone secretion
- d) Stock absorptions

16) The following is a part of the female reproductive tract:

- a) Vas deferens
- b) Vulva
- c) Spermatozoa
- d) Ovary

17) Where is oxygen produced in plants?

- a) A root hair cell
- b) Mesophyll cell
- c) Leaf epidermal cell
- d) Phloem cell

18) Rate of transpiration varies with all the following except

- a) Temperature
- b) Decreased number of chloroplasts
- c) Light
- d) Wind

19) The following are involved with plant respiration

- a) Stomata
- b) Chloroplasts
- c) Auxins
- d) Geotropism

20) Nematoda include all of the following except

- a) Hookworms
- b) Guinea worms
- c) Earthworms
- d) Ascaris

21) Arthropoda have all of the following except

- a) Hard exoskeleton
- b) Jointed legs
- c) Segmentation
- d) Closed circulation

22) Rod shaped bacteria are called

- a) Bacilli
- b) Plasmodia
- c) Cocci
- d) Spirilli

23) Most highly developed plants are

- a) Saprophytes
- b) Angiosperms
- c) Bryophytes
- d) Gymnosperms

24) When hybrids are crossed the genotype of the offspring is,

- a) 1:1
- b) 3:1
- c) 1:2:1
- d) 4:1

25) The following may cause mutation

- a) Malaria
- b) Tuberculosis
- c) Ultra violet rays
- d) Colchicines

26) Which one of the following is the major advantage of using a light microscope instead of an electron microscope?

- a) A superior resolving power
- b) Constant depth of focus
- c) Observation of living matter
- d) Use of very thin sections

27) Which one of the following possesses the most prominent Golgi apparatus

- a) An islet of Langerhans
- b) A retinal rod cell
- c) A striated muscle
- d) A xylem tracheid

28) Which of the following is a characteristic of mitochondria?

- a) They store proteolytic enzymes
- b) They provide sites for anaerobic respiration
- c) They produce cellular secretions
- d) They release energy via Krebs' cycle

29) Polysaccharides are synthesized in plants by the process of

- a) Hydrolysis
- b) Oxidation
- c) Condensation
- d) Glycolysis

30) Which one of the following is not immediately essential for protein synthesis?

- a) ATP
- b) Enzyme
- c) Glucose
- d) mRNA

SECTION II (CHEMISTRY)

1) Which one of the following gas diffuses most rapidly?

- a) Bromine
- b) Carbon
- c) Methane
- d) Nitrogen

2) Moist Ammonia Gas may be dried by passing it through

- a) Calcium Oxide
- b) Calcium Chloride
- c) Calcium Hydroxide
- d) Conc. Sulphuric Acid

3) Ordinary Air from the atmosphere contains about 21% Oxygen whereas the proportion of oxygen in the mixture released by boiling river is 30%. The best explanation of the increase in percentage of Oxygen is that?

- a) Oxygen is more soluble in water than nitrogen
- b) Carbon dioxide is more soluble in water than oxygen
- c) Nitrogen reacts with water
- d) The gases from boiled water contain no water vapour

4) For two substance to be separated by paper chromatography it is necessary that:

- a) They are both liquids
- b) They are both soluble in the same solvent
- c) They have different densities
- d) They have different colors

5) Which one of the following is not essential for photosynthesis?

- a) Carbon dioxide
- b) Chlorophyll
- c) Sugar
- d) Light

6) Crude Oil can be separated by fractional distillation because its constituents?

- a) Are compounds
- b) Are chemically similar
- c) Have different boiling points
- d) Have different densities

7) Which one of the following is a pure compound?

- a) Dry air
- b) Ethanol
- c) Steel
- d) Tap water

8) Whose e/m ratio resembles with that of electrons?

- a) Alpha Rays
- b) Beta Rays
- c) Gamma Rays
- d) X rays

9) What particles are present in solid lead (II) Bromide and in molten lead (III) Bromide?

Solid Lead Bromide	Molten Lead Bromide
A) Molecules	Atom
B) Molecules	Ions
C) Ions	Atoms
D) Ions	Ions

10) The ions X^+ contain 23 particles in the nucleus and 10 electrons outside the nucleus. What is the nucleus of the ion X^+ contain?

Protons	Neutrons
A) 9	14
B) 10	13
C) 11	12
D) 12	11

11) The Empirical Formula of a liquid compound to be C_2H_4O . What other information is needed to work out its molecular formula?

- a) The Percentage composition of the compound
- b) The Relative molecular mass of the compound
- c) The density of the compound
- d) The Volume occupied by 1 mole of the compound

12) Which Statement concerning energy change is false?

- a) Electrical Energy is produced by the reaction in a battery
- b) Light energy is absorbed during photosynthesis
- c) Light energy is produced when methane is burned in Oxygen
- d) Heat energy is absorbed when Sodium reacts with water

- 13) A sample of air was bubbled into pure water. The pH of water slowly changes from 7 to 6. Which gas in the sample caused this change?

a) Argon
b) Carbon Dioxide
c) Carbon Monoxide
d) Oxygen

- 14) Which statement about Carbon monoxide is correct?

a) It is involved in photosynthesis
b) It Produces Carbon when burnt
c) It is a pollutant
d) It is denser than air

- 15) Which statement provides the best information to confirm that two substances are allotropic forms of carbon?

a) They have different crystalline structure
b) They both reduce heated lead (II) Oxide to lead
c) Equal masses of both substances require equal masses of carbon Dioxide and no other product when completely burnt in oxygen.
d) Equal masses of both substances give equal masses of carbon dioxide and no other product when completely burnt in oxygen

- 16) What reacts at room temperature with ethanol and also with ethanoic acid?

a) Calcium Carbonate
b) Copper (II) Oxide
c) Methanol
d) Sodium

- 17) Which Statement concerning the ions of Group VII elements is correct?

a) Each contains an odd number of electrons
b) Each contains more protons than neutrons
c) Each contains more electrons than protons
d) Each has 7 electrons in the outer most shell

- 18) Which formula represents the oxide of an element Y in Group II of the periodic table?

a) YO
b) YO₂
c) YO₃
d) Y₂O

- 19) If four moles of sulphur Dioxide are oxidized to sulphur Trioxide. How many moles of oxygen are required?

a) 0.5
b) 1.0
c) 2.0
d) 4.0

- 20) What is the number of paired shared electrons in methane molecule?

a) 1
b) 2
c) 3
d) 4

- 21) The relative molecular mass of Carbon Dioxide is 44. What is the mass of 24dm³ of Carbon Dioxide at RTP?

a) 44
b) 22
c) 11
d) 12

- 22) When 0.24g of Magnesium is reacted with excess HCl, what volume of gas is liberated at RTP?

a) 120 cm³
b) 240 cm³
c) 360 cm³
d) 3600 cm³

- 23) Which of the following gases is least common in air?

a) Argon
b) Carbon dioxide
c) Nitrogen
d) Hydrogen

- 24) Which of the following element is the most abundant element in the universe?

a) Oxygen
b) Silicon
c) Hydrogen
d) Methane

- 25) Which of the following gases by the fractional distillation of liquid air?

a) Oxygen
b) Carbon Dioxide
c) Hydrogen
d) Methane

- 26) Which of the following is a not property of gamma particles?

a) They carry a charge of -1
b) They can penetrate Aluminium foil about 15cm – 20cm
c) They are a type of electromagnetic radiation
d) They have a mass approximately equal to a helium nucleus.

- 27) 10g of metal (relative atomic mass 40) are liberated by the passage of 0.5 moles of electrons through molten chloride. What is the formula of chloride?
- M₂Cl
 - MCl
 - MCl₄
 - MCl₂
- 28) In which of the following instances is there no change in the concentration of the solution during electrolysis?
- Concentrated HCl between Carbon Electrodes
 - Copper (ii) sulphate solution between platinum electrodes
 - Copper (ii) sulphate solution between copper electrodes
 - Dilute Sulphuric Acid between Platinum electrodes
- 29) Which one of the following correctly describes the process occurring at the electrodes when molten sodium chloride is electrolyzed?
- | Anode | Cathode |
|--------------|-----------|
| A) Oxidation | Oxidation |
| B) Reduction | Reduction |
| C) Oxidation | Reduction |
| D) Reduction | Oxidation |
- 30) Which one of the following compounds in a 1 mol/dm³ solution has the lowest pH value?
- Ammonia
 - Hydrochloric Acid
 - Ethanoic Acid
 - Sodium Hydroxide

SECTION III (PHYSICS)

- 1) What are the dimensions of work?
- (MLT⁻¹)
 - (M²LT⁻²)
 - (ML²T⁻²)
 - (MLT)
- 2) Resultant of two forces 30N and 40N acting perpendicular to each other is
- 70N
 - 10N
 - 50N
 - 40N

- 3) If the momentum of a body is doubled. Its K.E. will be
- Increase $\sqrt{2}$ times
 - Increase $\sqrt{3}$ times
 - Be doubled
 - Be quadrupled
- 4) First of all 'G' was measured in laboratory by
- Newton
 - Kepler
 - Cavendish
 - Plank
- 5) A body mass 'm' is dropped from the top of a tower during the first second it will fall through.
- 9.8 m
 - 4.9 m
 - 98 m
 - 49 m
- 6) Angular displacement is measured in
- Radians
 - Meters
 - Foot
 - Pound
- 7) A simple pendulum is oscillating. When the bob is at its highest point, the string suddenly breaks, the bob will
- Keep on moving in the circular path
 - Move horizontally
 - Move in a projectile motion
 - Move vertically downward with an acceleration of 9.8 ms^{-2}
- 8) A body is running along a circular path of circumference 44m and complete one round trip in 11 seconds. His displacement at the end of one trip is:
- 4 m
 - 44 m
 - Zero
 - 484 m
- 9) A ball 'A' is dropped vertically from the top of a building 49m high at the same time another ball 'B' is thrown horizontally with a speed of 21 m/s. (Neglecting the resistance of air)
- Ball 'A' will reach first than ball 'B'
 - Ball 'B' will reach first than ball 'A'
 - Both balls will reach the ground at the same time

10) If $\vec{A} \times \vec{B} = \begin{vmatrix} i & j & k \\ 2 & 3 & 4 \\ 1 & 5 & 2 \end{vmatrix}$ then vector $\vec{B}$ is:

- a) $2i + 3j + 4k$
- b) $i + 5j + 2k$
- c) $-142 + 7k$
- d) $2i + 5j + 4k$

11) Out of all the existing four forces the weakest force is:

- a) Gravitational force
- b) Strong force
- c) Electromagnetic force
- d) Weak Force

12) The relation between horse power and watt is:

- a) 1 hp = 546 watts
- b) 1 hp = 1000 watts
- c) 1 hp = 746 watts
- d) 1 hp = 476 watts

13) Velocity of Sound in Vacuum is:

- a) 332m/s
- b) 820 m/s
- c) 224 m/s
- d) Zero

14) One bel is equal to

- a) 10 decibel
- b) 100 decibel
- c) 1000 decibel
- d) 0.1 decibel

15) In Newton's ring when interference pattern is viewed from the above by means of reflected light, the central spot always appears:

- a) White
- b) Black
- c) Blue
- d) Red

16) The frequency of the Second pendulum is

- a) 2Hz
- b) 0.5Hz
- c) 20Hz
- d) 0.05Hz

17) The weight of 20kg of a body at the center of earth is

- a) 19.6 N
- b) 196 N
- c) 0 N
- d) 1.96 N

18) If the distance between two bodies is doubled and their masses are also doubled the gravitational force will:

- a) Increase four times
- b) Decrease four times
- c) Remain constant
- d) Increase two times

19) Weight of an object on the moon is 30N. Its weight on the earth will be

- a) 105 N
- b) 180 N
- c) 60 N
- d) 150 N

20) The power of lens is measured is

- a) Watt
- b) Joule
- c) Diopter
- d) Metre

21) SI unit of magnetic field is

- a) Gauss
- b) Tesla
- c) Weber/m²
- d) Ampere

22) The minimum charge on any object cannot be less than:

- a) 1 coulomb
- b) 1.6×10^{-19} coulomb
- c) 9.1×10^{-31} coulomb
- d) 8.8×10^{-12} coulomb

23) The number of neutron in ${}_{92}\text{U}^{235}$ is

- a) 92
- b) 235
- c) 143
- d) 327

- 24) the most appropriate device used to measure the EMF of a cell is
- Voltmeter
 - Potentiometer
 - Avometer
 - Ammeter
- 25) Which quantity can be measuring is unit of Joule/coulomb
- Current
 - Potential Difference
 - Power
 - Resistance
- 26) A 40 watt lamp turns half the electrical energy it uses into light energy. How much light energy does it give out is 10 seconds.
- 400J
 - 200J
 - 800J
 - 20J
- 27) What is the smallest total resistance which can be obtained using only 6Ω and 12Ω resistor.
- 2Ω
 - 4Ω
 - 6Ω
 - 3Ω
- 28) Two capacitors of $5\mu\text{f}$ and $10\mu\text{f}$ are connected in series. The total capacitance is:
- $15\mu\text{f}$
 - $5\mu\text{f}$
 - $3.3\mu\text{f}$
 - $1.11\mu\text{f}$
- 29) Which of the following is correct answer?
- $1\text{ F} > 1\text{ C}$
 - $1\text{ F} < 1\text{ C}$
 - $1\text{ F} = 1\text{ C}$
 - None of the above
- 30) A device which converts AC into DC voltage is called
- An amplifier
 - A rectifier
 - A voltmeter
 - An oscillator

SECTION IV (ENGLISH)

Select the lettered word or set of words that best completes the sentence:

- 1) Because of the _____ and prolonged nature of the _____ water must be carefully conserved and rationed
- Arid, Reservoir
 - Dire, Forecast
 - Severe, Drought
 - Negligent, Emergency
 - Miserly, Supply
- 2) Generally, fund raising parties are quite labor intensive and not very cost effective in other words putting in a great deal of _____ doesn't mean you will _____ a great deal of income:
- Restraint, obtain
 - Effort, generate
 - Successful, resistance
 - Challenged, discussion
 - Futile, Co-operation

Choose the lettered pair of words that is related in the same way as the pair in capital letters

- 3) Writer : Novelist
- Scientist: astronomer
 - Teacher: student
 - Physician: patient
 - Poet: searcher
 - Worker: Exertion
- 4) Cow: Calf
- Hog: Pork
 - Horse: Mule
 - Sheep: Lamb
 - Tiger: Stripe
 - Ram: Ewe

If there is an error select the one underlined part that must be changed to make the sentence correct.

- 5) Everyone of the shops in the town were closed on Tuesday because of the ten-inch rainfall that had fallen during the day. No Error
- A B C
D E

Dow Grads

- 6) Raising living costs together with escalating taxes, have proved to be a burden for everyone. No Error
- A B C D E

Change the Voice

- 7) He is not asking questions:
- Not asking questions he is
 - Questions are not asked by him
 - Questions are not being asked by him
 - Not asking questions is what he is doing
- 8) Were the crimes confessed by the criminals?
- Was the confession by the criminal made?
 - Were the crimes committed confessed by the criminals?
 - Were the criminals confessing to the crimes?
 - Did the criminals confess to the crimes?

Change the voice into indirect speech

- 9) Aneela Said, "Shahana is not doing her work properly".
- Aneela say that Shahana did not do her work properly
 - Aneela said that Shahana was not doing her work properly
 - Shahana did not do her work properly was said by Aneela
 - She is not doing her work properly is what Aneela said about Shahana
- 10) Fahad said, "All that glitters is not gold".
- Fahad said that all that glitters is not gold
 - All that glitters is not gold is said by Fahad
 - Fahad says that all that glitters is not gold
 - Fahad is saying for all that glitters is not gold

2001
ANSWER SHEET

BIOLOGY		CHEMISTRY		PHYSICS		ENGLISH	
1.	D	1.	C	1.	C	1.	D
2.	D	2.	D	2.	C	2.	A
3.	C	3.	B	3.	C	3.	D
4.	C	4.	C	4.	B	4.	B
5.	C	5.	C	5.	A	5.	A
6.	C	6.	C	6.	A	6.	C
7.	D	7.	B	7.	C	7.	D
8.	B	8.	B	8.	C	8.	C
9.	D	9.	D	9.	C	9.	A
10.	C	10.	C	10.	B	10.	A
11.	D	11.	B	11.	A		
12.	D	12.	D	12.	C		
13.	C	13.	C	13.	A		
14.	B	14.	C	14.	A		
15.	D	15.	D	15.	B		
16.	B	16.	D	16.	B		
17.	B	17.	D	17.	C		
18.	B	18.	A	18.	C		
19.	A	19.	C	19.	B		
20.	C	20.	D	20.	C		
21.	D	21.	A	21.	C		
22.	A	22.	A	22.	B		
23.	B	23.	A	23.	C		
24.	B	24.	B	24.	B		
25.	C	25.	B	25.	C		
26.	C	26.	D	26.	B		
27.	A	27.	D	27.	B		
28.	D	28.	A	28.	C		
29.	C	29.	C	29.	B		
30.	C	30.	B	30.	B		

10 YRS PAST PAPERS

BAHRIA
PAST PAPER - 2002

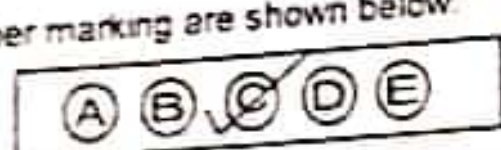
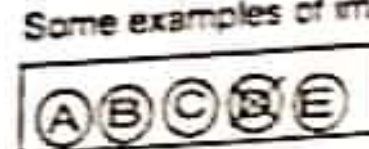
GENERAL INSTRUCTIONS **(PLEASE READ THE INSTRUCTIONS VERY CAREFULLY)**

- 1- Write in bold CAPITALS your Name and Seat Number as given in your Admit Card on allocated column of your Answer sheet.
- 2- Place your signature on the allocated column of the answer sheet.
- 3- This Question paper contains four parts.

Part - I	Biology	45 Questions
Part - II	Chemistry	30 Questions
Part - III	Physics	20 Questions
Part - IV	English	05 Questions
- 4- There will be negative marking in the test. Each correct Answer carries (4) marks. Each wrong Answer carries Minus one (-1) Mark.
- 5- The Answer sheet is made of special paper which gives itself a carbonized copy on the duplicate. All answers must be marked by filling the appropriate circle with pencil.
- 6- For every question in the paper, five choices are given. Please mark your choice by filling in the appropriate circle completely. Making it a dark circle as shown below:



Some examples of improper marking are shown below:



(The computer will mark improperly filled / dual marked circles as wrong answers)

- 8- You are NOT ALLOWED to take away any part of Question paper or note question elsewhere.
- 9- Use your time efficiently and at a premium. Do not spend too much time on one question; you may run short time for other questions.
- 10- Change of answer is not allowed. No marks will be given if two or more circles are filled.
- 11- All answers must be marked on the answer sheet, do not mark any answer on the question paper.

Seat No. _____

Signature of the Candidate _____

NOTE: STOP! DO NOT TURN THIS PAGE TILL FURTHER INSTRUCTIONS.

BAHARIA UNIVERSITY KARACHI 2002 **PART - I (BIOLOGY)**

- 1- The branch of biology, which deals with the life process, and function of processes is known as:
 - a) Physiology
 - b) Pathology
 - c) Histology
 - d) Evolution
- 2- _____ said that protoplasm is the physical basis of life.
 - a) Robert Brown
 - b) Robert Hook
 - c) Purkinje
 - d) Huxley
- 3- In foldings in the Mitochondrial inner membrane is called:
 - a) Cristae
 - b) Thyllakoid
 - c) Glyoxysome
 - d) Chromatic network
- 4- The bond present between carbohydrate units:
 - a) Peptide bond
 - b) Covalent bond
 - c) Glycosidic bond
 - d) Electrostatic bond
- 5- _____ discovered the microscopic creature under microscope in the red blood cells of malarial patient:
 - a) Ronald Ross
 - b) Laveran
 - c) Grassy
 - d) A.F.A. king
- 6- The non living part of protoplasm is:
 - a) Golgi bodies
 - b) Inclusion
 - c) Organelles
 - d) E.R
- 7- The movement of cytoplasm is called:
 - a) Exocytosis
 - b) Endocytosis
 - c) Pinocytosis
 - d) Cyclosis

8- Some bacteria live on dead organic matter they are called:

- a) Parasite
- b) Phages
- c) Saprophytes
- d) Symbionts

9- Sexual reproduction in bacteria is taken by:

- a) Binary fission
- b) Conidia
- c) Conjugation
- d) Zoospore

10- Cynobacteria is the typical example of:

- a) Chlorophyceae
- b) Cynophaceae
- c) Green algae
- d) Fungi

11- The blue pigment found only in blue green algae is:

- a) Chlorophyll
- b) Xanthophyll
- c) Carotses
- d) Phycocyanin

12- Saliva in mouth helps in digestion of:

- a) Bread
- b) Meat
- c) Fat
- d) Table salt

13- The major metabolic factory, detoxification center and storage organ of body is:

- a) Stomach
- b) Brain
- c) Kidney
- d) Liver

14- The active visual receptor of the eye are:

- a) Iris & Pupil
- b) Iris & Lens
- c) Rods & Cones
- d) Cornea & Retina

15- Hemoglobin is:

- a) Respiratory pigment
- b) Excretory pigment
- c) Digestive pigment
- d) Storage pigment

16- The excretory organs in insects are:

- a) Flame cell
- b) Melphigian tubules
- c) Contractile vacuules
- d) Kidneys

17- The kidney does not perform the function of:

- a) Regulation of blood pressure
- b) Secretion of bile
- c) Regulation of pH
- d) Homeostasis

18- Which of the following statement is true for Euglena:

- a) Autotrophic animal
- b) Unicellular Algae
- c) Prokaryotic autotrophy
- d) A-cellular animal

19- During inspiration, the path of air is:

- a) Nostrils – Pharynx – Bronchi – Trachea – Lungs
- b) Nostrils – Bronchi – Pharynx – Trachea – Lungs
- c) Nostrils – Pharynx – Trachea – Bronchi – Lungs
- d) Nostrils – Pharynx – Lungs – Trachea – Bronchi

20- The manufactured food is transported to various regions by a process called:

- a) Translocation
- b) Transpiration
- c) Osmosis
- d) Diffusion

21- The protoplasm is shrunk in the solution of:

- a) Normal saline solution
- b) Isotonic solution
- c) Hypotonic solution
- d) Hypertonic solution

- 22- The natural clock in the human body:
- Pineal body
 - Medulla
 - Pons
 - Cerebellum
- 23- The forebrain is divisible into two components:
- Cerebellum and Cerebral hemisphere
 - Medulla and Pons
 - Telencephalon and Diencephalons
 - Telencephalon and Medulla
- 24- Sugar level in blood is maintained by:
- Liver
 - Kidney
 - Adrenal gland
 - Islet of Langerhans
- 25- In meiosis I crossing over takes place in _____ stage.
- Pachytene
 - Diplotene
 - Zygotene
 - Diakinesis
- 26- Vertebral column is the part of _____ skeleton.
- Axial
 - Appendicular
 - Exoskeleton
 - Hydrostatic
- 27- _____ Muscle pulls the limb outward away from the body.
- Adductor
 - Protractor
 - Abductor
 - Extensor
- 28- _____ Hormone stimulates mammary gland to secrete milk and process of labor.
- Oxytocin
 - F.S.H.
 - L.H.
 - T.S.H.

- 29- _____ germinal layer give rise all the body muscles:
- Ectoderm
 - Endoderm
 - Mesoderm
 - Chorion
- 30- The science of aging is called:
- Ethnology
 - Gerontology
 - Embryology
 - Genetics
- 31- The possible result of crossing between heterozygous (T t) and homozygous dwarf (t t) is.
- 50% tall and 50% Dwarf
 - 100% tall
 - 100% dwarf
 - 75% tall and 25% dwarf
- 32- The tendency of gene in chromosomes to remain together is called:
- Crossing over
 - Synapsis
 - Terminalization
 - Linkage
- 33- The bond between adenine and thymine is:
- Single bond
 - Double bond
 - Triple bond
 - Covalent bond
- 34- _____ mono consists of deeper layer of water.
- Letic
 - Littoral
 - Limnetic
 - Profundal
- 35- Lichen is associated between:
- Fungus and Algae
 - Algae and Bacteria
 - Bacteria and Fungus
 - Bacteria and Virus

36- Extra circular chromosome present in bacteria is called:

- a) Centrosome
- b) Centro mere
- c) Plasmids
- d) D.N.A

37- The person with down's syndrome has:

- a) One extra chromosome
- b) One less chromosome
- c) Two extra chromosome
- d) Two less chromosome

38- The menstrual stage lasts for:

- a) 1 to 5 days
- b) 14 to 16 days
- c) 25 to 28 days
- d) 8 to 18 days

39- The eye spot in Euglena is sensitive for:

- a) Dark
- b) Light
- c) Red light
- d) Blue light

40- The whole tail of tadpole and its gills are lost during metamorphosis the change is:

- a) Progressive
- b) Retrograde
- c) Organizational
- d) None of the above

41- _____ numbers of ATP produce during Aerobic Respiration:

- a) 32 ATP
- b) 34 ATP
- c) 36 ATP
- d) 38 ATP

42- The dark band in muscle fibres are due to:

- a) Actin
- b) Myosin & Actin
- c) Trophomysin
- d) I - band

43- Neurospora is a fungus which is used to understand the principle of:

- a) Reproduction
- b) Locomotion
- c) Inheritance
- d) None of the above

44- The total Autosomes in human are:

- a) 44
- b) 46
- c) 22
- d) 23

45- The results of non - disjunction are:

- a) Down syndrome
- b) Turner syndrome
- c) Klinefelter syndrome
- d) All of the above

PART - II (CHEMISTRY) (2002)

In this part of the test you have 30 Questions, you will have 20 minutes to answer as many as you can.

- 1- The process in which a solid is directly converted to a gaseous state is called:
 - a) Boiling
 - b) Evaporation
 - c) Sublimation
 - d) Freezing
- 2- The no. of atoms nitrogen represented in the formula NH_4NO is,
 - a) 1
 - b) 2
 - c) 3
 - d) 4
- 3- Which of the following techniques could be used to separate the colored dyes from the black ink:
 - a) Chromatography
 - b) Filtration
 - c) Fractional distillation
 - d) Precipitation
- 4- The crystalline form of allotropes are:
 - a) Chemically same
 - b) Chemically different
 - c) Some time same and some time different
 - d) Depend upon substance
- 5- Alpha rays are:
 - a) Single negatively charged particles
 - b) Single positively charged particles
 - c) Double negatively charged particles
 - d) Double positively charged particles
- 6- Which is the lightest among following:
 - a) An atom of hydrogen
 - b) An electron
 - c) A neutron
 - d) A proton
- 7- "In an atom no two electrons can have same set of quantum numbers", this statement was given by:
 - a) Uncertainty principle
 - b) Pauli's exclusion principle
 - c) Hund's rule
 - d) Aufbau's principle

- 8- The bond formed by the electrostatic attraction between negative end of one molecule and positive end of other is called:
 - a) Covalent bond
 - b) Hydrogen bond
 - c) Ionic bond
 - d) Co-Ordinate covalent bond
- 9- The bond angle in a molecule having Sp^3 hybridization with no lone pair is:
 - a) 120 Angle
 - b) 180 Angle
 - c) 109 Angle
 - d) 90 Angle
- 10- Properties of system, which depend upon the amount of substance present in the system are called:
 - a) Chemical properties
 - b) Intensive properties
 - c) Extensive properties
 - d) Physical properties
- 11- Enthalpy is the expression for the:
 - a) Reaction rate
 - b) Heat content
 - c) Energy state
 - d) Activation state
- 12- The heat require to change 1 mole of Iodine (solid) to Iodine (gas) is called:
 - a) Heat of vaporization
 - b) Heat of sublimation
 - c) Heat of fusion
 - d) Melting point
- 13- In the electrolysis of CuCl_2 (Copper Chloride) the substance liberated at the anode:
 - a) Copper
 - b) Chloride
 - c) Hydrogen
 - d) Copper Chloride
- 14- Factors affecting the balance of a chemical equilibrium are following except:
 - a) Concentration
 - b) Temperature
 - c) Enthalpy
 - d) Pressure

- 15- The formula for general gas equation is:
- $P_1 V_1 = P_2 V_2$
 - $V_1/T_1 = V_2/T_2$
 - $E = mc^2$
 - $P_1 V_1/T_1 = P_2 V_2/T_2$
- 16- The electronic configurations of the noble gases are:
- ns², np⁵
 - ns², np⁶
 - ns², np⁴, nd⁸
 - ns², np⁶, nd¹⁰
- 17- The electro-negativity of the element is increased:
- Down the group
 - Across the period from right to left
 - Across the period from left to right
 - None of the above
- 18- Which of the following is the most suitable allotropy of sulphur:
- Plastic sulphur
 - Monoclinic sulphur
 - Rhombic sulphur
 - Laughing gas
- 19- S – black element have the following properties except:
- Reaction with water gives hydroxides
 - Reaction with water gives different oxides
 - Reaction with metals to give binary ionic salts
 - Reaction with non-metals to give binary ionic salts
- 20- Melting and Boiling points of transitive elements is high because:
- Half filled d-orbit
 - Metabolic bond
 - Small atomic size and valence electrons
 - None of the above
- 21- The compound formed by the action of the water on the calcium carbide is:
- Ethane
 - Ethene
 - Acetylene
 - Methane

- 22- The fat soluble vitamins is:
- Vitamin A
 - Vitamin B-12
 - Niacin
 - None of the above
- 23- The formula which indicates the relative numbers of different kinds of the atoms in a molecule is called:
- Structural formula
 - Empirical formula
 - Molecular formula
 - None of the above
- 24- Methane is obtained by which following method:
- Electrolysis of sodium acetate
 - Electrolysis of aluminum carbide
 - Heating sodium acetate with soda lime
 - All of these
- 25- The following reactions is an example of $C_2H_6 + \text{---} \xrightarrow{\text{heat}} C_2H_5Cl + 2HCl$
- Addition reaction
 - Reduction reaction
 - Oxidation reaction
 - Substitution reaction
- 26- In which of the following reaction the rate of nucleophilic substitution depend upon the concentration of the alkyl halide only:
- SN₁
 - SN₂
 - SN₃
 - Addition reaction
- 27- In the phenol:
- Alcoholic group attached on ring structure
 - Hydroxyl group attached on benzene ring
 - Alcoholic group attached on benzene ring
 - All of the above
- 28- P.V.C. prepared from:
- Ethene and hydrogen chloride
 - Ethane and hydrogen chloride
 - Methane and hydrogen chloride
 - Acetylene and hydrogen chloride

29- Methyl alcohol is also known as:

- a) Wood spirit
- b) Wood naphtha
- c) Methanol
- d) All of these

30- Commercially nitric acid is prepared by the:

- a) Oxidation of ammonia by Bayer's method
- b) Oxidation of ammonia by Oswald's method
- c) Reduction of ammonia by Oswald's method
- d) Oxidation of ammonia by Hoop's process

PART - III (PHYSICS) (2002)

In this part of the test you have 20 Questions, you will have 15 minutes to answer as many as you can.

1- Which of the following quantities are vector:

- a) Time and Mass
- b) Force and Displacement
- c) Length and Weight
- d) Temperature and Density

2- The momentum of the body increase by 20% what is the % increase in it's K.E.:

- a) 36
- b) 44
- c) 52
- d) 60

3- Whine one of the following statement is correct:

- a) A body has a constant velocity but varying speed
- b) A body has a constant speed but varying acceleration
- c) A body has a constant speed can not has a acceleration
- d) A body has a constant speed can has varying velocity

4- Centripetal force 'FC' is given by:

- a) mv^2/r
- b) mv/r
- c) m^2v/l
- d) mv/l

5- Value of 'G' is:

- a) $6.667 \times 10^{11} \text{ N-M}^2/\text{Kg}$
- b) $6.667 \times 10^{-11} \text{ N-M/Kg}^2$
- c) $6.667 \times 10^{13} \text{ N-M}^2/\text{Kg}$
- d) $6.667 \times 10^{-11} \text{ N-M}^2/\text{Kg}^2$

6- If we go away from the surface of the earth at a distance equal to radius of the earth then the value of 'g' will be:

- a) $1/4$
- b) $1/8$
- c) $1/2$
- d) $1/9$

7- The energy of a body at height 'h' from the center of earth to the which $g = 0$ is called:

- a) Kinetic energy
- b) Potential energy
- c) Gravitational energy
- d) Absolute potential energy

8- Which one of the following is not regarded as a fundamental quantity in physics:

- a) Length
- b) Time
- c) Mass
- d) Force

9- Acoustics is the study:

- a) Production of sound
- b) Production and properties of the sound
- c) Source of sound
- d) Intensity, Frequency and quality of sound

10- Which of the given condition is not a possible to discuss the Doppler's effect;

- a) When the listener is at moving and the source is at rest
- b) When the listener is at rest and the source is at moving
- c) When the listener and the source are at rest
- d) When the listener and source are at moving

11- Michelson interferometer is the instrument used to record the interference pattern of the light used to calculate:

- a) Interaction of light
- b) Wave length
- c) Refraction of light
- d) Diffraction of light

12- The specific heat time the mass of a body is called:

- a) Specific heat mass
- b) Thermal capacity
- c) Internal energy
- d) Heat capacity

13- The electric field intensity at a distance of 1m from a point of charge to 1 micro coulomb is:

- a) 1/4
- b) 9×10^9 N/C
- c) 9×10^6 N/C
- d) 9×10^3 N/C

14- The unit of rate of change of momentum is:

- a) N
- b) NC
- c) NS2
- d) N/S

15- Dot product of force and displacement is:

- a) Momentum
- b) Moment
- c) Power
- d) Work

16- Work done in unit time is:

- a) Pressure
- b) Work
- c) Energy
- d) Power

17- Erg is unit of energy in:

- a) MKS
- b) SI
- c) FPS
- d) CGS

18- Number of vibration in 1sec is:

- a) Time period
- b) Frequency
- c) Amplitude
- d) None of these

19- Total energy of vibrating body is:

- a) Increase
- b) Decrease
- c) Constant
- d) None of these

20- The waves in which motion of particles of medium and propagation of wave are in same direction are:

- a) Longitudinal
- b) Transverse
- c) Electromagnetic
- d) None of these

PART - IV ENGLISH 2002

In this part of the test you have 05 Questions, you will have 5 minutes to answer as many as you can.

Note: Read the paragraph carefully and gives the best answer of the given questions:

College is the practicing ground for the student. It is not sufficient that he gives himself up to studies. He must develop the noble qualities of showing respect to elders and teachers create a team spirit and participate in the various extra curricular activities and maintain college discipline to raise the prestige of his college:

1- What part does college life play in the training of the student?

- a) College life groom and prepares a student for life
- b) Prepare the student in academics
- c) Develops noble qualities in academics
- d) Help the students to take part in extra curricular activities

2- Choose the suitable preposition:

Rich lady found grate pleasure _____ showing off her riches.

- a) In
- b) For
- c) At
- d) By

3- BANE (Synonym)

- a) Banter
- b) Mystery
- c) Exile
- d) Affliction

4- TENOUNS (Antonym)

- a) Extreme thinness
- b) Heavy
- c) Long lasting
- d) Strong

5- LUCRATIVE (Synonym)

- a) Very profitable
- b) Generous
- c) Over Crowded
- d) Defeatism

BAHRIA UNIVERSITY
September 2002
ANSWER SHEET

Part - I 45 Questions				Part - II 30 Questions		Part - III 20 Questions		Part - IV 05 Questions	
BIOLOGY				CHEMISTRY		PHYSICS		ENGLISH	
1	A	31	A	1	C	1	B	1	A
2	D	32	D	2	B	2	B	2	A
3	A	33	B	3	A	3	D	3	D
4	C	34	D	4	A	4	A	4	D
5	B	35	A	5	D	5	D	5	A
6	B	36	C	6	B	6	A		
7	D	37	A	7	B	7	D		
8	C	38	A	8	B	8	D		
9	C	39	B	9	C	9	B		
10	B	40	B	10	C	10	C		
11	D	41	D	11	B	11	B		
12	A	42	B	12	B	12	D		
13	D	43	C	13	B	13	D		
14	C	44	A	14	C	14	A		
15	A	45	D	15	D	15	D		
16	B			16	B	16	D		
17	B			17	C	17	D		
18	A			18	C	18	B		
19	C			19	C	19	C		
20	A			20	B	20	A		
21	D			21	C				
22	A			22	A				
23	C			23	C				
24	D			24	C				
25	B			25	D				
26	A			26	A				
27	C			27	D				
28	A			28	D				
29	C			29	D				
30	B			30	B				

10 YRS PAST PAPERS

BAHRIA PAST PAPER - 2003

GENERAL INSTRUCTIONS

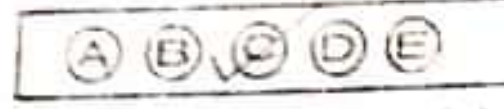
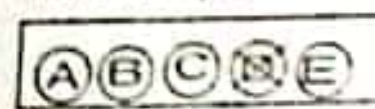
(PLEASE READ THE INSTRUCTIONS VERY CAREFULLY)

- 1- Write in bold CAPITALS your Name and Seat Number as given in your Admit Card on allocated column of your Answer sheet.
- 2- Place your signature on the allocated column of the answer sheet.
- 3- This Question paper contains four parts.

Part - I	Biology	30 Questions
Part - II	Chemistry	30 Questions
Part - III	Physics	30 Questions
Part - IV	English	10 Questions
- 4- There will be negative marking in the test. Each correct Answer carries (+1) marks. Each wrong Answer carries Minus one (-1) Mark.
- 5- The Answer sheet is made of special paper, which gives itself a carbonized copy on the duplicate. All answers must be marked by filling the appropriate circle with pencil.
- 6- For every question in the paper, five choices are given. Please mark your choice by filling in the appropriate circle completely. Making it a dark circle as shown below.



- 7- Some examples of improper marking are shown below.



(The computer will mark improperly filled / dual marked circles as wrong answers)

- 8- You are NOT ALLOWED to take away any part of Question paper or note question elsewhere.
- 9- Use your time efficiently and at a premium. Do not spend too much time on one question, you may run short of time for other questions.
- 10- Change of answer is not allowed. No marks will be given if two or more circles are filled.
- 11- All answers must be marked on the answer sheet, do not mark any answer on the question paper.

Seat No. _____

Signature of the Candidate _____

NOTE: STOP! DO NOT TURN THIS PAGE TILL FURTHER INSTRUCTIONS.

2013

SECTION-I Biology

1. Human liver secretes _____.
a) Bile b) Pepsin c) Vit D d) Cholecystokinin
2. Dalbergia sisso is botanical name of _____.
a) Sesbania b) Edible pea
c) Sweet pea d) Red wood
3. Digestion of cell contents by lysosomes is termed as _____.
a) Endocytosis b) Autophagy
c) Immunolysis d) Exocytosis
4. Coniferous forests are found at _____.
a) High altitudes & latitudes
b) Low altitudes & latitudes
c) High longitudes & latitudes
d) High altitudes & longitudes
5. Fungi cause many diseases in humans such as _____.
a) Elephantiasis b) Leshmaniasis
c) Warts d) Moniasis
6. Depletion of ozone layer is due to CFC which release _____.
a) Cl atoms b) CO c) F d) CCl₄
7. Fertilization in mammals takes place in _____.
a) Uterus b) Ovary c) Vagina d) Fallopian tube
8. Extremely small amount of yolk are called _____.
a) Alecithal b) Polylecithal c) Oligolecithal
9. Ectoderm give rise to _____.
a) Muscular system b) Vascular system
c) Nervous system d) Digestive system
10. Science of aging is called _____.
a) Palaentology b) Entomology c) Gerantology d) Parasitology

11. The organism in which both extracellular and intracellular digestion takes place is _____.

- a) Hydra b) Amoeba c) Paramecium d) Planaria

12. Hydra and corals are classified in _____.

- a) Coelenterate b) Porifera
c) Echinodermata d) Mollusca

13. Snail is included in the class _____.

- a) Gastropoda b) Bivalvia c) Arachnida d) Myriapoda

14. Ascaris lives in the _____ of man.

- a) Intestine b) Liver c) Blood d) Bladder

15. Sea star moves with the help of _____.

- a) Pseudopodia b) Tube feet c) Cilia d) Flagellum

16. Pike and trout are:

- a) Jawless fishes b) Bony fishes.
c) Cartilaginous fishes. d) All of the above

17. Defensive cells of cnidarians are called _____.

- a) Nematocysts b) Gonocytes c) Oocytes d) Gastrocytes

18. Growth of pollen tube towards ovary is an example of _____ movement.

- a) Phototropic b) Chemotactic
c) Sensory d) Mechanical

19. Cells which are absent in gymnosperms but present in angiosperms are _____.

- a) Vessels b) Tracheids
c) Collenchyma d) Parenchyma

20. Under secretion of thyroxine in infants causes _____.

- a) Cretinism b) Dwarfism c) Gigantism d) Osteoporosis

21. Protein synthesis takes place in _____.

- a) Ribosomes b) Mitochondria
c) Mesosomes d) Golgi bodies

22. Major oil & gas fields in Pakistan are located in _____.

- a) Potohar b) Ghotki c) D/I Khan d) FATA

23. The humidity in air is inversely proportional to _____ process in plants.

- a) Transpiration b) Respiration
c) Photosynthesis d) None of the above

24. Shrinkage of protoplasm from cell wall under hypertonic solution is called _____.

- a) Plasmolysis b) Endosmosis c) Protolysis d) Torsion

25. Acetyl CoA loses CoA and combines with a 4-Carbon compound which is:

- a) Oxaloacetic acid) b) Citric acid)
c) Succinic acid) d) Malic acid

26. Product of anaerobic respiration in animals is _____.

- a) Pyruvic acid b) Lactic acid
c) Alcohol d) NADPH₂

27. Seedless fruits which grow without fertilization are called _____.

- a) Angiosperms b) Parthenocarpus
c) Ovulatory d) Messangial

28. If temperature is reduced to near or below freezing point enzymes become:

- a) Denatured b) Dormant
c) Renatured d) All of these

29. The feathers which help the birds in flying are:

- a) Quill feathers b) Hair feathers
c) Down feathers d) Plumulae

30. Sugar is stored in the liver as:

- a) Starch b) Glycogen c) Maltose d) Cellulose

SECTION-II CHEMISTRY

31. Platinum or polonium are used as catalysts in:-

- a) Haber process b) Contact Process c) Oxidation of NH₃

32. For "f" orbital value of l is:-

- A) 3 B) 2 C) 0 D) 1

33. Characteristic of number 1000 (in log) is:-

- A) 3 B) 2 C) 4 D) 0

34. Perrie Curie and Maire Curie isolated a new radioactive element which was
A) Radium B) Thorium C) Uranium
35. Faraday's experiment leads to the discovery of
A) Electron B) Positron C) Nucleus D) Proton
36. S.I unit of viscosity is:-
A) Ns m^{-1} B) Ns m^{-2} C) $\text{Ns}^{-1} \text{m}^2$
37. 1 poise can be fined as:-
A) 1 gm/m B) 1 gm/cm C) 1 gm/mm
38. Excited state of C atom can be represented as:-
A) $1s^2 2s^2 2p_x^1 2p_y^1 2p_z^1$
B) $1s^1 2s^2 2p_x^1 2p_y^1 2p_z^1$
C) $1s^2 2s^2 2p_x^1 2p_y^1 2p_z^1$
39. Except sucrose all oligosaccharides are :-
A) Reducing B) Non reducing C) none
40. Ketone can be oxidized by using:-
A) Tollen's reagent B) Fehling solution
C) $\text{K}_2\text{Cr}_2\text{O}_7$ with H_2SO_4 D) $\text{K}_2\text{Cr}_2\text{O}_7$ with HNO_3
41. $(\text{BeH}_2)_n$ is an example of
A) Border line Hydride B) Covalent Hydride
C) Ionic Hydride D) Polymeric Hydride
42. S- block elements are powerful:-
A) Oxidizing agents B) Reducing agents
C) None D) Both
43. "The electrons are filled to the orbitals of lowest energy in sequence, two to each orbital", this is statement of:-
A) Pauli's exclusion principle B) Aufbau Principle
C) $(n+l)$ Rule D) Hund's Rule
44. Anomalous behavior & high B)P of H_2O is due to:-
A) Strong covalent bond B) Hydrogen bonding
C) Ionic bonding D) Waver val forces

45. Two substances having same crystal structure are said to be:-
A) Polymorphous B) Isomorphous
C) Homomorphous D) Synmorphous
46. Molarity of 10 gm of NaOH dissolved in 1 dm³ of solution is:-
A) 2 M B) 0.25 M C) 10 M D) 20 M
47. Which one is correct:-
A) Glass is not a true solid b/c molecules are not arranged in definite manner
B) Glass is a true solid b/c molecules are not arranged in a definite manner
48. A solution which tends to resist change in pH is:-
A) Boric solution B) Basic solution
C) Buffer solution D) Neutral solution
49. A substance which does not lower energy of activation but combines with reactant molecule is called:-
A) Catalyst B) Negative catalyst
C) Positive catalyst D) none
50. Which one is not correct:-
A) Alpha particles are negatively charged
B) Alpha particles are helium nuclei
C) Alpha particles are positively charged
D) Alpha particles move towards anode
51. Which one shows alpha- decay:-
a) ${}_Z^AX^A \longrightarrow {}_Z^AX^{A-4} + {}_2^4\text{He}^4$
b) ${}_Z^AX^A \longrightarrow {}_Z^AX^{A+4} + {}_2^4\text{He}^4$
c) ${}_Z^AX^A \longrightarrow {}_Z^AX^{A-4} + {}_2^4\text{He}^4$
d) ${}_Z^AX^A \longrightarrow {}_Z^AX^{A+4} + {}_2^4\text{He}^4$
52. MgCl_2 is surrounded by _____ number of water molecules:-
A) 2 B) 6 C) 5 D) 8
53. Gas constant "R" is independent of:-
A) Pressure of gas B) Volume of gas
C) Nature of gas D) Temperature
54. Which has shortest C-C bond length:-
A) C_2H_2 B) C_2H_4
C) CCl_4 D) $\text{C}_2\text{H}_5\text{OH}$

55. Cyclohexane is an example of:-

- A) Acyclic aromatic compound)
- B) Alicyclic aliphatic compound)
- C) Alicyclic homocyclic compound)
- D) none of the above

56. Which of the following is an example of decrease in entropy:-

- A) Evaporation of H_2O
- B) Sublimation of ice

57. Which indicator has green colour in neutral solution .

- A) Phenolphthalein
- B) Universal indicator
- C) Methyl orange
- D) acid-base indicator

58. An example of covalent hydride is:

- A) NaH
- B) CaH_2
- C) SrH_2
- D) AsH_3

59. Which is the most basic of the following oxides:

- A) Na_2O
- B) BaO
- C) As_2O_3
- D) Al_2O_3

60. Corrosion of a metal produces:

- A) Chloride
- B) Sulphate
- C) Nitrite
- D) Oxide

SECTION-III PHYSICS

61. A car starts from rest and acceleration at 4 m/s^2 find its final velocity after 5 seconds.

- A) 25 ms^{-1}
- B) 15 ms^{-1}
- C) 20 ms^{-1}
- D) 15 ms^{-1}

62. 1 Kwatt hr =

- A) $36 \times 10^5 \text{ Joule}$
- B) $3.6 \times 10^5 \text{ Joule}$
- C) $3.6 \times 10^5 \text{ Joule}$

63. A field in which work done along a closed path is zero is called)

- a) Gravitational field)
- b) Conservative field)
- c) Electric field)
- d) Magnetic field)

64. Absolute P.E of a mass "m" lying at the surface of earth is:-

- A) $\frac{G M m}{R}$
- B) $\frac{G M m}{R^2}$
- C) $\frac{G M}{R}$
- D) $-\frac{R M m}{G}$

65. Work done by centripetal force is

- A) Maximum
- B) Zero
- C) Negative

66. Two vectors are lying at x and y plane, their resultant will be directed along

- A) x - axis
- B) y - axis
- C) z - axis
- D) None of the above

67. A boy standing in a free falling lift drops a ball, then ball will:-

- A) Fall down
- B) Goes up
- C) Remain stationary
- D) None of the above

68. The point at which whole mass of the body is concentrated is called:-

- A) Centre of gravity
- B) centre of mass
- C) Centre of action
- D) centre of body

69. Stokes observed that an upward retarding force acts on a body falling in a liquid is given by:-

- A) $F = 6 \pi \eta r^2 v$
- B) $F = 6 \pi \eta r v^2$
- C) $F = 6 \pi \eta r v^2$
- D) $F = 6 \pi \eta r v$

70. Unit of electric intensity is:-

- A) J/C)
- B) N/C)
- C) J/A
- D) N/A

71. Alpha - particles are:-

- A) Hydrogen nuclei
- B) Helium nuclei
- C) Oxygen nuclei
- D) None

72. e/m ratio of an electron is:-

- A) $1.75 \times 10^{11} \text{ C/kg}$
- B) $1.75 \times 10^5 \text{ C/kg}$
- C) $1.75 \times 10^5 \text{ C/kg}$

73. Vector product of a vector by itself is :-

- A) Maximum
- B) Zero
- C) Double
- D) None

74. The frictional force of a car taking turn on a circular track is equal to.

- A) Centripetal Force
- B) Centrifugal force
- C) Magnetic force
- D) Gravitational force

75. 1 H.P = _____ watt

- A) 750 watt
- B) 746 watts
- C) 756 watt
- D) 740 watt

76. Two forces equal in magnitude, opposite in direction and not acting along same line constitute a:

- A) Torque B) Couple
C) Moment of arm D) None

77. Points of zero amplitude are called:

- A) Nodes B) Antinodes C) Both D) None

78. Two waves of same frequency and amplitude travelling in opposite direction give rise to:-

- A) Transverse waves B) Longitudinal waves
C) Standing waves D) Electromagnetic waves

79. Two vectors of magnitude 3 & 4 have resultant of 5, the angle between vectors is:-

- A) 90° B) 120° C) 45° D) 60°

80. Layer of soap is coloured due to:-

- A) Interference B) Polarization
C) Transmission D) None

81. In Young's slit experiment, 2 waves interfere constructively, if path difference is:-

- a) $m\lambda$ b) $(m-1)\lambda$ c) $(m+1)\lambda$ d) $(m-1)\lambda/2$

82. If length of pendulum become 4 times, time period will be:-

- A) $T/2$ B) $4T$ C) $2T$ D) $T/4$

83. A projectile is launched at an angle θ to the horizontal with an initial velocity v . In the absence of air resistance, which of the following statements is correct?

- a) The horizontal velocity increases during flight
b) The horizontal velocity remains constant during flight
c) The horizontal velocity decreases during flight
d) The vertical acceleration increases during flight
e) The vertical acceleration decreases during flight

84. ^{235}U nucleus is from a ^{238}U nucleus in that it contains three more

- a) Neutrons
b) Alpha particles
c) Protons
d) Electrons
e) Positrons

85. Motor operates from a 120-volt electrical source and uses 4 amperes of current. What is the power generated by the motor?

- a) 30 W
b) 48 W
c) 120 W
d) 480 W
e) 640 W

86. The vector quantity in following is

- a) Distance
b) Impulse
c) Energy
d) Temperature

87. The angle of projection of projectile for which its maximum height is equal to one fourth of its horizontal range is

- a) 30°
b) 45°
c) 60°
d) 75°

88. Reactions in which heavy nuclei split into nuclei of intermediate mass number are called:

- a) Fusion reaction
b) Fission reaction
c) Meson reaction
d) Annihilation reactions
e) Chemical reactions

89. Two vectors each of magnitude F makes angle of 0° and 270° with x-axis their resultant is:

- a) 0 b) $2F$
c) $2\sqrt{F}$ d) $F\sqrt{2}$

90. Choose the relation which is not correct:

- a) Ampere = coulomb per second
b) Volt = coulomb per joule
c) Ohm = Volt per ampere
d) 1 coulomb = charge of 6.24×10^{18} electrons

SECTION-IV **ENGLISH**

91. She _____ an accident last year, she _____ driven since.
A) has, has not
B) had, has not
C) have, had not
D) had, had not

92. He _____ a prize last year, since he _____ his personality.
A) wins, has change B) won, has changed

93. He did _____ excellent job while was among us.
A) a B) an C) the D) x

94. Flee is
A) An insect
B) A bird
C) An animal

95. Wander means:-
A) Foolish B) Slow C) Roam.

SYNONYMS

96. Novice:
A) Doyen
C) Professional
B) Expert
D) Neophyte

97. Incendiary:
A) Motive
C) Arsonist
B) Goad
D) Insensitive

98. Incense:
A) Cause
C) Irritated
B) Exciting
D) Enrage

ANTONYMS

99. Extol:
A) Defame B) Alienate
C) Conceal D) Refrain E) Reject

100. Efface:
A) Trust B) Restore
C) Embody D) Contract E) Appreciate

RESPONSE SHEET



DOW GRADS

DO IT FROM THOSE WHO HAVE DONE IT NOT FROM THOSE WHO CLAIM!

A large grid of 100 small squares, each containing a letter, arranged in 10 rows and 10 columns. A blue arrow points from the bottom left towards the center of the grid.

College

Percentage	
------------	--

Contact

E-mail _____

DOW Signatory

01. [A] [B] [C] [D] [E] 34. [A] [B] [C] [D] [E]

- | | | | | | |
|-----|-----------------|----|-----------------|-----|-----------------|
| 01. | [A][B][C][D][E] | 34 | [A][B][C][D][E] | 68 | [A][B][C][D][E] |
| 02 | [A][B][C][D][E] | 35 | [A][B][C][D][E] | 69 | [A][B][C][D][E] |
| 03 | [A][B][C][D][E] | 36 | [A][B][C][D][E] | 70 | [A][B][C][D][E] |
| 04 | [A][B][C][D][E] | 37 | [A][B][C][D][E] | 71 | [A][B][C][D][E] |
| 05 | [A][B][C][D][E] | 38 | [A][B][C][D][E] | 72 | [A][B][C][D][E] |
| 06 | [A][B][C][D][E] | 39 | [A][B][C][D][E] | 73 | [A][B][C][D][E] |
| 07 | [A][B][C][D][E] | 40 | [A][B][C][D][E] | 74 | [A][B][C][D][E] |
| 08 | [A][B][C][D][E] | 41 | [A][B][C][D][E] | 75 | [A][B][C][D][E] |
| 09 | [A][B][C][D][E] | 42 | [A][B][C][D][E] | 76 | [A][B][C][D][E] |
| 10 | [A][B][C][D][E] | 43 | [A][B][C][D][E] | 77 | [A][B][C][D][E] |
| 11 | [A][B][C][D][E] | 44 | [A][B][C][D][E] | 78 | [A][B][C][D][E] |
| 12 | [A][B][C][D][E] | 45 | [A][B][C][D][E] | 79 | [A][B][C][D][E] |
| 13 | [A][B][C][D][E] | 46 | [A][B][C][D][E] | 80 | [A][B][C][D][E] |
| 14 | [A][B][C][D][E] | 47 | [A][B][C][D][E] | 81 | [A][B][C][D][E] |
| 15 | [A][B][C][D][E] | 48 | [A][B][C][D][E] | 82 | [A][B][C][D][E] |
| 16 | [A][B][C][D][E] | 49 | [A][B][C][D][E] | 83 | [A][B][C][D][E] |
| 17 | [A][B][C][D][E] | 50 | [A][B][C][D][E] | 84 | [A][B][C][D][E] |
| 18 | [A][B][C][D][E] | 51 | [A][B][C][D][E] | 85 | [A][B][C][D][E] |
| 19 | [A][B][C][D][E] | 52 | [A][B][C][D][E] | 86 | [A][B][C][D][E] |
| 20 | [A][B][C][D][E] | 53 | [A][B][C][D][E] | 87 | [A][B][C][D][E] |
| 21 | [A][B][C][D][E] | 54 | [A][B][C][D][E] | 88 | [A][B][C][D][E] |
| 22 | [A][B][C][D][E] | 55 | [A][B][C][D][E] | 89 | [A][B][C][D][E] |
| 23 | [A][B][C][D][E] | 56 | [A][B][C][D][E] | 90 | [A][B][C][D][E] |
| 24 | [A][B][C][D][E] | 57 | [A][B][C][D][E] | 91 | [A][B][C][D][E] |
| 25 | [A][B][C][D][E] | 58 | [A][B][C][D][E] | 92 | [A][B][C][D][E] |
| 26 | [A][B][C][D][E] | 59 | [A][B][C][D][E] | 93 | [A][B][C][D][E] |
| 27 | [A][B][C][D][E] | 60 | [A][B][C][D][E] | 94 | [A][B][C][D][E] |
| 28 | [A][B][C][D][E] | 61 | [A][B][C][D][E] | 95 | [A][B][C][D][E] |
| 29 | [A][B][C][D][E] | 62 | [A][B][C][D][E] | 96 | [A][B][C][D][E] |
| 30 | [A][B][C][D][E] | 63 | [A][B][C][D][E] | 97 | [A][B][C][D][E] |
| 31 | [A][B][C][D][E] | 64 | [A][B][C][D][E] | 98 | [A][B][C][D][E] |
| 32 | [A][B][C][D][E] | 65 | [A][B][C][D][E] | 99 | [A][B][C][D][E] |
| 33 | [A][B][C][D][E] | 66 | [A][B][C][D][E] | 100 | [A][B][C][D][E] |

SECTION I BIOLOGY		SECTION II CHEMISTRY		SECTION III PHYSICS		SECTION IV ENGLISH	
01	A	31	C	61	C	91	B
02	D	32	A	62	B	92	B
03	B	33	A	63	B	93	B
04	B	34	A	64	A	94	A
05	D	35	A	65	B	95	C
06	A	36	B	66	D	96	D
07	D	37	B	67	C	97	C
08	C	38	C	68	B	98	C
09	C	39	A	69	D	99	B
10	C	40	C	70	B	100	B
11	A	41	D	71	B		
12	A	42	B	72	A		
13	A	43	B	73	B		
14	A	44	B	74	B		
15	B	45	B	75	B		
16	C	46	B	76	B		
17	A	47	A	77	A		
18	B	48	C	78	C		
19	D	49	B	79	A		
20	A	50	A	80	A		
21	A	51	B	81	B		
22	C	52	B	82	C		
23	A	53	C	83	B		
24	A	54	A	84	A		
25	A	55	C	85	D		
26	B	56	D	86	B		
27	B	57	B	87	B		
28	B	58	B	88	B		
29	D	59	A	89	D		
30	B	60	D	90	D		

10 YRS PAST PAPERS

BAHRIA PAST PAPER - 2004

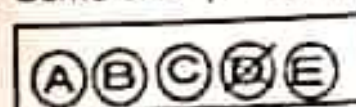
GENERAL INSTRUCTIONS (PLEASE READ THE INSTRUCTIONS VERY CAREFULLY)

- 1- Write in bold CAPITALS your Name and Seat Number as given in your Admit Card on allocated column of your Answer sheet.
- 2- Place your signature on the allocated column of the answer sheet.
- 3- This Question paper contains four parts.

Part - I	Biology	30 Questions
Part - II	Chemistry	30 Questions
Part - III	Physics	30 Questions
Part - IV	English	10 Questions
- 4- There will be negative marking in the test. Each correct Answer carries (4) marks. Each wrong Answer carries Minus one (-1) Mark.
- 5- The Answer sheet is made of special paper. Which gives itself a carbonized copy on the duplicate. All answers must be marked by filling the appropriate circle with pencil.
- 6- For every question in the paper, five choices are given. Please mark your choice by filling in the appropriate circle completely. Making it a dark circle as shown below:



- 7- Some examples of improper marking are shown below



(The computer will mark improperly filled / dual marked circles as wrong answers)

- 8- You are NOT ALLOWED to take away any part of Question paper or note question elsewhere.
- 9- Use your time efficiently and at a premium. Do not spend too much time on one question. you may run short of time for other questions.
- 10- Change of answer is not allowed. No marks will be given if two or more circles are filled.
- 11- All answers must be marked on the answer sheet, do not mark any answer on the question paper.

Seat No. _____

Signature of the Candidate _____

NOTE: STOP! DO NOT TURN THIS PAGE TILL FURTHER INSTRUCTIONS.

BAHRIA UNIVERSITY KARACHI 2004

PART - I (BIOLOGY)

- 1- Which of the following regarding true fungi is correct:
 - a) They are autotrophs
 - b) They have chlorophyll
 - c) Less number of species
 - d) They have mycelium and hyphae
- 2- The number of chromosomes in man is:
 - a) 44
 - b) 48
 - c) 46
 - d) 23
- 3- According to Hayflick the aging is due to loss of:
 - a) Cell function
 - b) Immunity
 - c) Stability
 - d) A O T
- 4- Locomotion by pseudopodium is found in:
 - a) Amoeba
 - b) Chlamydomonas
 - c) Ciliates
 - d) Euglena
- 5- In catkin inflorescence the flowers are:
 - a) Motile
 - b) Sessile
 - c) Stalked
 - d) Branched
- 6- Eggs of amphioxus:
 - a) Alecithal
 - b) Telolecithal
 - c) Meiolecithal
 - d) Mesolecithal
- 7- The condition in which there is one extra chromosome is:
 - a) Monosomic
 - b) Trisomic
 - c) Disomic
 - d) Nullisomic

Dow Grads

- 8- The plants and animals of an area which live successfully and reproduce in particular surrounding constitute:
- Community
 - Population
 - Ecology
 - Biosphere
- 9- The temperature deciduous forest are characterized by plant:
- Maple
 - Pine
 - Fir
 - Spruce
- 10- The proportion of different alleles of a gene in a population is termed as:
- Allele frequency
 - Genetic drift
 - Gene frequency
 - Allelomorph
- 11- The primitive vascular plant rhynia is:
- Pteridophyte
 - Bryophyte
 - Spermatophyte
 - Gametophyte
- 12- The number of chromosomes in sporophytic *ulva*:
- 13
 - 25
 - 24
 - 12
- 13- The botanical name of mustard is:
- Brassica oleracea*
 - Brassica campestris*
 - Brassica napus*
 - Raphanus sativus*
- 14- Some plants form embryo without fertilization called:
- Apomixis
 - Epimixis
 - Apomixes
 - Endomixis
- 15- Circulatory system in grass hopper is of:
- Hemi open type
 - Open type
 - close type
 - Hemi close type

Dow Grads

- 16- When solid particles are ingested by cell through plasma membrane:
- Pinocytosis
 - Phagocytosis
 - Endosmosis
 - Exosmosis
- 17- Who discovered protozoan in the red blood cells of human:
- Leveran
 - King
 - Ronald Ross
 - Grassi
- 18- Chloroplast contain substantial amount of _____ and are capable of programming synthesis of some of their new structural components:
- Chlorophyll
 - DNA
 - RNA
 - Chlorophyll a
- 19- Centres of cellular respiration in a cell is:
- Mitochondrion
 - Cytoplasm
 - Golgi bodies
 - Ribosome
- 20- In earthworm the absorptive ridge which runs along the dorsal side of intestine is:
- Villi
 - Typhlosole
 - Vessel
 - None
- 21- Movement in protozoan specially in sarcodinians is:
- Amoebid
 - Euglenoid
 - Flagellary
 - Brachiation
- 22- Rods and cones (photoreceptor cells) are present in:
- Iris
 - Pupil
 - Cornea
 - retina
- 23- An automatic, preprogrammed and genetically inherited behavior is:
- Learning
 - Instinctive
 - Parental
 - Genetical

24- A division without formation of spindle is:

- a) Mitosis
- b) Meiosis
- c) Amitosis
- d) Karyokinesis

25- In pond ecosystem the macronutrient include:

- a) Iron
- b) Manganese
- c) Copper
- d) Magnesium

26- The % of lipids in protoplasm is about:

- a) 2 to 3%
- b) 2 to 4%
- c) 2 to 5%
- d) 10%

27- Flowering plants which produce seeds within fruits called:

- a) Angiosperms
- b) Gymnosperms
- c) Cryptogams
- d) Pteridophytes

28- The pressure which cell contents exert on cell wall:

- a) Turgor pressure
- b) Wall pressure
- c) Suction pressure
- d) Osmotic pressure

29- The muscles which rotate the whole or part of limb at one of its joint:

- a) Protractor
- b) Retractor
- c) Rotractor
- d) Rotator

30- Leucoplast are:

- a) Found in underground parts of the plant
- b) Found in aerial parts of plants
- c) They have chloroplast
- d) They are found exogenously

PART - II (CHEMISTRY)

Electronic configuration $1s^2, 2s^2, 2p^6, 3s^1$, is of

- a) Sodium
- b) Magnesium
- c) Neon
- d) Calcium

2- In borax $\text{Na}_2\text{B}_4\text{O}_7 \cdot X \text{H}_2\text{O}$, Where $X =$ _____?

- a) 10
- b) 12
- c) 14
- d) 8

3- NaBH_4 is _____ hydride:

- a) Polymeric
- b) Complex
- c) Covalent
- d) Metallic

4- The properties of elements are periodic functions of their atomic number called:

- a) Moseley law
- b) Periodic law
- c) Doberniers law
- d) Newlands law

5- Hydrogen is placed in various groups sometimes in IV-A, VII-A, by different authors. Its actual position is:

- a) Unknown
- b) Undetected
- c) Undecided
- d) None

6- s - block elements are highly:

- a) Oxidizing agents
- b) Reducing agents
- c) Low reacting metalloids
- d) Dehydrating

7- d - block elements are:

- a) Outer transition
- b) Inner transition
- c) Transition
- d) Reactive metals

8- According to I.U.P.A.C. the $\text{CH}_3\text{CH}_2\text{OH}$ is:

- a) Ethanal
- b) Ethanol
- c) Methanol
- d) Alcohol

9- The rate at unity $1\text{mol}/1\text{dm}^3 \cdot 1\text{sec}$ than K is known as:

- a) Rate of reaction
- b) Specific rate constant
- c) Rate constant
- d) Velocity of reaction

10- $2\text{NO} + \text{O}_2 \rightarrow 2\text{NO}_2$ Where as $\frac{dx}{dt} = k[\text{NO}]^2[\text{O}_2]$ then reaction is:

- a) 1st order reaction
- b) 2nd order reaction
- c) 3rd order reaction
- d) 0 order reaction

11- In a reaction:
 $\text{Ca(OH)}_2 + \text{Cl}_2 \rightarrow \text{H}_2\text{O}$

- a) CaO, Cl
- b) HCl
- c) CaO, Cl_2
- d) HOCI

12- In M-shell maximum number of electrons:

- a) 16
- b) 15
- c) 12
- d) 14

13- The heat absorbed or evolved in a given reaction must be independent of particular manner in which reaction takes place.

- a) Hess law
- b) Law of conservation of energy
- c) Law of thermodynamics
- d) Law of mass action

14- The change of enthalpy when 1gm mol of substance is formed from its elements is:

- a) Heat of reaction
- b) Heat of combustion
- c) Heat of formation
- d) Heat of condensation

15- ZnSO_4 and NiSO_4 are _____ crystals.

- a) Cubic
- b) Trigonal
- c) Orthorhombic
- d) Hexagonal

16- The temperature at which rhombic and monoclinic are in equilibrium called:

- a) Critical temperature
- b) Transition temperature
- c) Absolute temperature
- d) None

17- The percentage of HCl in gastric juice is:

- a) 15%
- b) 2%
- c) 0.5%
- d) 3%

18- A group of water soluble vitamin obtained from liver and yeast called:

- a) Vitamin C
- b) Vitamin B complex
- c) Vitamin K
- d) Vitamin A

19- Inorganic salts which contain some elements like N, P, K which are very essential for growth of plants are:

- a) Proteins
- b) Carbohydrates
- c) Fertilizers
- d) Minerals

20- The number of elements essential for life is about:

- a) 23
- b) 22
- c) 18
- d) 37

21- In which compound co-ordinate covalent bond is present:

- a) CH_3OH
- b) POCl_3
- c) H_2O
- d) CHCl_3

22- Atoms of group of atoms present in an organic compound confers the properties of organic compound to that atom or group of atoms called:

- a) Sulphonating group
- b) Nitrating group
- c) Functional group
- d) Hydrating group

23- The molecular mass of SO_2 is 64 and molecular mass of He is 4 then He diffuses how many times greater than SO_2 .

- a) 2 times
- b) 4 times
- c) 16 times
- d) 12 times

24- $[\text{Co(en)}_3(\text{NO}_3)_3]$ is:

- a) Triamine trinitro cobalt III
- b) Tris (ethylene diamine) Cobalt III nitrate
- c) Diamine tetra ethyl Cobalt III
- d) Tri Nitro Trisethylene diamine cobalt III

25- The method was used purification of Bauxite containing excess of silica:

- a) Habers process
- b) Halls Method
- c) Bayers method
- d) Serpex method

26- In the reaction $\text{SO}_2 + \text{O}_2 \rightarrow 2\text{SO}$ $\Delta H = 45 \text{ K.CAL}$ is:

- a) Endothermic reaction
- b) Exothermic reaction
- c) Constitutional reaction
- d) Both A and B

27- In inhalation and alkylation of benzene the process involved is:

- a) Substitution
- b) Addition
- c) Friedel crafts reaction
- d) None of these

28- In a reaction $\text{NaCl} + \text{NH}_4\text{HCO}_3 \rightarrow \text{_____} + \text{NH}_4\text{Cl}$

- a) NaHCO_3
- b) Na_2CO_3
- c) H_2CO_3
- d) $(\text{NH}_4)_2\text{CO}_3$

29- In reaction $2\text{MnO}_4 + 5\text{H}_2\text{C}_2\text{O}_4 + 6\text{H} \rightarrow 2\text{Mn} + \text{_____} + 8\text{H}_2\text{O}$

- a) 10MnCO_3
- b) $10\text{H}_2\text{CO}_3$
- c) 10CO_2
- d) 10MnO_2

30- Maximum number of electrons try to occupy orbitals singly only when all the orbitals are singly occupied only then pairing of electrons commences is:

- a) Hund's rule
- b) Pauli's exclusion principle
- c) N-L rule
- d) Aufbau principle

SECTION - III (PHYSICS) (2004)

1- The diameter of molecule considered to be sphere is:

- a) $3 \times 10^{-10} \text{ m}$
- b) $5 \times 10^{-10} \text{ m}$
- c) $4 \times 10^{-10} \text{ m}$
- d) $7 \times 10^{-10} \text{ m}$

2- When 100w bulb is operated at 240 volts the current flows through is:

- a) 0.416A
- b) 0.466A
- c) 0.496A
- d) 0.455A

3- Wheat stone bridge is used to measure:

- a) Resistance
- b) Current
- c) Voltage
- d) Emf

4- A process in which momentum of system is conserved the law is known as;

- a) Law of conservation of energy
- b) Law of conservation of momentum
- c) Law of inertia
- d) Elastic collision

5- Magnetic lines of is equal to:

- a) $B = \Delta\phi_n / \Delta A_n$
- b) $B = \Delta\phi_n / \Delta A \cos\theta$
- c) $\Delta\phi_n / B \times \Delta A$
- d) $\Delta\phi_n / B \cdot \Delta A$

6- The distance between two consecutive dark or bright fringes is called:

- a) λ
- b) $\lambda/2$
- c) $\lambda/4$
- d) Fringe spacing

7- Electromagnetic waves are:

- a) Transverse
- b) Longitudinal
- c) Mechanical
- d) Material dependent

8- Dopped semiconductors are also known as:

- a) Extrinsic
- b) Intrinsic
- c) Mixed
- d) None of these

9- A material consisting of fissionable isotope called:

- a) Coolants
- b) Reactor fuel
- c) Moderator
- d) Control rods

10- α - particles when shoot up from nucleus with speed of:

- a) $0.1 \times 10^8 \text{ m/s}$
- b) $0.4 \times 10^8 \text{ m/s}$
- c) $0.9 \times 10^8 \text{ m/s}$
- d) $0.6 \times 10^8 \text{ m/s}$

11- Isobaric process takes place at constant:

- a) Pressure
- b) Temperature
- c) Volume
- d) Heat

12- Electromagnetic waves speed depends upon:

- a) Permeability of medium
- b) Permeability of free space
- c) Permeability of free space
- d) Both a and b

13- Electric flux is zero when angle is:

- a) $\theta > 90$
- b) $\theta < 90$
- c) $\theta = 90$
- d) $\theta = 0$

14- Natural earth's heat is referred as:

- a) Tidal energy
- b) Geothermal energy
- c) Mechanical energy
- d) Heat energy

15- In step up transformer the condition is:

- a) $N_s < N_p$
- b) $N_s > N_p$
- c) $N_s = N_p$
- d) None of these

16- .1 Tesla is equal to:

- a) $1\text{N}/1\text{A} \times 1\text{cm}$
- b) $1\text{N}/1\text{A} \times 1\text{m}$
- c) $1\text{A} \times 1\text{m}/1\text{N}$
- d) $1\text{A} \times 1\text{cm}/1\text{N}$

17- Maximum no of beats heard by human ear is:

- a) 7 beats/min
- b) 12 beats/sec
- c) 16 beats/sec
- d) 7 beats/sec

18- According to Laplace the compressions and rarefactions of sound waves take place:

- a) Isothermally
- b) Adiabatically
- c) Isobarically
- d) Isochorically

19- According to Newton's speed of sound waves depends upon:

- a) Compressibility and inertia
- b) Pressure
- c) Mass and energy
- d) Density of medium

20- Single isolated disturbance traveling in a system:

- a) Wave length
- b) Wave fronts
- c) Wave pulse
- d) Wave frequency

21- Unit of power of lens:

- a) Capacitance
- b) Diopeter
- c) Watt
- d) Horse power

22- Time of upward motion of projectile when reached at maximum height:

- a) $T = V_0 \sin 2\theta / g$
- b) $T = V_0 \sin \theta / g$
- c) $T = V_0 \sin \theta / g$
- d) $T = V_0 \cos \theta / g$

23- The process in which the momentum and energy of system remains constant:

- a) Elastic collision
- b) Inelastic collision
- c) Both
- d) None

24- S.I. unit of luminous intensity is:

- a) Kelvin
- b) Candela / sec.
- c) Candela
- d) Watt / sec.

25- The distance between node and antinode in standing wave is:

- a) $\lambda/2$
- b) $\lambda/4$
- c) $\lambda/3$
- d) λ

26- Vector which can be displaced to itself:

- a) Free vector
- b) Null vector
- c) Position vector
- d) Zero vector

- 27- When two vectors are in a plane than their vector product so formed is:
- Parallel
 - Perpendicular
 - In x-axis
 - In y-axis

- 28- Total energy of oscillating body is:
- $\frac{1}{2} Kx_0^2$
 - $\frac{1}{2} Kx_0 - x_0^2$
 - $\frac{1}{2} Kx_0$
 - $\frac{1}{2} (Kx_0 + 2x_0)$

- 29- The exact number of molecules in 1 mole of substance:
- 6.022×10^{23} molecules/mol
 - 6.029×10^{23} molecules/mol
 - 6.024×10^{24} molecules/mol
 - 6.032×10^{24} molecules/mol

- 30- In a photoconductive cell infrared radiations has wavelength of the order of:
- 10^6 m
 - 10^7 m
 - 10^8 m
 - 10^9 m

PART IV ENGLISH 2004

- 1- Put an appropriate word:

1. She always _____ her pet dog with herself
- Took
 - Takes
 - Taken
 - Has taken

2. Come in! I _____ have a cup of tea.

- Is
- Am having
- Have
- Has

3. Who _____ that terrible noise

- Is making
- Is producing
- Develops
- Developing

- 2- Put a pair of correct words:

- 4- Convict _____ a long prisonment, when he _____
- Was serving, escaped
 - Served, escape
 - Serving, escape
 - Serves, escaped

- 5- No one _____ television, so my father _____ it off.
- Watched, turned
 - Was watching, turns
 - Was watching turned
 - Watching, turned

- 3- Give meaning of the following words:

- 6- Recoil means:
- Remorse
 - Withdraw
 - Secure
 - Reconciling

- 7- Cynic means:
- Polite
 - Bad person
 - Disbeliever
 - Extremist

- 8- Ingenuity means:
- Unfairness
 - Laziness
 - Cleverness
 - Hopelessness

- 9- Torrent means:
- A name of tree
 - A rushing stream
 - River
 - A violent storm of wind

- 10- Indolence means:
- Threatening
 - Frankness
 - Impudence
 - Licentious



DOW GRADS

DO IT FROM THOSE WHO HAVE DONE IT NOT FROM THOSE WHO CLAIM!

College _____

Percentage _____

Contact _____

E-mail _____

DOW Sign _____

01	A	01	A	01	A	01	A	01	A
02	A	02	A	02	A	02	A	02	A
03	A	03	A	03	A	03	A	03	A
04	A	04	A	04	A	04	A	04	A
05	A	05	A	05	A	05	A	05	A
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16	A	16	A	16	A	16	A	16	A
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23	A	23	A	23	A	23	A	23	A
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25	A	25	A	25	A	25	A	25	A
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61	A	61	A	61	A	61	A	61	A
62	A	62	A	62	A	62	A	62	A
63	A	63	A	63	A	63	A	63	A
64	A	64	A	64	A	64	A	64	A
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98	A	98	A	98	A	98	A	98	A
99	A	99	A	99	A	99	A	99	A
100	A	100	A	100	A	100	A	100	A

BAHARIA UNIVERSITY 2004 ANSWER SHEET

Part - I 30 Questions BIOLOGY		Part - II 30 Questions CHEMISTRY		Part - III 30 Questions PHYSICS		Part - IV 10 Questions ENGLISH	
1	D	1	A	1	A	1	B
2	C	2	A	2	A	2	B
3	A	3	B	3	A	3	A
4	A	4	B	4	B	4	A
5	B	5	C	5	A	5	C
6	C	6	B	6	D	6	B
7	B	7	A	7	A	7	B
8	A	8	B	8	A	8	C
9	A	9	B	9	B	9	B
10	C	10	C	10	A	10	C
11	A	11	C	11	A		
12	B	12	B	12	D		
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14	A	14	C	14	B		
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27	A	27	C	27	B		
28	A	28	A	28	A		
29	D	29	C	29	A		
30	A	30	A	30	A		

10 YRS PAST PAPERS

D.M.C. & S.M.C.

PAST PAPER - 2005

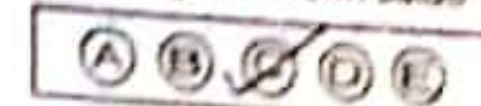
GENERAL INSTRUCTIONS (PLEASE READ THE INSTRUCTIONS VERY CAREFULLY)

1. Write in bold CAPITALS your Name and Seat Number as given in your Admit Card on allocated column of your Answer sheet.
2. Place your signature on the allocated column of the answer sheet.
3. This Question paper contains four parts.

Part - I	Biology	30 Questions
Part - II	Chemistry	30 Questions
Part - III	Physics	25 Questions
Part - IV	English	05 Questions
4. There will be negative marking in the test. Each correct Answer carries (4) marks. Each wrong Answer carries Minus one (-1) Mark.
5. The Answer sheet is made of special paper which gives itself a carbonized copy on the duplicate. All answers must be marked by filling the appropriate circle with pencil.
6. For every question in the paper, five choices are given. Please mark your choice by filling in the appropriate circle completely. Making a dark circle as shown below.



Some examples of improper marking are shown below



(The computer will mark improperly filled / checked marked circles as wrong answers)

9. You are NOT ALLOWED to take away any part of Question paper or note question elsewhere.
10. Use your time efficiently and at a premium. Do not spend too much time on one question, you may run short of time for other questions.
11. Change of answer is not allowed. No marks will be given if two or more circles are filled.
12. All answers must be marked on the answer sheet, do not mark any answer on the question paper.

Seat No. _____

Signature of the Candidate _____

NOTE: STOP! DO NOT TURN THIS PAGE TILL FURTHER INSTRUCTIONS.

- 1- Ringworm is caused by:
 - a) Round worm
 - b) Fungus
 - c) Protozoa
 - d) None of the above
- 2- Adrenal glands are found adjacent to:
 - a) Kidney
 - b) Heart
 - c) Lungs
 - d) Liver
- 3- When a diploid egg is formed without fertilization.
 - a) Parthenogenesis
 - b) Apomixis
 - c) Facultative apomixes
 - d) Reverse transcription
- 4- The drastic change in climate over a past century has been illustrated by:
 - a) Increase in rainfall
 - b) Decrease in rainfall
 - c) Colder winter
 - d) Warmer winter
- 5- The uptake of minerals in a plant occurs through:
 - a) Root hair
 - b) Root cap
 - c) Root tip
 - d) Stomata
 - e) Guard cells
- 6- Positive geotropism by the:
 - a) Root
 - b) Stem
 - c) Leave
 - d) Flower
- 7- The internal maintenance in human:
 - a) Homeostasis
 - b) Peliploidy
 - c) Excretion
 - d) Epistaxis
- 8- Animal with open circulatory system:
 - a) Cockroach
 - b) Frog
 - c) Snake
 - d) Duck billed platypus

- 9- The ability of cell to swell after absorption of H_2O :
 - a) Inhibition
 - b) Deplasmolysis
 - c) Guttation
 - d) Transpiration
- 10- Sensors that receive stimulus:
 - a) Receptor
 - b) Neuron
 - c) Effector
 - d) All of the above
- 11- A lipid molecule is composed of glycerol fatty acid in a ratio of:
 - a) 1:1
 - b) 1:3
 - c) 1:4
 - d) 1:2
- 12- Which of the following is not a reflex action?
 - a) Eye blink
 - b) Knee jerk
 - c) An animal salivating spontaneously as the bell rings without given food
 - d) Watering of mouth at the sight of food
- 13- 50% of blood volume is:
 - a) Plasma
 - b) Formed elements
 - c) Cells
 - d) Both b and c
- 14- Plasmodium can be founding the mosquito at:
 - a) Salivary gland
 - b) CBC
 - c) Stomach
 - d) Both A & B
- 15- The formula of glyceraldehyde is:
 - a) $C_3H_6O_4$
 - b) $C_3H_6O_3$
 - c) $C_6H_{12}O_6$
 - d) $C_7H_{14}O_7$
- 16- A green single celled organism in the protista kingdom:
 - a) Algae
 - b) Bacteria
 - c) Euglena
 - d) Amoeba

A green
single
celled

17- Ribosomes are attached to:

- a) One another
- b) ER
- c) Golgi bodies
- d) Centromeres

18- The reproductive structure of ferns is:

- a) Cones
- b) Needles
- c) Sporophyll
- d) Gametophylls

19- Solanacea is also called:

- a) Legume family
- b) Potato family
- c) Night Shade Family
- d) b & c both

20- In mitosis the no. of chromosomes in daughter cells:

- a) $2n$
- b) n
- c) $1/2n$
- d) None of the above

21- A purely marine phylum:

- a) Molluska
- b) Echinodermata
- c) Annelids
- d) Arthropods

22- Which animal uses highly vascularized skin for respiration:

- a) Fish
- b) Frog
- c) Snake
- d) A new borne mammal

23- The end product of glycolysis:

- a) Citric acid
- b) Pyruvic acid
- c) Lactic acid
- d) Acetyl CoA

24- Process of transferring bacterium from a donor bacterium to a receive bacterium.

- a) Conjugation
- b) Transduction
- c) Transformation
- d) Regressive Mutation

25- Lamark identified evolution as:

- a) Natural selection
- b) Acquired characters
- c) Disuse / use of organs
- d) Mutation Theory

26- pH of blood is about:

- a) 7.35
- b) 7.0
- c) 7.85
- d) 7.66

7.35 → blood PH

27- An ecosystem is:

- a) Population living together
- b) Species living together
- c) A community with its non living components
- d) A community with its living & non living components

28- movement is due to touch response:

- a) Themonastic
- b) Photonastic
- c) Seismonastic
- d) Thigmotropism

29- Ascent of sap is governed by the activity of the living cell in the form of pulsatory. This theory is called:

- a) Physical force theories
- b) Passive force theories
- c) Vital force theories
- d) Active force theories

30- In coughing the air is forcibly blown larynx while in sneezing it is blown through:

- a) Mouth
- b) Nasal cavities
- c) Both of these
- d) None of these

SECTION-II (CHEMISTRY)

1- What will make an electrolyte:

- a) Salt dissolved in water
- b) Sugar dissolved in water
- c) Salt dissolve inorganic solvent
- d) All of the above

2- Chemistry is the study of:

- a) Chemicals
- b) Acids
- c) Base
- d) Water
- e) None of the above

3- For Boyle's law to be true, which of the following must be fixed:

- a) Isothermal
- b) Isochoric
- c) Adiabatic
- d) Isobaric

4- Which of the following statements about fluorine is not true:

- a) It is the most electronegative element
- b) 19 proton in nucleus
- c) Compound can engage in H bond
- d) Oxidation state -1
- e) Found in nature as diatomic gas

5- Which of the following molecules has the strongest carbon to carbon bond.

- a) C_2H_2
- b) C_2H_4
- c) C_2H_6
- d) C_3H_8

6- Gases are good conductors at:

- a) High pressure
- b) Low temperature
- c) Low pressure
- d) None of the above

7- Which of the following will increase the rate of a reaction:

- a) Lowering the temp
- b) Increasing volume
- c) Reducing activation energy
- d) Increasing activation energy

8- Bituminous coal can produce by destructive distillation

- a) Coke
- b) Coal gas
- c) Coal tar
- d) Coal tear

9- Period three elements have:

- a) Same atomic no.
- b) Same atomic mass
- c) Principal quantum no.
- d) Azimuthal No.

10- The tabular arrangement of _____ principle is called long term of the periodic table

- a) $(n+l)$ rule
- b) Pauli's Exclusion principle
- c) Aufbau principle
- d) Hund's rule

11- The K.E. of molecules is:

- a) Linearly proportional to Temperature
- b) Inversely proportional to Temperature
- c) Linearly proportional to temperature
- d) Inversely proportional to square root of temperature

12- The K.E. of molecular theory does not state:

- a) There are intermolecular forces w/n gases
- b) There is no volume in gases
- c) Collision course is elastic
- d) All of the above

13- Capillary action is due to:

- a) Surface tension
- b) Cohesion
- c) Adhesion
- d) Viscosity

14- Which is the most electronegative element.

- a) Cl
- b) Br
- c) Ca
- d) K

15- The order of which period 3 gases, in order of increasing atomic weight:

- a) K, Ca, Br, Kr
- b) Br₂, Ca, Kr, Ca
- c) N₂, O₂, Cl₂, Ar
- d) F₂, N₂, O₂, Ne

16- What happens when no of moles of gas are doubled at a fixed volume:

- a) Pressure doubles
- b) Pressure is reduced to 1/2
- c) Pressure is reduced by 1/4
- d) Pressure stays the same

17- Which of these is organic:

- a) SO₂
- b) CH₄
- c) H₂O
- d) HCl

18- Alkenes is the first:

- a) Saturated hydrocarbon
- b) Unsaturated hydrocarbon
- c) Aromatic Hydrocarbon
- d) Super saturated Hydrocarbon

- 19- Which of the following has highest B.P.
 a) Ethyl alcohol
 b) Diethyl ether
 c) H_2O
 d) Acetone
- 20- The first alkane is:
 a) Methane
 b) Ethane
 c) Halothane
 d) Methylene
- 21- The density of Methane at 27°C and 2 atm is?
 a) 0.6 gm/cm³
 b) 9.1 gm/cm³
 c) 1.8 gm/cm³
 d) 0.91 gm/cm³
- 22- Which of the following is not a chemical change:
 a) Rusting of iron
 b) Refractive index
 c) Boiling of water
 d) Electrolysis of Cu^{++}
- 23- What is true about water density:
 a) It is less than that of ice
 b) It is more than that of ice
 c) It is equal
 d) None of the above
- 24- _____ periods, contain equal elements:
 a) 1st & 2nd
 b) 2nd & 3rd
 c) 2nd & 4th
 d) 6th & 7th
- 25- The elements which have only one valence electron are called:
 a) Coinage metals
 b) Alkaline earth metals
 c) Alkali metals
 d) Heavy metals
- 26- The position of hydrogen in periodic table is:
 a) IA
 b) IVA
 c) VIIA
 d) Still undecided
- 27- Liquefying point of N_2 gas is:
 a) $-252^\circ C$
 b) $+252^\circ C$
 c) $-196^\circ C$
 d) $+196^\circ C$

- 28- M.P of NaCl is $804^\circ C$. In down's process it is reduced to $600^\circ C$ by using:
 a) $MgCl$
 b) $CaCl$
 c) $MgSO_4$
 d) $CaSO_4$
- 29- Which one of the following sets of atomic numbers represents only alkali metals:
 a) 2, 4, 10
 b) 3, 11, 19
 c) 2, 10, 18
 d) 9, 17, 35
- 30- The I.U.P.A.C. name of $K_4[Fe(CN)_6]$:
 a) Potassium hexacyano ferrate (II)
 b) Potassium hexacyano ferrate (III)
 c) Potassium ferrocyanate
 d) Potassium ferricyanide

SECTION-III (PHYSICS)

- 1- In a NPN transistor, P work as a:
 a) Collector
 b) Base
 c) Emitter
 d) None of these
- 2- A transformer works according to:
 a) Mutual Induction
 b) Lenz's law
 c) Faraday's law
 d) All of the above
- 3- Work has same units as:
 a) Torque
 b) Power
 c) Momentum
 d) Angular Momentum
- 4- Two mass each of 60g are attached to a frictionless pulley. When 10g are added to one of the mass, what is its downward acc in cm/s^2 .
 a) 32
 b) 80
 c) 160
 d) 320
- 5- Compton effect describes the nature of light as a:
 a) Photon
 b) Wave
 c) Dual nature
 d) Longitudinal waves

6- A bomb blasts on the moon, its sound will be heard on earth after:

- a) Sound will not be heard
- b) 11-2 min later
- c) 138 minutes later
- d) 88 minutes later

7- Longitudinal waves cannot travel in:

- a) Water
- b) Solid
- c) Vacuum
- d) Air

8- By changing the temperature of air which of the following quantities will not change:

- a) Wavelength
- b) Frequency
- c) Intensity
- d) Amplitude

9- Torque describes:

- a) Acceleration
- b) Angular acceleration
- c) Linear acceleration
- d) Direction of movement

10- Which of the following is not an adiabatic process:

- a) Rapid expansion of air
- b) Burst of fire
- c) Cloud formation
- d) Freezing of liquid water into ice

A radioactive element will not emit:

- a) Electron
- b) Proton
- c) Beta particle
- d) Helium

A 20Ω and a 60Ω are connected in series to a DC generator. The $I = 4.0A$, What will be the voltage across 60Ω resistor.

- a) 100
- b) 120
- c) 240
- d) 24

The deflection velocity (angular) by linear velocity of an object moving in a circle is always

- a) 90°
- b) 180°
- c) 270°
- d) 100°

Dow Grads

107

10 Years Past Papers

14- Doppler's effect applies to:

- a) Sound waves only
- b) Light waves only
- c) Both sound waves by light waves only
- d) Magnetic waves only

15- The distance of a monochromatic source of light is tripled from the surface, what happens to intensity:

- a) Doubled
- b) Tripled
- c) Reduced to $1/3$
- d) Reduced to $1/9$

16- α -rays are:

- a) Hydrogen nuclei
- b) Helium nuclei
- c) Hydrogen electrons
- d) negatively charged electrons

17- The thing needed / required in the medium for light wave propagation:

- a) Elasticity
- b) Resistance
- c) Density
- d) None of the above

18- An ideal current has resistance:

- a) Zero
- b) Finite but zero
- c) Infinite
- d) 1 Ohm

19- An emf introduced in the A.C generator depends on:

- a) Thickness of armature
- b) Thickness of Shaft
- c) The strength of magnetic field
- d) Capacitance of inductor

20- Unit of Conductance is:

- a) Siemens
- b) Ohm
- c) MHO
- d) Diopes

21- An object is placed, 30cm from a convex lens whose focal length is 15cm, What is the height of image as compared to object.

- a) Same
- b) Smaller
- c) Larger
- d) Double the size

- 100
- Dow Grade**
- 22- A convex lens produces virtual image when the object is placed
- a) at f
 - b) b/w principle focus & centre of curvature
 - c) b/w principle focus & Optical center
 - d) Beyond $2f$
- is maximum:

23. In SHM, the velocity is maximum:
- Extreme position
 - Mean position
 - In the middle
 - None of the above

- 24- Capacitance depends on:
- a) Density of wire
 - b) Thickness of wire
 - c) Resistance of wire
 - d) All of the above

25. Vector addition follows:
- a) Commutative law
 - b) Associative law
 - c) Distributive law
 - d) All of these.

ENGLISH

- 1- Jubilant Antonym:
- a) Triumphant
 - b) Dejected
 - c) Rejected
 - d) Perceptive

2. Transitory synonym:
a) Get through
b) Hesitant
c) Short lived
d) Interval

- 3- **Elegant synonymy:**
 a) Graceful
 b) Ugly
 c) Diligent
 d) Mercurial.

- 4- Find the error:

Will I go now or later?
a b c d e

5. Global warming is the term Referred to:

- c) The temp of the earth is rising
- b) The temp of the universe is rising
- c) The temp of the specific regions is rising
- d) The temp of the ozone layer is rising

RESPONSE SHEET



DOW GRADS

DO IT FROM THOSE WHO HAVE DONE IT NOT FROM THOSE WHO CLAIM!

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
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College	
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Percentage

Contact E-mail

DOW Signatory

Fill the appropriate [] completely LIKE this [X] Tick or improper mark will be marked incorrect by the machine

- | | | | | | |
|-----|-----------------|-----|-----------------|------|-----------------|
| 01. | [A][B][C][D][E] | 34. | [A][B][C][D][E] | 68. | [A][B][C][D][E] |
| 02. | [A][B][C][D][E] | 35. | [A][B][C][D][E] | 69. | [A][B][C][D][E] |
| 03. | [A][B][C][D][E] | 36. | [A][B][C][D][E] | 70. | [A][B][C][D][E] |
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| 09. | [A][B][C][D][E] | 42. | [A][B][C][D][E] | 76. | [A][B][C][D][E] |
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| 12. | [A][B][C][D][E] | 45. | [A][B][C][D][E] | 79. | [A][B][C][D][E] |
| 13. | [A][B][C][D][E] | 46. | [A][B][C][D][E] | 80. | [A][B][C][D][E] |
| 14. | [A][B][C][D][E] | 47. | [A][B][C][D][E] | 81. | [A][B][C][D][E] |
| 15. | [A][B][C][D][E] | 48. | [A][B][C][D][E] | 82. | [A][B][C][D][E] |
| 16. | [A][B][C][D][E] | 49. | [A][B][C][D][E] | 83. | [A][B][C][D][E] |
| 17. | [A][B][C][D][E] | 50. | [A][B][C][D][E] | 84. | [A][B][C][D][E] |
| 18. | [A][B][C][D][E] | 51. | [A][B][C][D][E] | 85. | [A][B][C][D][E] |
| 19. | [A][B][C][D][E] | 52. | [A][B][C][D][E] | 86. | [A][B][C][D][E] |
| 20. | [A][B][C][D][E] | 53. | [A][B][C][D][E] | 87. | [A][B][C][D][E] |
| 21. | [A][B][C][D][E] | 54. | [A][B][C][D][E] | 88. | [A][B][C][D][E] |
| 22. | [A][B][C][D][E] | 55. | [A][B][C][D][E] | 89. | [A][B][C][D][E] |
| 23. | [A][B][C][D][E] | 56. | [A][B][C][D][E] | 90. | [A][B][C][D][E] |
| 24. | [A][B][C][D][E] | 57. | [A][B][C][D][E] | 91. | [A][B][C][D][E] |
| 25. | [A][B][C][D][E] | 58. | [A][B][C][D][E] | 92. | [A][B][C][D][E] |
| 26. | [A][B][C][D][E] | 59. | [A][B][C][D][E] | 93. | [A][B][C][D][E] |
| 27. | [A][B][C][D][E] | 60. | [A][B][C][D][E] | 94. | [A][B][C][D][E] |
| 28. | [A][B][C][D][E] | 61. | [A][B][C][D][E] | 95. | [A][B][C][D][E] |
| 29. | [A][B][C][D][E] | 62. | [A][B][C][D][E] | 96. | [A][B][C][D][E] |
| 30. | [A][B][C][D][E] | 63. | [A][B][C][D][E] | 97. | [A][B][C][D][E] |
| 31. | [A][B][C][D][E] | 64. | [A][B][C][D][E] | 98. | [A][B][C][D][E] |
| 32. | [A][B][C][D][E] | 65. | [A][B][C][D][E] | 99. | [A][B][C][D][E] |
| 33. | [A][B][C][D][E] | 66. | [A][B][C][D][E] | 100. | [A][B][C][D][E] |
| | | 67. | [A][B][C][D][E] | | |

SECTION I BIOLOGY		SECTION II CHEMISTRY		SECTION III PHYSICS		SECTION IV ENGLISH	
01	B	01	A	01	B	01	C
02	A	02	A	02	A	02	C
03	A	03	A	03	A	03	A
04	D	04	B	04	B	04	A
05	A	05	A	05	A	05	A
06	A	06	A	06	A		
07	A	07	C	07	C		
08	A	08	D	08	C		
09	B	09	C	09	B		
10	B	10	C	10	D		
11	B	11	A	11	B		
12	C	12	A	12	C		
13	A	13	A	13	C		
14	A	14	A	14	C		
15	B	15	C	15	C		
16	C	16	A	16	B		
17	B	17	B	17	A		
18	C	18	B	18	B		
19	D	19	C	19	A		
20	A	20	A	20	C		
21	B	21	B	21	A		
22	B	22	C	22	C		
23	B	23	B	23	B		
24	A	24	B	24	D		
25	B	25	C	25	D		
26	A	26	D				
27	D	27	C				
28	D	28	B				
29	A	29	B				
30	C	30	A				

10 YRS PAST PAPERS

N.T.S.

PAST PAPER - 2006

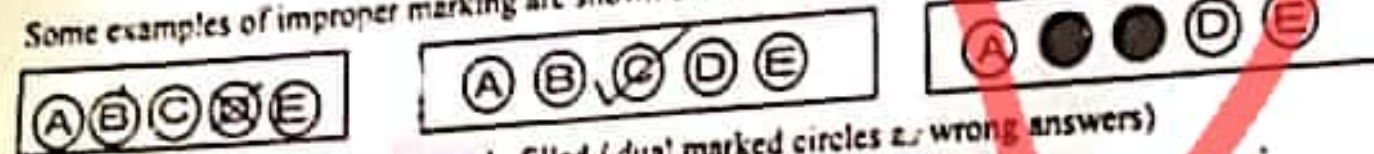
GENERAL INSTRUCTIONS (PLEASE READ THE INSTRUCTIONS VERY CAREFULLY)

- 1- Write in bold CAPITAL S your Name and Seat Number as given in your Admit Card on allocated column of your Answer sheet.
- 2- Place your signature on the allocated column of the answer sheet.
- 3- This Question paper contains four parts:

Part - I	English	05 Questions
Part - II	Physics	30 Questions
Part - III	Chemistry	30 Questions
Part - IV	Biology	30 Questions
- 4- There will be negative marking in the test. Each correct Answer carries (4) marks. Each wrong Answer carries Minus one (-1) Mark.
- 5- The Answer sheet is made of special paper. Which gives itself a carbonized copy on the duplicate. All answers must be marked by filling the appropriate circle with pencil.
- 6- For every question in the paper, five choices are given. Please mark your choice by filling in the appropriate circle completely. Making it a dark circle as shown below:



7- Some examples of improper marking are shown below:



(The computer will mark improperly filled / dual marked circles as wrong answers)

- 8- You are NOT ALLOWED to take away any part of Question paper or note question elsewhere.
- 9- Use your time efficiently and at a premium. Do not spend too much time on one question, you may run short of time for other questions.
- 10- Change of answer is not allowed. No marks will be given if two or more circles are filled.
- 11- All answers must be marked on the answer sheet, do not mark any answer on the question paper.

Seat No. _____

Signature of the Candidate _____

NOTE: STOP! DO NOT TURN THIS PAGE TILL FURTHER INSTRUCTIONS.

APTITUDE TEST 2006

SECTION-I (ENGLISH)

Synonyms:

1. loathsome
 - a) Handsome
 - b) Unpleasant
 - c) Brilliant
 - d) Compare
2. Comprise
 - a) Consists of
 - b) Forms
 - c) To agree
 - d) Accumulate

Antonyms:

3. Bliss
 - a) Happiness
 - b) Misery
 - c) Comfort
 - d) Pleasure
4. Exhibit
 - a) Conceal
 - b) Showy
 - c) To sell
 - d) Conceive
5. 'Inflict pain' to other means:
 - a) To injure
 - b) To harness
 - c) Feeling Pain
 - d) Giving pain

SECTION-II (PHYSICS)

1. In which we add more impurity in transistor?
 - a) base
 - b) Emitter
 - c) Collector
 - d) None of these
2. Special theory of Relativity solves problems with?
 - a) Inertial Frame
 - b) Accelerating frame
 - c) both
 - d) none
3. Object displaced from 5 m height reaches ground at a distance of 10m. What is its final velocity?
 - a) 10 m/s
 - b) $10\sqrt{2}$ m/s
 - c) 20 m/s
 - d) $20\sqrt{2}$ m/s
4. Temperature of a gas is directly proportional to?
 - a) Vibrational K.E. of molecules
 - b) Rotational K.E. of molecules
 - c) Angular transitional K.E. of molecules
 - d) None of the above
5. Gamma-rays are:
 - a) Longitudinal Waves
 - b) Transverse Waves
 - c) Mechanical Waves
 - d) None
6. In electrostatics, Newton's laws are:
 - a) All applicable
 - b) 1st law is not
 - c) 2nd law is not
 - d) 3rd law is not
7. Vectors of same length, how they must be having angle so we get resultant twice of either?
 - a) 30°
 - b) 0°
 - c) 60°
 - d) 45°

8. When we move from pole to equator, weight of the body:
 - a) Increases
 - b) Decreases
 - c) Remains same
 - d) None of these
9. Loudness of the sound is:
 - a) Inversely proportional to amplitude
 - b) Directly proportional to amplitude
 - c) Inversely proportional to frequency
 - d) Directly proportional to frequency
10. In Newton's Rings experiment, the width of the air film:
 - a) Gradually increases
 - b) Gradually decreases
 - c) Suddenly increases
 - d) Suddenly decreases
11. An object from the bridge falls, at the end of 3rd second of motion its velocity is?
 - a) 2 m/s
 - b) 10000 mm/s
 - c) 3000 cm/s
 - d) 10 m/s
12. Induced e.m.f. is directly proportional to?
 - a) Change in e.m.f.
 - b) Change in inertial flux
 - c) Rate of change of flux
 - d) Both 'a' and 'b'
 - e. None
13. Faraday's Law of electromagnetic induction is:
 - a) $\Delta\Phi / \Delta t$
 - b) $N\Delta\Phi / \Delta t$
 - c) $-N\Delta\Phi / \Delta t$
 - d) $-N\Delta\Phi / \Delta t$
14. A)C) current can be produced by:
 - a) Turbine
 - b) Motor
 - c) Generator
 - d) Transformer
 - e. None
15. The graph formed by Ohm's Law:
 - a) Straight Line
 - b) Parabola
 - c) Hyperbola
 - d) Sigmoid

16. Red cloth seems red because:
 a) Absorbs red light
 b) Reflects red light
 c) Absorbs white light
 d) Annuls red light
17. 1st Law of Thermodynamics can be given by:
 a) $\Delta E = Q + \Delta W$
 b) $\Delta H = E$
 c) $\Delta Q = \Delta E$
 d) None of the above
18. Two vectors A and B make an angle θ between them, the projection of B on A can be given as:
 a) $A \cos \theta$
 b) $\frac{A \cdot B}{A}$
 c) $\frac{A \cdot B}{B}$
 d) $B \cos \theta$
19. If 42 J heat is supplied to a system and work done by the system is 32 J. What is the change in internal energy?
 a) 1 J
 b) 70 J
 c) 10 J
 d) 8 J
20. The unit of angular acceleration is:
 a) radian s^{-1}
 b) $\text{radian} \cdot 3\text{s}^{-1}$
 c) radian s^{-2}
 d) radian
21. Which of the following quantities have the same dimensions?
 I. Power II. Energy III. Torque
 a) I and II
 b) I and III
 c) II and III
 d) I, II and III
22. If $A \cdot B = 0$, we conclude that;
 a) either of the two vectors is a null vector
 b) both of the vectors are null
 c) the vectors are mutually perpendicular
 d) all of the above
23. The scalar product of two vectors A and B at an angle θ with each other is:
 a) $A \cdot B \sin \theta$
 b) AB
 c) $AB \cos \theta$
 d) $AB \sec \theta$
24. The cross product of a vector A with itself ($A \times A$) is equal to:
 a) A
 b) 1
 c) 2A
 d) Zero
25. Which one of the following process obeys Boyle's Law?
 a) Isothermal
 b) Isochoric
 c) Adiabatic
 d) Non-Adiabatic
 e) All of the above
26. Charge on each plate of a parallel plate capacitor is 6 μC and capacitance is 4 μF . What is the potential difference between plates?
 a) 2.5 coulomb
 b) 3.8 Farad
 c) 1.8 Volt
 d) 1.5 Volt
 e) 3.5 Joule
27. Lenz Law is accordance with law of conservation of:
 a) Momentum
 b) Charge
 c) Angular momentum
 d) Energy
 e) Mass
28. Which of the following purpose is served by the transformer?
 a) Converts alternating current into direct current
 b) Converts direct current into alternating current
 c) Changes the voltage of the A.C.
 d) Decrease the power supply
 e) Increase the power supply
29. Which of the following statement is incorrect in perspective of modern physics?
 a) Energy has inertia
 b) Time dilation phenomenon does not occur in daily life
 c) Light is electromagnetic radiation
 d) Universe is continuously contracting

30. Visible spectrum of Light lies in:
- Paschen series
 - Balmer series
 - Lyman series
 - Bracket series
 - Pfund series

SECTION-II (CHEMISTRY)

- Which has high M.P. and B.P. amongst VI-A given:
 - Pb
 - Tl
 - Se
 - Bi
- Which is radioactive Halogen?
 - I₂
 - Cl₂
 - Br₂
 - None
- Which has highest K.E. amongst following?
 - H₂
 - O₂
 - CH₄
 - All of these
- Which is not a normal cracking phenomenon?
 - Thermal cracking
 - Catalytic cracking
 - Steam cracking
 - Normal cracking
- What is the property of oxidizing agent?
 - Electron gainer
 - Electron donor
 - Negative charge
 - Loose hydrogen
- $H_2 + Br \rightarrow 2HBr$
The rate expression is: $Rate = K [H_2] [Br]^n$, the order of the reaction is:
 - 2
 - 1.5
 - 1
 - Zero

- Which is not a nucleophile:
 - H₂O
 - H₂S
 - BF₃
 - NH₃
 - All of these
- Which alcohol can form a ketone?
 - Primary Alcohol
 - Secondary Alcohol
 - Tertiary Alcohol
 - Methanol
 - None of the above
- Which is not an allotrope of carbon?
 - Diamond
 - Coke
 - Methane
 - Graphite
- What are the raw materials in the formation of urea:
 - Steam, methane and ammonia
 - Steam and ammonia
 - Methane and steam
 - None of the above
- In a reactor:

$$NO + O_2 \xrightarrow{\quad\quad\quad} NO_2 + \text{heat}$$
 Le Chatlier's principle can be favoured by:
 - High pressure – High temperature
 - High pressure – Low temperature
 - Low pressure – Low temperature
 - Low pressure – High temperature
- The compounds which appose enzyme activity are:
 - Co-enzymes
 - Catalysts
 - Inhibitors
 - Negative Enzymes
 - c and d
- Decay of metal with interaction with the surrounding is called:
 - Radioactivity
 - Corrosion
 - Alloying
 - None of these

DPT

14. 18 gm of glucose in 1000 gm of solution of glucose. The molar concentration of Glucose is:

- a) 1M
- b) 0.1M
- c) 0.02M
- d) 2M

15. 1gm of hydrogen contains how many number of molecules:

- a) 6.02×10^{23} molecules
- b) 3.01×10^{23} molecules
- c) 0 molecules
- d) 1.04×10^{23} molecules

16. Alkanes are also known as:

- a) Saturated Hydrocarbons
- b) Unsaturated Hydrocarbons
- c) Paraffins
- d) None of the above

17. Isotopes differ in:

- a) Proton number
- b) Neutron number
- c) Atomic Number
- d) Electron number

18. Benzene doesn't show any one of the following:

- a) Oxidation
- b) Addition
- c) Elimination
- d) Substitution

19. Noble gases have electronic configuration of:

- a) $ns^2 np^4$
- b) $ns^2 np^6$
- c) $ns^2 np^6$
- d) ns, np

20. Which of the following is/are electropositive elements:

- a) Alkali Metals
- b) Alkaline Earth metals
- c) Halogens
- d) Both 'a' and 'b'

21. The ore of the aluminium is:

- a) Pyrite
- b) Bauxite
- c) Galena
- d) Azurite

22. The correct sequence of high power vapour pressure is:

- a) Water > Ethanol > Acetone > Ether
- b) Ether > Acetone > Ethanol > Water
- c) Ether > Ethanol > Acetone > Water
- d) Water > Ethanol > Acetone > Ether

23. Boric Acid cannot be used as:

- a) Antiseptic
- b) Eye washer
- c) Soda bottle formula
- d) Glazing tiles and ceramics

24. At constant temperature the volume of gas is 95 cm^3 at a pressure of $9.961 \times 10^4 \text{ N/m}^2$. What would be its volume at $10.13 \times 10^4 \text{ N/m}^2$?

- a) 1 cm^3
- b) 94 cm^3
- c) 93 cm^3
- d) 10 cm^3

25. Which one of the following is water soluble vitamin?

- a) Biotin
- b) Vit. A
- c) Vit. D
- d) Vit. K

26. Honey is a mixture of _____.

- a) Glucose + Fructose
- b) Vitamins
- c) Minerals
- d) All of the above

27. The ratio of the rate of diffusion of two gases is 1:3, then ratio of their molecular wt. is:

- a) 1:3
- b) 3:1
- c) 1:9
- d) 9:1

28. Which one of the following bond has the most polar character:

- a) C-O
- b) C-F
- c) C-Br
- d) C-S

29. Point the incorrect statement in case of equilibrium state:

- a) It is dynamic in nature
- b) It readjusts itself in the changing conditions
- c) It can be attained from the side of reactants only
- d) It can be attained from either side of the reaction

30. A catalyst can't initiate the reaction but speeds up a reaction which is possible:

- a) Physically
- b) Thermodynamically
- c) Chemically
- d) In laboratory

SECTION-IV (BIOLOGY)

1. Different Food Chains interact with each other to form:

- a) Food Pyramid
- b) Energy Graph
- c) Food Web
- d) Energy Pyramid
- e) Food Chain Complex

2. Physical basis of life is:

- a) Nucleotide and proteins
- b) Nucleus
- c) Cell
- d) Protoplasm

3. Rough Endoplasmic Reticulum are involved in:

- a) Protein synthesis
- b) DNA synthesis
- c) RNA synthesis
- d) Protein Modification

4. Many ribosomes attached to Endoplasmic reticulum form?

- a) Abnormal Function
- b) Polyribosomes
- c) Intense protein synthesis
- d) Cell becomes malignant

5. Earth Worms gaseous exchange occurs through:

- a) Lungs
- b) Skin
- c) Blood
- d) Nephridia

6. Amoeba is:

- a) Acellular
- b) Prokaryote
- c) Multinucleated
- d) None of the above

Nostoc contains following except:

- a) Cellulose
- b) Ribosomes
- c) Golgi Apparatus
- d) Chloroplasts

8. Mitochondria can be found in:

- a) Virus
- b) Bacteria
- c) Protozoan
- d) Nostoc
- e) None of the above

9. Echinoderms have a brain:

- a) Well-developed
- b) Rudimentary
- c) Absent
- d) None of the above

10. Male honey bees are:

- a) Diploid
- b) Haploid
- c) Triploid
- d) None

11. Spores are:

- a) Diploid
- b) Haploid
- c) Triploid
- d) None

12. Most optimum environment for the growth of fungus is:

- a) Dry
- b) Moist
- c) Shady
- d) All of the above

13. Maize Endosperm is:

- a) Diploid
- b) Haploid
- c) Triploid
- d) None

14. Which of the following have an edible seed coat?

- a) Guava
- b) Mango
- c) Pomegranate
- d) Water Melon

15. Malaria is caused by:

- a) Mosquito
- b) Virus
- c) Protozoa
- d) Bacteria

Which of the following is not a bacterial disease?

- a) Typhoid and T B)
- b) Rabies and Polio
- c) Diphtheria and Cholera
- d) Dysentery and Diarrhea

17. How many mitotic divisions produce 12 cells?

- a) 3 Divisions
- b) 6 Divisions
- c) 4 Divisions
- d) 12 Divisions

18. Chloroplast can be traced in:

- a) Root
- b) Stem
- c) Leaf
- d) None of these

19. Guttation is due to:

- a) High Transpiration
- b) Low Relative Humidity
- c) Negative Root Pressure
- d) Positive root Pressure

20. When your hand touches the hot body, which path is taken by stimulus to withdraw your hand?

- a) Sensory Neurons → Spinal Cord → Brain → Muscle
- b) Sensory Neurons → Spinal Cord → Muscle
- c) Sensory Neurons → Brain → Spinal Cord → Motor Neuron
- d) Hot Object → Spinal Cord → Hand

A man with blood group 'A' marries a woman with blood group 'B', child can have:

- a) A and B
- b) B and AB
- c) A and AB
- d) A, B, AB and O

In a population 36% people have 'Blue Eyes'; allele frequency for blue is:

- a) 36%
- b) 40%
- c) 60%
- d) 64%

Which of the following is not a requirement for seed germination?

- a) Oxygen
- b) Water
- c) Abicissic Acid
- d) Temperature

24. Reduction division in Meiosis is:

- a) Anaphase - II
- b) Anaphase - I
- c) Diakinesis
- d) Diplotene

25) The life was evolved first:

- a) On Earth
- b) In water
- c) In sky
- d) On the hill

26) The _____ structure was first proposed by Robertson:

- a) Cellulo-protein
- b) Star protein
- c) Cyto protein
- d) Lipo protein

27) Enzymes associated with Kreb's cycle are found in:

- a) Outer layer of mitochondrion
- b) Inner layer of mitochondrion
- c) Inner membrane space of mitochondrion
- d) Matrix of mitochondrion

28) Xanthophyll is a pigment having:

- a) Yellow color
- b) Red color
- c) Green color
- d) Blue color

29) Outer membrane of mitochondrion and endoplasmic reticulum are similar in having:

- a) Similar structure and constituents substance.
- b) The enzyme monoamine oxidase
- c) The enzyme NADH cytochrome reductase.
- d) Identical metabolic function

30) There are different types of chlorophyll like chlorophyll a, and:

- a) b c d and e
- b) b
- c) Xanthophyll
- d) Only 'c'



DOW GRADS

DO IT FROM THOSE WHO HAVE DONE IT NOT FROM THOSE WHO CLAIM!

College

Percentage

Contact

E-mail

DOW Signatory

Fill the appropriate [] completely LIKE this [] Tick or improper mark will be marked incorrect by the machine

- 01 [A][B][C][D][E] 34 [A][B][C][D][E] 68 [A][B][C][D][E]
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APTITUDE TEST 2006 ANSWER SHEET

Part - I 05 Questions ENGLISH		Part - II 30 Questions PHYSICS		Part - III 30 Questions CHEMISTRY		Part - IV 30 Questions BIOLOGY	
1.	B	1.	B	1.	A	1.	C
2.	A	2.	A	2.	D	2.	C
3.	B	3.	B	3.	D	3.	A
4.	A	4.	D	4.	D	4.	C
5.	D	5.	B	5.	A	5.	B
		6.	A	6.	B	6.	D
		7.	B	7.	C	7.	C
		8.	C	8.	D	8.	C
		9.	D	9.	C	9.	C
		10.	A	10.	D	10.	A
		11.	C	11.	B	11.	A
		12.	C	12.	C	12.	B
		13.	C	13.	B	13.	A
		14.	C	14.	B	14.	A
		15.	A	15.	B	15.	C
		16.	B	16.	A	16.	B
		17.	D	17.	B	17.	C
		18.	B	18.	B	18.	C
		19.	C	19.	C	19.	D
		20.	C	20.	D	20.	B
		21.	C	21.	B	21.	D
		22.	D	22.	A	22.	A
		23.	C	23.	C	23.	C
		24.	D	24.	C	24.	B
		25.	A	25.	A	25.	B
		26.	D	26.	A	26.	A
		27.	D	27.	D	27.	D
		28.	C	28.	B	28.	A
		29.	D	29.	C	29.	A
		30.	B	30.	C	30.	B

N.T.S. PAST PAPER 2007

GENERAL INSTRUCTIONS
(PLEASE READ THE INSTRUCTIONS VERY CAREFULLY)

1. Write in bold CAPITALS your Name and Seat Number as given in your Admit Card on allocated column of your Answer sheet

2. Place your signature on the allocated column of the answer sheet.

3. This Question paper contains four parts.

Part - I	English	10 Questions
Part - II	Biology	30 Questions
Part - III	Chemistry	30 Questions
Part - IV	Physics	30 Questions

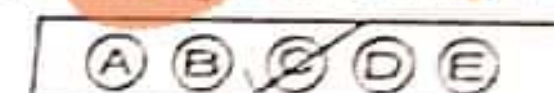
There will be negative marking in the test. Each correct Answer carries (4) marks. Each wrong Answer carries Minus one (-1) Mark.

The Answer sheet is made of special paper. Which gives itself a carbonized copy on the duplicate. All answers must be marked by filling the appropriate circle with pencil.

For every question in the paper, five choices are given. Please mark your choice by filling in the appropriate circle completely. Making it a dark circle as shown below:



Some examples of improper marking are shown below:



(The computer will mark improperly filled / dual marked circles as wrong answers)

You are NOT ALLOWED to take away any part of Question paper or note/question elsewhere. Use your time efficiently and at a premium. Do not spend too much time on one question, you may run short of time for other questions.

Change of answer is not allowed. No marks will be given if two or more circles are filled.

All answers must be marked on the answer sheet, do not mark any answer on the question paper.

Seat No. _____

Signature of the Candidate

NOTE: STOP! DO NOT TURN THIS PAGE TILL FURTHER INSTRUCTIONS.

SECTION - I ENGLISH - 2007

1. The word closest in meaning to 'Competent' is:
 a. Experienced
 b. Skilful
 c. Knowledgeable
 d. Dynamic
2. People get tensed during difficult situations. But this condition is not only with common people, but also the very much experienced psychologists get worried during the problems.
 The idea that the author is producing in this passage is:
 a. Everyone gets worried during their problems
 b. Even psychologists get worried during their personal problems
 c. Difficult situations are faced by everyone
 d. Common people face a lot of problems
3. The reason they gave was _____, and they explained it _____.
 a. Correct, correct
 b. Correct, correctly
 c. Correctly, correct
 d. Correctly, correctly
4. Monotonous Synonym :
 a. Limited
 b. Conclusive
 c. Extremist
 d. Repeatedly
5. INTENSIVE (Antonym)
 a. Casual
 b. Regressive
 c. Progressive
 d. Attentive
6. Presice (antonym)
 a. Consize
 b. Comprehensive
 c. Exaggerate
 d. Explanatory

7. We heard on TV that the weather is not fine. We say that we should not go out?
 What must be changed in order to make the sentences correct?
 a. We [in the first sentence]
 b. We [in the second sentence]
 c. That the weather
 d. We should not go out
9. We have not to smoke because smoking can harm out lungs.
 a. Have not
 b. To smoke
 c. Smoking
 d. Can harm
10. Many Businessmen waste their time in checking their competitors businesses. Instead what they should do is try to improve their own style of working and bring the amendments that are required to make their business prosper.
 a. Businessmen need to look after their competitors business as well.
 b. Many businessmen just waste their time by poking into their competitors businesses.
 c. Businessmen can improve their business by improving their business strategies.
 d. Handwork can make any business successful.

SECTION-II (BIOLOGY)

1. Which one of the following is a voluntary muscle?
 a. Biceps
 b. Cardiac
 c. Stomach
 d. Vessels
2. Which one of the following plants store water in their parenchymatous cells?
 a. Succulent Plants
 b. Angiosperms Plants
 c. Gymnosperms Plants
 d. Xerophytic Plants
3. Which type of respiration is the most effective in frog?
 a. Coetaneous
 b. Lung
 c. Pulmo-cutaneous
 d. Through gills

4. ALL of the following are symptoms of hepatitis, except:
- Deficit Immunity
 - Enlargement of liver
 - Fever and fatigue
 - Jaundice
5. O Group is universal donor because
- It has both A & B Antigens
 - No Antigens
 - A antigen
 - B Antigen
6. Most of the fruits belong to Family:
- Poaceae
 - Fabaceae
 - Rosaceae
 - Solanaceae
7. Oxygen reacts with RuBP in the absence of CO₂ to form:
- PGA
 - PGA and phosphoglycolate
 - Oxalo acetate
 - ATP
9. Ring Worm is caused by:
- Bacteria
 - Virus
 - Neutrophils
 - Fungus
10. The phytohormone responsible for Fruit development, bud sprouting, development of pollen tube and parthenocarpy is:
- Oxin
 - Gibrellin
 - Cytokinin
 - Thallomycin
11. Which protozoan class is totally parasite:
- Protozoa
 - Plasmodium
 - Ciliata
 - Sporozoa

12. Which structure in bacteria composed of polysaccharides and amino acids that prevents bacteria from antibiotics and increases its pathogenacity?
- Flagella
 - Capsule
 - Cell Membrane
 - Cyto skeleton
13. Opening and closing of stomata is controlled by which of the following taking place in the guard cells?
- Tubidity
 - Turgidity
 - Flacidity
 - Turgor Force
14. Guanine always pairs with:
- Thymine
 - Adenine
 - Cytosine
 - Uracin
15. Formation of mRNA from DNA is called:
- Translation
 - Transcription
 - Genetics
 - Mutation
16. In sickle cell anemia
- Valine is replaced by Glutamic Acid
 - Glutamic Acid is replaced by Valine
 - Glutamic Acid is replaced by Adenine
 - B-chain is replaced by Valine
17. Photosynthetic Bacteria use which of the following instead of H₂O?
- H₂S
 - H₂SO₄
 - H₂ Gas
 - Atomic Hydrogen
18. Which of the following is a bacterial disease::
- Leishmeniasis
 - Tetnas
 - Measels
 - Polio

1489 61
- 614
1461, 67
- 606

19. In vitro fertilization is:
- Fertilization of egg outside the body in lab
 - Fertilization within the female
 - Fertilization in water exclusively
 - Giving nutrition to fertilized egg

20. Which statement about fungi is not true?
- They have cellulose
 - They are heterotrophs
 - They have starch
 - They are coenocytic

21. In AMYLOPECTIN the linkage is:
- 1.3
 - 1.6
 - 2.6
 - 6.6

22. Formation of annual ring is the result of the activity of:
- Cork Cambium
 - Cork Cambium and Vascular cambium
 - Intracalary meristem
 - lateral cambium

23. Following are the characteristics of arteries except:
- They are Deep
 - They don't have valves
 - They have narrow lumen
 - They are thin walled

24. Forgetfulness, loss of memory and losing track of even the most familiar place are the progressive characteristics of which disease:
- Huntington's Disease
 - Parkinson's disease
 - Alzheimer's Disease
 - Copoli's Sarcoma

25. Nitrogen is the ultimate product of:
- Fats
 - Carbohydrates
 - Proteins
 - Organic salts

26. Club Mosses belong to the sub-division:
- Spermopsida
 - Pteropsida
 - Lycopsida
 - Psilopsida

27. Which one of the following undergoes complete metamorphosis?
- Grasshopper
 - Butterfly
 - Crab
 - Scorpion

28. The TMV first appeared on the leaves of Tobacco plant and was characterised by:
- Wrinkling of leaves
 - Yellow patches on lamina surface
 - Falling off of the leaves
 - Transmission from one organism to another

29. Poikilotherms are the animals:
- Whose arteries contract in cold temperature
 - Who have lot of stored body fats
 - Who maintain their body temperature constant
 - Hibernate during winter

30. Skin color, Height and intelligence vary in different people due to:
- Pleiotropy
 - Epistasis
 - polygenic inheritance
 - Crossing over

SECTION-III (CHEMISTRY)

1. How many Carbons are there in STEARIC ACID?
- 15
 - 18
 - 05
 - 07

2. What will be the product of Specific Heat and molecular formula?
- Molar specific heat
 - Molar mass
 - Heat capacity
 - Molar heat

3. Which one is different from the others?

- a. Propane
- b. Benzene
- c. Ethane
- d. Methane

4. Dilute CO_2 is used in cure of?

- a. H_2SO_4
- b. NaOH
- c. KOH
- d. H_2S poisoning

5. Who observed radioactivity first?

- a. Henry Backwell
- b. Rutherford
- c. Newton
- d. Bohr

6. Unit of Specific Rate Constant for zero order reaction is :

- a. s^{-1}
- b. $\text{mol dm}^{-3} \text{s}^{-2}$
- c. dm^6
- d. $\text{mol}^2 \text{dm}^{-6} \text{s}^{-1}$

7. Which rule is violated in this $1s^2, 2s^2, 2p^6, 3p^6, 4s^2$

- a. Aufbau's Principle
- b. Hund's Rule
- c. $n+1$ rule
- d. Pauli's Exclusion Principle

8. Atoms of sulphur and 6 oxygen atoms are held in a thionate molecule by?

- a. Ionic Bond
- b. Covalent Bond
- c. Dative Bond
- d. Polar Bond

9. How many lone pairs are present in H_2O ?

- a. 2
- b. 4
- c. 1
- d. 3

10. How many C atoms are in Valeric Acid?

- a. 7
- b. 5
- c. 6
- d. 11

11. Pyrex Glass is also called

- a. Brosilicate
- b. Potassium silicate
- c. Calcium silicate
- d. Sodium silicate

12. On oxidation with KMnO_4 ; ethyne does not form which of the following:

- a. Formic Acid
- b. Oxalic Acid
- c. Glycol
- d. CO_2

13. Rate of reaction depends on what in $\text{S}_\text{N}2$ reactions?

- a. $R = k [\text{R-X}]^{n-4}$
- b. $R = k [\text{R-X}]^2$
- c. $R = K [n^4] -$
- d. $R = K [n] [\text{R-X}]$

14. Which one of the following compound dissociate more readily?

- a. ChloroAcetic Acid
- b. Formic Acid
- c. HCl
- d. Ether

15. In Triads what would be the mass of Bromine according to the law of triads $\text{Cl} = 35.5$ $\text{I} = 126.5$?

- a. 81 a.m.u
- b. 90 a.m.u
- c. 79.5 a.m.u
- d. 82 a.m.u

16. Enzymes are completely destroyed at which temperature?

- a. 37°C
- b. 98.6°F
- c. 102°C
- d. 100°C

17. Modern Periodic law is based on what?
- No. of Protons and neutrons
 - No Electrons
 - Mass No
 - Charge density

18. The s-s bond is weaker than s-p or p-p because:
- Radius of s orbits is short
 - Radial distribution of charge in s orbital
 - Distribution of charge at the Axis
 - Distribution of charge at Nodal Planes

19. Hess's Law is specially applied in the process?
- To find out the heat of formation
 - To find out the specific heat of formation
 - To find the total heat involved in the process where it cannot be directly be calculated
 - To determine the rate of reaction

20. Because of the spilling of H₂O Molecules in our body; which of the following structure may be affected?

- Minerals
- Carbohydrates
- Proteins
- Fats

21. 80 gm Oxygen occupies the volume of

- 55dm³
- 20dm³
- 58.5dm³
- 22.4dm³

22. NH₃ dissolved in water will act as:

- Proton acceptor
- Proton donor
- Electron donor
- Both proton acceptor and electron donor

23. An electron can use Excitation Energy in the form of all of the following except
- Photochemistry
 - Florence
 - Heat
 - Radiation

24. Zn has only one oxidation state because?
- Completely filled 4d-orbital
 - Half filled 4s orbital
 - Completely filled 3d orbital
 - Half filled 3s orbital

25. The energy that the reactants should gain in order to enter into reacting phase is called?
- Minimum Energy
 - Activation Energy
 - Threshold Frequency
 - Kinetic Energy

26. In the preparation of NaOH which of the following element is involve?
- Fe
 - Pb
 - Hg
 - Se

27. Urea was prepared in the laboratory by Wohler first time in?
- 1918
 - 1928
 - 1938
 - 1948

28. Product of Specific Heat and Molecular Formula is?
- Molar Specific Heat
 - Molar Heat
 - Heat capacity
 - Specific Heat Capacity

29. $N_2 + O_2 \rightarrow 2NO$; Heat = +ve, the yield of N₂ can be increased in this reaction by:
- Increasing the pressure
 - Increasing the temperature
 - Decreasing the temperature
 - Decreasing the pressure

30. Primary Alkyl Halides undergo E2 Reactions in:
- Polar solvent
 - Non-polar solvents
 - Electrolytic Solutions
 - Polar and non-polar solution

SECTION-IV (PHYSICS)

1. $ABA = BAB$ is according to which property?
 a. Closure property of multiplication
 b. Commutative property of multiplication
 c. Distributive property of multiplication
 d. Associative property of multiplication
2. A current that flows from the high concentration towards a low concentration the charge is called:
 a. Conventional current
 b. Electric Current
 c. Fluctuating Current
 d. Alternating Current
3. A massive Ball collides with a light ball. What will be the velocity of the heavy ball after collision?
 a. Doubled
 b. Halved
 c. It will remain stationary
 d. Lighter ball will rebound with double velocity
4. A current of 3 Amp is passing through a galvanometer. If a high resistance connected in series to it, the total current that will pass through the resistance will be
 a. 6 Amp
 b. 1.5 Amp
 c. 3 Amp
 d. 2 Amp
5. In x-ray Production L, M, N come into K the rays formed are:
 a. $L\alpha, L\beta, L\gamma$
 b. $M\alpha, M\beta, M\gamma$
 c. $N\alpha, N\beta, N\gamma$
 d. $K\alpha, K\beta$
6. If an image is formed at a distance of 24 cm and is 4 times magnified than actual object, the object must have been placed at a distance of?
 a. 12 cm
 b. 6 cm
 c. 24 cm
 d. 18 cm

7. Relation b/w visual angle and apparent size is?
 a. Inversely proportional
 b. Directly proportional
 c. Proportional to its square
 d. Proportional to the square-root
8. Diffraction grating is used to study:
 a. Newton's rings
 b. Polarization Effect
 c. Interference
 d. Diffraction
9. Standing waves frequency in a pipe in which one side is closed and the other side is opened is the;
 a. Even Multiples of $v/2L$
 b. Odd Multiples of $v/2L$
 c. Even Multiples of $v/4L$
 d. Odd Multiples of $v/4L$
10. If the radius of the earth is halved, keeping its mass constant, then the value of g will become:
 a. $1/2$
 b. $2g$
 c. g
 d. $1/4$
11. The unit of momentum is;
 a. N.s
 b. N.m
 c. N.s⁻¹
 d. N.m²
12. If the net torque acting on a body is zero; then which quantity remains constant;
 a. angular Momentum
 b. linear momentum
 c. 1st condition of equilibrium
 d. 2nd condition of equilibrium
13. The corresponding frequency of 'n' loops is given by:
 a. nf
 b. $v/2L$
 c. $f/2L$
 d. $v/2L$

14. A ball is moving on the table which energy does it possess:

- a. potential energy
- b. kinetic energy
- c. Both potential and kinetic
- d. none

15. Find the dimension of C in $s=c/t^3$

- a. $[1/T^3]$
- b. $[LT^{-3}]$
- c. LT^2
- d. LT^3

16. If a body is immersed into water, the weight of the body becomes equal to:

- a. Weight of the water displaced by the body
- b. Weight of body - acceleration due to gravity
- c. Weight of the body
- d. Mass of the body

17. A ball thrown upwards with a velocity of 19.6 m/s. will come momentarily in rest:

- a. 2s
- b. 3.5
- c. 2.8
- d. 0s

18. Potential energy of an electron of mass 'm' moving with a velocity 'v' in a radius of r:

- a. $-mv/r$
- b. $-Ze^2/r$
- c. mv/r^2
- d. mZe^2/r

19. If a body is moving with uniform velocity in the direction of curling of fingers of right hand, then the thumb will point at:

- a. Angular velocity
- b. Tangential velocity
- c. Angular Acceleration
- d. Linear Velocity

20. If a spring is pulled in the right direction, the force that will cause the spring to further in left direction, after passing through the mean point, will be:

- a. Elastic restoring force
- b. Applied force
- c. Inertial force
- d. Gravitational force

21. Which device produces Electromagnetic waves?

- a. Rectifier
- b. Oscillator
- c. Vibrator
- d. Amplifier

22. A particle moving with a speed of an electromagnetic wave possess K.E. 4.5×10^{16} exp 16. What mass can be estimated of the moving particle?

- a. 10 kg
- b. 1 kg
- c. 1gm
- d. 0.1 kg

Solution

KE is given = 4.5×10^{16} exp 16
and we have to put velocity = $v = c = 3 \times 10^8$ exp. 8
then this value was to be put into the KE formula
 $KE = \frac{1}{2} m v^2$

$$4.5 \times 10^{16} \text{ exp } 16 = \frac{1}{2} m \times 9 \times 10^{16} \text{ exp } 16$$

$$4.5 \times 10^{16} \text{ exp } 16 \times 2 = m \times 9 \times 10^{16} \text{ exp } 16$$

$$9 \times 10^{16} \text{ exp } 16$$

$$\text{then, } m = 1 \text{ kg. ans.}$$

23. As the particle moves in from infinity, the value of Absolute gravitational potential energy:

- a. increases
- b. decreases
- c. remains same
- d. varies at different heights

24. If there is a vector X, then what should be added to it in order to make its magnitude as well as its direction equivalent to null value?

- a. vector 'X'
- b. Vector '-X'
- c. a null vector
- d. a unit vector

25. Mass of a body in an elevator which is moving up with an acceleration that is equal to the the acceration due to gravity in that place will be:

- a. Remains same
- b. increases
- c. decreases
- d. becomes zero

DOW GRADS

DO IT FROM THOSE WHO HAVE DONE IT NOT FROM THOSE WHO CLAIM

[illegible]

DOW Signatory

27. The force on a charge carrying conductor in the magnetic field of induction is:
- a $F = I [l \times B]$
 - b $F = I \times l \times B$
 - c $F = I \times [l \times B]$
 - d $F = I B$
28. Meter bridge, wheatstone bridge, potentiometer, P.O. Box all help in the determination of:
- a Resistance
 - b Potential Difference
 - c Voltage
 - d Current
29. For Forward Biasing:
- a P should be connected to negative terminal of the battery
 - b P should be connected to the negative while N to positive terminal of the battery
 - c N should be connected to the negative while P to positive terminal of the battery
 - d N should be connected to any high resistance
30. If the Kelvin's condition becomes un-applicable, then:
- a A refrigerator will work more efficiently
 - b It will consume less energy and do more work
 - c It will rather work on Clausius condition
 - d NOT

Fill the appropriate [] completely LIKE this [X]. Tick or improper mark will be marked incorrect by the machine.

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| 24 | [A][B][C][D][E] | 57 | [A][B][C][D][E] | 91 | [A][B][C][D][E] |
| 25 | [A][B][C][D][E] | 58 | [A][B][C][D][E] | 92 | [A][B][C][D][E] |
| 26 | [A][B][C][D][E] | 59 | [A][B][C][D][E] | 93 | [A][B][C][D][E] |
| 27 | [A][B][C][D][E] | 60 | [A][B][C][D][E] | 94 | [A][B][C][D][E] |
| 28 | [A][B][C][D][E] | 61 | [A][B][C][D][E] | 95 | [A][B][C][D][E] |
| 29 | [A][B][C][D][E] | 62 | [A][B][C][D][E] | 96 | [A][B][C][D][E] |
| 30 | [A][B][C][D][E] | 63 | [A][B][C][D][E] | 97 | [A][B][C][D][E] |
| 31 | [A][B][C][D][E] | 64 | [A][B][C][D][E] | 98 | [A][B][C][D][E] |
| 32 | [A][B][C][D][E] | 65 | [A][B][C][D][E] | 99 | [A][B][C][D][E] |
| 33 | [A][B][C][D][E] | 66 | [A][B][C][D][E] | 100 | [A][B][C][D][E] |
| | | 67 | [A][B][C][D][E] | | |

ENGLISH		BIOLOGY		CHEMISTRY		PHYSICS	
01	B	01	A	01	B	01	B
02	A	02	A	02	A	02	A
03	A	03	A	03	B	03	D
04	A	04	B	04	D	04	B
05	A	05	C	05	A	05	A
06	C	06	B	06	A	06	B
07	E	07	D	07	A	07	D
08	A	08	D	08	A	08	C
09	A	09	B	09	B	09	D
10	C	10	D	10	A	10	D
		11	B	11	C	11	A
		12	B	12	D	12	A
		13	B	13	A	13	A
		14	C	14	A	14	C
		15	B	15	A	15	D
		16	A	16	D	16	A
		17	A	17	B	17	A
		18	B	18	B	18	B
		19	A	19	C	19	A
		20	A	20	B	20	A
		21	B	21	C	21	B
		22	B	22	A	22	B
		23	D	23	A	23	C
		24	C	24	C	24	B
		25	C	25	B	25	A
		26	C	26	C	26	A
		27	B	27	B	27	A
		28	A	28	A	28	A
		29	D	29	B	29	C
		30	C	30	B	30	C

10 YRS PAST PAPERS

D.U.H.S.

PAST PAPER - 2008

GENERAL INSTRUCTIONS
(PLEASE READ THE INSTRUCTIONS VERY CAREFULLY)

Write in bold capital letters your Name and Seat Number as given in your Form Card on allocated column of the Answer sheet.

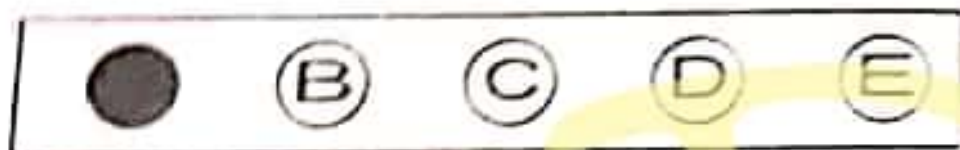
Sign your signature on the allocated column of the answer sheet.

This question paper contains four parts:

Part - I	Biology	30 Questions
Part - II	Chemistry	30 Questions
Part - III	Physics	30 Questions
Part - IV	English	10 Questions

There will be negative marking in the test. Each correct Answer carries (4) marks. Each wrong Answer carries (1) Mark.

The Answer sheet is made of special paper. Which gives itself a carbonized copy on the duplicate. All answers must be marked by filling the appropriate circle with pencil. For every question in the paper, five choices are given. Please mark your choice by filling in the appropriate circle completely. Making it a dark circle as shown below.



Some examples of improper marking are shown below.



The computer will mark improperly filled / dual marked circles as wrong answers.

You are NOT ALLOWED to take away any part of Question paper or note question elsewhere. Use your time efficiently and at a premium. Do not spend too much time on one question, you may run out of time for other questions.

Change of answer is not allowed. No marks will be given if two or more circles are filled. All answers must be marked on the answer sheet, do not mark any answer on the question paper.

at No. _____

Signature of the Candidate _____

NOTE: STOP! DO NOT TURN THIS PAGE TILL FURTHER INSTRUCTIONS.

TEST CONDUCTED BY DOW UNIVERSITY OF HEALTH SCIENCES
PAPER - 2008

Section - I (Biology)

- 1- In which of the following exoskeleton of glasslike silica is found:
 - a) Molluscs
 - b) Arthropoda
 - c) Diatoms
 - d) Star Fish
- 2- The food reserved CO_2 and H_2 are stored in:
 - a) Plastids
 - b) Mitochondria
 - c) Ribosomes
 - d) Nucleus
- 3- Which of the following replaced by thymine:
 - a) Uracil
 - b) Cytosine
 - c) Guanine
 - d) Adenosine
- 4- Which of the following is not an enzyme:
 - a) Polymerase
 - b) Hydroxylase
 - c) Isomerase
 - d) Enzymate
- 5- Which of the following is the most efficient way of respiration in frog:
 - a) Pulmonary
 - b) Bucco-pharyngeal
 - c) Both bucco and pharyngeal
 - d) Cutaneous
- 6- Plants convert solar energy into chemical energy present in the bonds of chemical substances namely:
 - a) NADPH₂
 - b) Nucleotide
 - c) ATP
 - d) RNA
- 7- The hormone which helps in the contraction of uterus at time of birth is:
 - a) Prolactin
 - b) Oxytocin
 - c) Gonadotrophin
 - d) Antidiuretic hormone

- 8- Subcutaneous fat accumulation is essential for:
 a) Seal
 b) Shark
 c) Elephant
 d) Kangaroo
- 9- Hirudin anti clotting factor is found in:
 a) Leech
 b) Earthworm
 c) Vaucheria
 d) Neries
- 10- Animals and plants characters are found in:
 a) Eugleena
 b) Paramecium
 c) Bacteria
 d) Virus
- 11- The photogenic bacteria transfer DNA material to another bacteria by:
 a) Transduction
 b) Transformation
 c) Conjugation
 d) Fragmentation
- 12- When a prosthetic group join with enzymes it forms:
 a) Co-Factor
 b) Co-Enzyme
 c) Holoenzyme
 d) Apo enzyme
- 13- What does not take place in meiosis:
 a) Chromosome multiplied
 b) Separation of chromosomes
 c) Production of hormones
 d) Production of enzymes
- 14- When phenotypically tall plant is crossed with pure dwarf plant what cross is:
 a) Test
 b) Monohybrid
 c) Dihybrid
 d) Multiple cross
- 15- In chick embryo the space between 2 layers is:
 a) Coelome
 b) Blastocoel
 c) Primitive Streak
 d) Gasrocoel

- 16- In which region organic material and shells are found:
 a) Littoral
 b) Abassyal
 c) Euphotic
 d) Bathyal
- 17- Which parasite cant survive in artificial medium:
 a) Endoparasite
 b) Ectoparasite
 c) Obligate
 d) Facultative
- 18- In light antennae complex _____ is not found:
 a) Haemoglobin
 b) Chlorophyll a
 c) Chlorophyll b
 d) Cortenoid
- 19- If the level of CO_2 and H^+ ions increase in the body the rate of breathing will:
 a) Increase
 b) Decrease
 c) Remains constant
 d) unstrated and voluntary
- 20- Lacteals requires which muscle to flow:
 a) Unstriated and involuntary
 b) Striated and voluntary
 c) Striated and involuntary
 d) Unstriated and voluntar,
- 21- The theory of inheritance of acquired characters was proposed by:
 a) Devaries
 b) Darwin
 c) Lamark
 d) Weisman
- 22- Which is associated with long term memory.
 a) Hippocampus
 b) Hypothalamus
 c) Brain Stem
 d) Pons
- 23- Pituitary gland is controlled by which of the following:
 a) Adrenal Cortex
 b) ACTH
 c) Hypothalamus
 d) Cytoxin

24- Bacteria takes rDNA in the presence of:

- a) NaOH
- b) HCL
- c) Calcium Chloride
- d) NaCl

25- Centromere at the end of Chromosomes:

- a) Meta centric
- b) Aerocentric
- c) Sub Meta centric
- d) None of above

26- In plant cell, cell wall and cell plate is the product of:

- a) Golgi bodies
- b) Mitochondria
- c) Endoplasmic reticulum
- d) Plastids

27- Tissue fluid don't contain:

- a) WBC
- b) Glucose
- c) Amino acid
- d) Fibrinogen

28- In which cycle of plasmodium which of the following does not effected, changed:

- a) Sporozoid
- b) Merozooids
- c) Cryptozoid
- d) Trophozooids

29- Which enzyme is used to cut DNA into small pieces:

- a) DNA polymerase
- b) Vector
- c) Restriction enzyme
- d) DNA Ligase

30- Number of chromosomes in human cell are:

- a) 48
- b) 46
- c) 44
- d) 56

Section - II (Chemistry)

1- Grignard reagent on reaction with methanol gives:

- a) Primary alcohol
- b) Secondary alcohol
- c) Ethane
- d) Tertiary alcohol

2- A compound X is oxidize in the presence of $K_2Cr_2O_7$ of conc. H_2SO_4 to form a compound Y which on reaction with gives a sweet / pleasant smell:

- a) Aldehyde
- b) Ester
- c) Ether
- d) Acetone

3- Total pressure is equal to the sum of partial pressure of all the individual gases due to following reason:

- a) Collisions with the walls of the container
- b) Effective collision of all individual gases
- c) Collision with the bottom of the container
- d) None of the above

4- Metallic character of Al^{+3} is greater than boron b/c:

- a) Due to less electron population
- b) The number of neighboring atoms in lattice increase
- c) Both of the above
- d) None of the above

5- $NaNO_2 + \text{heat} \rightarrow NaNO_2 + O_2$ in his reaction:

- a) $K_p = K_c$
- b) $K_p > K_c$
- c) $K_p < K_c$
- d) $K_p K_c$

6- Following are the metals of group IIA:

- a) Mg, Be, Sr, Ra
- b) Be, Ra, Rb, Ba
- c) Mg, Be, Ca, Sr
- d) None of the above

7- When the reaction is in equilibrium:

- a) Production is equal to reaction
- b) Rate of forward reaction is equal to backward reaction
- c) Product is not equal to reaction
- d) All of the above

8- Hydrogen has ground state energy of:

- a) 1.5 e.v
- b) 13.6 e.v
- c) -0.8 e.v
- d) 23.3 e.v

24-

9- Hydrogen is not absorbed by:

a) Ni
b) Pt
c) Pd
d) Fe

25-

10- Bond energy is needed to know:

a) Nature of compound
b) Nature of ionic compound
c) Nature of Reaction
d) Nature of Bonding

26-

11- 1 ounce pressure of gas at STP contains:

a) 22400 cubic meter
b) 4 cubic meter
c) 4400 cubic meter
d) 224 cubic meter

27-

12- Swollen tongue, eyes and mouth are caused by:

a) B₁
b) B₂
c) B₆
d) B₁₂

28-

13- 66 Nylon is a polymer of:

a) Vinyl chloride
b) Aldenylde
c) Adipic acid + Hexamethylene Diamine
d) Both a and b

29-

14- The traid contain chlorine 35.5 and iodine give the atomic mass of bromine:

a) 80
b) 81
c) 72
d) 90

30-

15- Anode mud, in blistening of copper is:

a) Impurity left
b) Pure copper
c) Electrolyte copper sulphate
d) Blisten copper

16- Reforming is a process in:

- a) Which smaller compound into large compounds
b) Straight chain into branches chain
c) Less volatile compound into high volatile
d) Branches chain into cyclic compound

17- In destructive distillation of coal all of the following are obtained except:

- a) Coal liquor
b) Coal tar
c) Colle
d) Coal gas

18- Azimuthal quantum number shows:

- a) Orientation if in space
b) Size of the orbital
c) Shape of orbital
d) None of the above

19- In ethane, which of the following is correct:

- a) Its p orbital is hybridized
b) Its s orbital is hybridized
c) Its p and s orbital both are
d) Its p and s both are un hybridized

20- In SN² reaction the rate of reaction is dependent upon:

- a) Nucleophile
b) Substrate
c) Electrophile
d) Both substrate and Nucleophile

21- Unsaturated compounds are formed by the following:

- a) Elimination substitution
b) Substitution
c) Nucleophile Substitution
d) Electrophilic

22- Glucose and fructose are linked by _____ linkage>:

- a) 1 2 glucosidic linkage
b) 2 2 4 glycoside linkage
c) 1 4 glycoside linkage
d) 1 4 glycoside linkage

23- Compound having different forms this statement is related to:

- a) Allotropy
b) Isomerism
c) Polymorphism
d) Isomorphism

24- The heat absorbed at constant volume is equal to:

- a) External pressure
b) Energy change
c) Internal pressure
d) All of the above

- 25- Equilibrium is established when:
 a) Rate of forward reaction = rate backward reaction
 b) Rate of forward reaction > rate backward reaction
 c) Rate of forward reaction < rate of backward reaction
 d) Rate of forward reaction rate of backward reaction

- 26- Rate of reaction depends upon all of the following except:
 a) Concentration
 b) Pressure
 c) Temperature
 d) Molecularity

- 27- Products of exothermic reaction are:
 a) Unstable
 b) Sometimes stable and sometimes unstable
 c) Stable
 d) None of the above

- 28- Cathode does not depend upon:
 a) Neither on gas nor the material of the cathode.
 b) Neither on H_2SO_4 nor on the liquid used
 c) All of the above
 d) None of the above

- 29- The electronic configuration of noble gases is:
 a) ns^2, np^6
 b) ns^2, np^6
 c) ns^2, np^6, nd^8
 d) None of the above

- 30- Rutherford discovered that the nucleus of an atom has _____ charge.
 a) No
 b) Negative
 c) Positive
 d) Very less

Section + III (Physics)

- 1- If T of string is increased 4 times what will be the speed of wave?
 a) $V/2$
 b) $V/4$
 c) $2V$
 d) $4V$

- 2- What is the formula of concave lens without convention of sign:
 a) $1/q - 1/p = 1/f$
 b) $1/p + 1/q = -1/f$ = $CA - q$
 c) $1/p - 1/p = 1/f$
 d) $1/p - 1/q = 1/f$

- 3- "Sone" is the unit of loudness. What is the relation of sone with db:
 a) 1 sone = 40db
 b) 1 sone = 40db at 100Hz
 c) 1 sone = 40db at 1000Hz
 d) 1 sone = 40db at 10,000Hz

- 4- Three sources each of 1.5 volt are connected in series what is the net emf.
 a) 3 volt
 b) 4.5 volt
 c) 1.5 volt
 d) 6 volt

- 5- Electromagnetic wave depends upon:
 a) Permeability of the medium only
 b) Permeability of the medium only
 c) Both permeability of permeability of medium
 d) Nature of medium

- 6- If light enters from air to denser medium of "n" refractive index, then wavelength
 a) Increases "n" times
 b) Decreases "n" times
 c) Decreases
 d) Remains same

- Shadow is formed due to the phenomenon of:
 a) Interference
 b) Polarization
 c) Diffraction
 d) Both interference and diffraction

- If A.B. = 0, then between them is:
 a) 0°
 b) 45°
 c) 90°
 d) 60°

If power of a lens is 2 diopter, of other is -4 diopter, what is focal length of combination of lenses:

- 9-
a) 20cm
b) 50cm
c) -50cm
d) 100cm

- 10- Energy of hydrogen atom in ground state:
a) -13.6 eV
b) -0.85 eV
c) -10.2 eV
d) -0.25 eV

- 11- Dot product of force and velocity:
a) Rate of Energy
b) Rate of Work
c) Rate of Power
d) Rate of change of moment

- 12- If " λ " is the decay const. then half life is:
a) 0.693
b) 0.693/ λ
c) λ 0.693
d) 0.95/ λ

- 13- When a body moves downward on inclined plane in the presence of friction " f ", force is.
a) $W \sin -f$
b) $W \cos -f$
c) $W \cos$
d) $F - W \sin$

- 14- Graph of Isobaric process is:
a) $P \text{ --- } V$
b) $P \text{ --- } V$
c) $P \text{ --- } V$
d) $P \text{ --- } V$

- 15- According to thermodynamic, at constant volume $Q = ?$
a) $\Delta U = \Delta Q$
b) $P \Delta U = \Delta W$
c) $\Delta U = \Delta W$
d) $Q = 0$

- 16- Velocity of sound is:
a) Directly proportional to absolute temperature
b) Directly proportional to square root of absolute temperature
c) Directly proportional to square root of absolute temperature
d) Directly proportional to pressure

- 17- If the axis of rotation passes through the body and the body moves around itself then the motion is called:
a) Spin motion
b) Orbital
c) Rectilinear motion
d) Translatory motion

- 18- If radius of earth decrease by half and mass remains unchanged, then the value of " g " is:
a) 4g
b) 2g
c) $g/4$
d) $g/2$

- 19- Which instrument is used to detect path of visible rays?
a) Geiger counter
b) Wilson Cloud Detector
c) Solid State Detector
d) None of the above

- 20- Light having range of 7.6×10^{-7} cm to 4.0×10^{-7} cm lies in the region.
a) Infrared
b) X-rays
c) Visible
d) U V light

- 21- Relation between liner velocity and angular velocity:
a) $v = rw$
b) $r = vw$
c) $s = r$
d) $v = \frac{\Delta s}{\Delta t}$

- 22- Instrument having very high magnification used to see microorganisms clearly:
a) Simple Microscope
b) Compound Microscope
c) Telescope
d) Magnifying Glass

- 23- Gauss's law can be used to find electric field intensity.
a) Charge enclosed in a sphere
b) Charge placed between two opposite charged surfaces
c) Charge on an infinite sheet
d) All of the above

Ratio of Emf in a potentiometer is directly prop to ratio of:

- 24-
a) Balancing length
b) Balancing width
c) Balancing Height
d) None of the above

25- Number of lines originating from q to the surface area of sphere is:

- a) Flux intensity
b) Flux linkage
c) Flux density
d) Flux region

26- In solid state detector:

- a) Sealed tube is used
b) Air alcohol and argon is used
c) p-n junction is used
d) All of the above can be used

27- The velocity of alpha particles is:

- a) 0.8×10^8 m/s
b) 0.5×10^8 m/s
c) 0.3×10^8 m/s
d) 0.1×10^8 m/s

28- 2nd Harmonic is also known as:

- a) Fundamental frequently
b) 1st overtone
c) 2nd overtone
d) Basic overtone

29- The carrier waves are over modulated to be distorted version of m_a^3 modulating signal when:

- a) $m_a = 1$
b) $m_a < 1$
c) $m_a > 1$
d) $m_a \neq 1$

30- The frequency acquired to produce artificial gravity in satellite is:

- a) 8.5 rev / min
b) 9.5 rev / min
c) 9.8 rev / min
d) 10.0 rev / min

Section - IV (English)

151

10 Years Past Papers

1- Destitute (Give Synonym)

- a) Disable
c) Poor

- b) Isolated
d) Emangled

2- Orthodox (Give Antonym)

- a) Emangled
c) Conventional

- b) Unconventional
d) Traditional

3- Economical (Antonym)

- a) Maddest
c) Extravagant

- b) Efficient
d) Expensive

4- Limitations (Synonym)

- a) Short comings
c) Vicinity

- b) Restriction
d) Inability

5- Expedient (Synonym)

- a) Scapegoat
c) Suitable

- b) Heterogeneous
d) Apathetic

6- Voracious (Synonym)

- a) Reject
c) Honest

- b) Excitable
d) Greedy

7- Hamper (Antonym)

- a) Ogle
c) Assist

- b) Clarify
d) Uncover

Monotonous (Synonym)

- a) Limited
c) Extremist

- b) Conclusive
d) Repeatedly

1- Intensive: (Antonym)

- a) Casual
c) Progressive

- b) Regressive
d) Attentive

0- Presice (Synonym)

- a) Concise
c) Exaggerate

- b) Comprehensive
d) Explanatory

DO IT FROM THOSE WHO HAVE DONE IT NOT FROM THOSE WHO CLAIM

College

will be marked incorrect by the machine

BIOLOGY		CHEMISTRY		PHYSICS		ENGLISH	
01	C	01	C	01	C	01	C
02	A	02	B	02	B	02	B
03	A	03	A	03	C	03	D
04	D	04	D	04	B	04	B
05	A	05	B	05	C	05	B
06	C	06	C	06	A	06	D
07	B	07	B	07	D	07	C
08	B	08	B	08	C	08	D
09	A	09	A	09	B	09	A
10	A	10	D	10	A	10	C
11	A	11	B	11	C		
12	C	12	B	12	B		
13	C	13	C	13	A		
14	C	14	B	14	A		
15	A	15	A	15	A		
16	B	16	B	16	B		
17	A	17	C	17	A		
18	A	18	C	18	A		
19	A	19	C	19	B		
20	A	20	D	20	C		
21	C	21	A	21	A		
22	A	22	A	22	B		
23	C	23	C	23	D		
24	C	24	B	24	A		
25	D	25	A	25	C		
26	A	26	D	26	C		
27	D	27	C	27	D		
28	D	28	A	28	B		
29	C	29	B	29	C		
30	B	30	C	30	B		

11 YRS PAST PAPERS

MODEL TEST
PAST PAPERS 2010

SECTION-I (BIOLOGY)

1. The chemical formula of Uric acid is:

- a) $(\text{NH}_2)_2\text{CO}$
- b) $\text{NH}_2\text{CO NH}_2$
- c) $\text{C}_5\text{O}_3\text{N}_4\text{H}_4$
- d) $\text{C}_6\text{O}_4\text{N}_2\text{H}_2$

2. How many metacarpal bones are present in a lower limb of a human being:

- a) 5
- b) 7
- c) 2
- d) None of these

3. Cardiac muscles are:

- a) Involuntary & Untreated
- b) Voluntary striated
- c) Involuntary & Striated
- d) Skeletal muscles

4. The supply line b/w embryo & mother is:

- a) Placenta
- b) Umbilical cord
- c) Amnion
- d) Chorion

5. Which of the following disease if not treated can cause blindness in new born:

- a) Syphilis
- b) AIDS
- c) Hepatitis
- d) Gonorrhea

6. The sperms are stored in:

- a) Seminiferous tubules
- b) Vas Deferens
- c) Epididymus
- d) Testes

7. Exchange of genetic material takes place in:

- a) Meiosis
- b) Prophase I
- c) Diplotene
- d) Pachytene

The genotype of a Turner's syndrome patient is:

- a) XXY
- b) Trisomy 21
- c) XO
- d) OXY

9. Conception in human female takes place at:

- a) Uterus
- b) Vagina
- c) Ovary
- d) Fallopian tube

10. Which secretion neutralizes the acidity of urethra?

- a) Urine
- b) Testosterone
- c) Sperms
- d) Semens

11. The example of gliding joints is:

- a) Ankle & Wrist
- b) Ribs
- c) Vertebrae
- d) Ball & Socket joint

12. The intra muscular injection at hip should be injected in:

- a) Inner & lower quadrant
- b) Outer & upper quadrant
- c) Sciatic nerve
- d) Iliac vein

13. Abductor muscular allow the arm to move:

- a) Towards the body
- b) Away from the body
- c) Upwards
- d) Downwards

14. Renal vessels leave kidney at:

- a) Hypocotyl
- b) Hilus
- c) Vasa Recta
- d) Pyramids

15. Urine contains urea in a concentration of about:

- a) 95%
- b) 0.3 - 0.7%
- c) 2%
- d) 5%

16. If a colour blind person marries a homozygous normal female, what are the chances of sons to be colour blind?

- a) 50%
- b) 25%
- c) 100%
- d) None of these

17. The largest part of fore brain is:
 a) Diencephalon
 b) Corpus callosum
 c) Telencephalon
 d) Cerebrum
18. The chemical attractant which directs the pollen tube to enter the ovary to route microphyle is:
 a) Seminal fluid
 b) Abscission acid
 c) Auxin
 d) Calcium
19. Alpha cells of pancreas secrete:
 a) Amylase
 b) Lipase
 c) Insulin
 d) Glucagon
20. The physiological or functional contact b/w two neurons is called:
 a) Neuroglia
 b) Action potential
 c) Synapse
 d) R.M.T.
21. All are the bones of axial skeleton except:
 a) Scapula
 b) Sternum
 c) Ribs
 d) Vertebrae
22. Sliding filament theory of muscle contraction was given by:
 a) Rudolf Virchow
 b) Sheldon & Schwann
 c) Huxley
 d) Robert Brown
23. The process of Haemodialysis takes about:
 a) 1-2 hrs
 b) 24 hrs
 c) 6-10 hrs
 d) More than 24 hrs
24. What is the connection b/w Renal pelvis & urinary bladder:
 a) Pyramids of medulla
 b) Renal vessels
 c) Ureter
 d) Urethra

25. Thermo regulation is done by sweat glands in mammals especially, but some lack glands so they do:
 a) Guttation
 b) Transpiration
 c) Panting
 d) Hydathodes
26. The process in which fruit privation takes place without the formation of seeds, is called:
 a) Pathogenesis
 b) Regeneration
 c) Apomixis
 d) Parthenocarpy
27. Every statement is true about fraternal twins except:
 a) They can be of different sex
 b) They are dizygotic
 c) They are developed sexually
 d) The genotype of each zygote is sense
28. About urethra each statement is correct except:
 a) It is the last part of urinary system
 b) It is the extension of urinary bladder
 c) It is a known as urogenital duct in females
 d) Contains prostatic gland at its tip
29. About menstruation phase: pick the odd one out:
 a) Onset off bleeding and debris of uterus through vagina
 b) Lasts about 5 days
 c) Progesterone secretion stops
 d) Blood & Myometrium flows out through vagina.
30. Amitosis takes place in:
 a) Germ cells
 b) Somatic cells
 c) Skin cells
 d) Bacterial cells

SECTION II (CHEMISTRY)

1. Final zeros to the right of the decimal point are:
- Non-significant
 - Necessarily not significant
 - Significant
 - b and c

2. The property of matter that is independent of its surrounding conditions is:
- Volume
 - Density
 - Mass
 - Weight

3. In s-block elements low ionization enthalpy, low electron affinity values the elements:
- Readily lose their valence electrons
 - Form anions
 - Form cations
 - a and c

4. Elements were arranged in ascending order in the periodic table by:
- Mendeleev
 - Dobereiner
 - Newland
 - Moseley

5. The percentage of nitrogen in ammonia is given by the expression:
- $\frac{14}{17} \times 100$
 - $\frac{7}{17} \times 100$
 - $\frac{4}{17} \times 100$
 - $\frac{17}{14} \times 100$

6. Elastic collision involves:
- Loss of energy
 - Gain of energy
 - NO gain, no loss of energy
 - None of the above

7. Natural gas contains about:
- 80% Methane
 - 75% Methane
 - 60% Methane
 - 50% Methane

8. An oxide of nitrogen contains 30.4% of nitrogen, what is the mole ratio of:
- 5
 - 4
 - 2
 - 7

9. On the upper right corner of periodic table is:
- Non-metal
 - Metals
 - Metalloids
 - Solids

10. The density of a substance is 1.5g/ml. what will 100ml of its weight be?
- 150 g
 - 100 g
 - 200 g
 - 300 g

11. Elements of a same group have:
- Same size
 - Same reactivity
 - Same chemical properties
 - Same physical properties

12. Water is said to be permanently hard when it contains:
- Chlorides of Ca^{+2} & Mg^{+2}
 - Carbonate of Ca^{+2} & Mg^{+2}
 - Bicarbonate of Ca^{+2} & Mg^{+2}
 - None

13. No. of moles of CO_2 which contain 8.0g of oxygen.
- 0.25
 - 0.5
 - 1
 - 1.5

14. Rutherford's atomic model failed b/c:
- It did not account for the stability of atom
 - It did not account for the attraction b/w protons and neutrons
 - The atom did not have neutrons
 - None of these

15. The second and third periods contain:
- 18 elements each
 - 32 elements each
 - 8 elements each
 - 14 elements each

16. Cathode rays travel in discharge tube from:
- Anode to cathode
 - Cathode to anode
 - Glass tube to vacuum pump
 - Cathode through air

17. The subshell which has only one orbital is identified by the letter:

- a) s
- b) d
- c) p
- d) f

18. Sodium amalgam is an alloy of:

- a) Sodium and mercury metals
- b) Sodium and lead
- c) Sodium and zinc
- d) None

19. Elements which have completely filled outermost shell and do not combine with other elements are called:

- a) Noble elements
- b) Reactive elements
- c) Unstable elements
- d) Rare earth elements

20. Washing soda is:

- a) Na_2CO_3
- b) $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$
- c) NaOH
- d) $\text{Na}_2\text{CO}_3 \cdot \text{H}_2\text{O}$

21. By standard temperature we mean:

- a) 0°C
- b) 293°C
- c) 35°C
- d) -0°C

22. Which solid is called dry ice?

- a) CO_2
- b) $\text{H}_2 + \text{CO}$
- c) H_2
- d) None

23. 23g of sodium is present in:

- a) 1 atom of sodium
- b) 23 atoms of sodium
- c) 1 gram atom of sodium
- d) 1 mole of sodium
- e) c and d

24. Alkali metals are strongly:

- a) Reducing agents
- b) Oxidizing agents
- c) Both of these
- d) None of these

25. Hydrogen forms salt hydrides with the element of:

- a) IA and IIA
- b) IIIA and IVA
- c) VA and VIA
- d) VIIA

26. Super oxides are expressed as:

- a) O^{2-}
- b) O_2^{2-}
- c) O_2^{1-}
- d) O^{1-}

27. Hexagonal system has axes:

- a) $a = b = c$
- b) $a \neq b = c$
- c) $a \neq b \neq c$
- d) $a = b \neq c$

28. Three dimensional array of points is called:

- a) Unit cell
- b) Crystal
- c) Crystal lattice
- d) a and b

29. Standard pressure:

- a) 760 torr
- b) 1 atm of Hg
- c) 760 mmHg
- d) All of the above

30. SiO_2 is known as:

- a) c and d
- b) rust
- c) Silicon dioxide
- d) Sand or silica

SECTION-IV (PHYSICS)

1. A 2.0×10^3 kilogram car travels at constant speed of 12 meters per second around circular curve of radius 30 meters. As the car goes around the curve, the centripetal force is directed:

a) Toward the center of the circular curve.
 b) Away from the center of the circular curve.
 c) Tangent to the curve in the direction of motion.
 d) Tangent to the curve opposite the direction of motion.

2. The amount of work done against friction to slide a box in a straight line across uniform horizontal floor depends most on the:

a) Time taken to move the box
 b) Distance the box is moved
 c) Speed of the box
 d) Direction of the box's motion

3. A car travels 90 meters due north in 15 seconds. Then the car turns around and travels 40 meters due south in 5.0 seconds. What is the magnitude of the average velocity of the car during this 20 second interval?

a) 2.5 m/s
 b) 5.0 m/s
 c) 6.5 m/s
 d) 7.0 m/s

4. At the top of the trajectory of a projectile, the directions of its velocity and acceleration are:

a) Parallel to each other.
 b) Perpendicular to each other
 c) Inclined at 45° to the horizontal
 d) Inclined at 45° to each other

5. A bag is dropped from an aeroplane flying horizontally at a constant speed. Neglecting air resistance, where will the aeroplane be when the bag reaches the ground.

a) Directly above the bag.
 b) Ahead of the bag.
 c) Behind the bag.
 d) Data is not sufficient.

6. The angle subtended at the centre of a circle by an arc of length equal to its radius is equal to:

a) One rotation
 b) Half rotation
 c) One degree
 d) One radian

7. A car is moving with a uniform speed of 2 m/s in a circle of radius 0.4m. Its angular speed is given in rad/sec is:

a) 0.8
 b) 1.6
 c) 4
 d) 5

8. The necessary centripetal force to the moving car around a corner track is provided by:

a) Gravitation force
 b) Coulomb force
 c) Force of Friction
 d) Centrifugal force

9. If a body of mass "m" is moving with a constant speed "V" (also called Tangential Velocity) in a circular path of radius "r", its tangential acceleration is:

a) rv^2
 b) v^2/r
 c) Zero
 d) None of these

10. If a body moves with a constant speed in a circle:

a) No work is done on it.
 b) No force acts on it.
 c) No acceleration is produced in it.
 d) Its velocity remains constant.

11. A body is called rigid Body, if under the action of the force, distance between two points:

a) Changes
 b) Remains the same
 c) Increases
 d) Decreases

12. If the line of action of force passes through the axis of rotation then torque is:

a) Zero
 b) Maximum
 c) Minimum
 d) One

13. When the first condition of equilibrium is satisfied then the body has no:

a) Linear Velocity
 b) Linear acceleration
 c) Rotational Acceleration
 d) Motion

14. When the second condition of equilibrium is satisfied then the body has no:

a) Linear Velocity
 b) Linear acceleration
 c) Rotational acceleration
 d) Motion

15. If the distance between two mass particles is doubled, the gravitational force between them:

- a) Is reduced to half of the original value
- b) Is reduced to one fourth of the original value
- c) Becomes Double
- d) Becomes four times

16. If the distance between two mass particles is halved, the gravitational force between them becomes:

- a) Four times
- b) Three times
- c) Double
- d) Half

17. If the distance between two mass particles is doubled and their masses are also doubled, the gravitational force:

- a) Is doubled
- b) Is Halved
- c) Becomes four times
- d) Remains unchanged (constant)

18. The minimum velocity at which a body must be projected to go out from the earth's gravitational pull is called:

- a) Angular Velocity
- b) Escape Velocity
- c) Terminal Velocity
- d) Root mean square Velocity

19. The intensity of the earth's gravitational field is maximum at:

- a) The centre of the earth
- b) The Equator
- c) The poles
- d) None of the above

20. A body weights 100N on the surface of the earth. Its weight at a depth equal to half the radius of the earth is:

- a) 44.44 N
- b) 50 N
- c) 25 N
- d) 75 N

21. The Value of acceleration due to gravity "g" at the centre of the earth is:

- a) 1.96 m/s^2
- b) 9.8 m/s^2
- c) 4.9 m/s^2
- d) Zero

22. The Artificial Gravity is created in a satellite overcome the state of weightlessness, experienced by the astronauts in it by:

- a) Decreasing the orbital speed of the satellite
- b) By increasing the Orbital speed of the satellite
- c) By causing the satellite to rotate about its own axis
- d) None of these

23. Weightlessness experienced in a spaceship is due to:

- a) Absence of inertia
- b) Absence of gravity
- c) Absence of Accelerating force
- d) Free fall of the spaceship

24. If the force and displacement are the right angle to each other, then the workdone is:

- a) Zero
- b) Negative
- c) Maximum
- d) Positive

25. A body is displaced from $X = X_1$ to $X = X_2$ by a force of $2X$. The work done is:

- a) $2X(X_2 - X_1)$
- b) $2X^2(X_2 - X_1)$
- c) $X_2^2 - X_1^2$
- d) $(X_2 - X_1)^2$

26. The Unit of power in British Engineering System is:

- a) Watt
- b) Kilo-watt
- c) Joule
- d) Horse power

27. An Electric Motor creates a tension of 45N in a hosting cable and reels it in at the rate of 2m/s. The power of the motor is:

- a) 15 KW
- b) 90 Watt
- c) 225 KW
- d) 9000 hp

28. The Kinetic Energy of a 100gm Bullet moving at a speed of 100 m/s is:

- a) 2500 J
- b) 500 J
- c) 1250 J
- d) 250 J

29. Electron volt is the unit of:

- a) Power
- b) Energy
- c) ~~Work~~ p.d
- d) Momentum

30. Watt is the unit of power. Kilowatt-hour is the unit of:

- a) Power
- b) Work
- c) Momentum
- d) Energy

SECTION-IV (ENGLISH)

1. Nexus:

- a) Shameless
- b) Influence
- c) Connection
- d) Love

2. Bennett:

- a) To seek something
- b) A person's family name
- c) To look after
- d) Amaze

3. Mob:

- a) A web
- b) Disorderly crowd
- c) Exhaust
- d) None of these

4. Dabble:

- a) Involve one self without serious effort
- b) To train some one for some work
- c) Excellent work
- d) To enjoy the work

5. Vice:

- a) Sensible
- b) Senseless
- c) A tool with two jaws
- d) A man of good health

6. Don't post these letters (change the voice)

- a) Let these letters be posted
- b) Post these letters
- c) Don't let these letters be posted
- d) Didn't post these letters

7. He asked me why I was going to Karachi. (Change the form of Narration)

- a) He said to me, "why are you going to Karachi".
- b) He said to me, "why I am going to Karachi".
- c) He said to me, "Why had you gone to Karachi".
- d) He said to me, "Why had you gone to Karachi".

8. What are you doing? (Change into Passive voice)

- a) What you have done?
- b) What is being done by you?
- c) What is done by you?
- d) What have been done by you?

9. H is known _____ every lady (Insert preposition).

- a) by
- b) to
- c) for
- d) about

10. _____ Punjab is _____ fertile province of Pakistan. (Insert articles)

- a) The a
- b) The X
- c) X a
- d) X X

DOW GRADS

DO IT FROM THOSE WHO HAVE DONE IT NOT FROM THOSE WHO CLAIM!

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85															

College

Percentage

Contact

Environ Biol Fish

DOW Signatory

[illegible]

MODEL TEST 2009 ANSWER KEY

BIOLOGY		CHEMISTRY		PHYSICS		ENGLISH	
01	C	01	C	01	A	01	C
02	D	02	C	02	B	02	B
03	C	03	D	03	A	03	B
04	B	04	C	04	B	04	A
05	D	05	A	05	A	05	C
06	C	06	C	06	D	06	C
07	C	07	A	07	D	07	A
08	C	08	C	08	C	08	B
09	A	09	A	09	C	09	B
10	D	10	A	10	A	10	C
11	C	11	C	11	B		
12	B	12	A	12	A		
13	B	13	A	13	B		
14	B	14	A	14	C		
15	C	15	C	15	B		
16	A	16	B	16	A		
17	C	17	A	17	D		
18	D	18	A	18	B		
19	D	19	A	19	C		
20	C	20	B	20	B		
21	A	21	A	21	D		
22	C	22	A	22	C		
23	C	23	C	23	D		
24	C	24	A	24	A		
25	C	25	A	25	A		
26	D	26	C	26	D		
27	D	27	D	27	B		
28	C	28	C	28	B		
29	D	29	D	29	C		
30	D	30	A	30	D		

11 YRS PAST PAPERS

DURU

N.T.S

PAST PAPER - 2009

Read the passage to answer question 1-2:
It is of course a trifle absurd to speak of Asia as a unity, and only opposition to Western Imperialism has caused people to think in these terms. Asia contains half the population of the world and at least three very distinct civilizations: that of Islam, that of India, and that of China. These differ from each other just as much as they differ from the civilization of Christendom, and there is not the faintest reason to expect them all to act in unison. What is to be hoped is, not an endeavor after cultural or political unity, but a determination to uphold independence at home and to respect it elsewhere – and when I speak of independence, I am not thinking only of politics, but also of culture. There is a great danger of too much cultural uniformity. NO great civilization has ever been cosmopolitan.

ENGLISH

- 1- In the view of the author, what is the harm of cultural uniformity?
- It will not lead to the formation of a great civilization
 - It will lead to the formation of a great civilization
 - It will lead to the formation of a rich culture
 - It will lead to the change of environment
 - It will lead to the destruction of values

- 2- The author is in favor of:
- Cultural uniformity
 - Cosmopolitanism
 - Political and cultural independence
 - Unity of Asia
 - None of the above

Complete the sentences by choosing the most appropriate word, from the given lettered choices (A to E) below each:

- 3- Nearly everyone dreams of building _____ ideal house.
- its
 - their
 - his
 - him
 - them

- 4- Life _____ water, light, a moderate temperature and a variety of chemical elements.
- finds
 - reveals
 - designs
 - calculates
 - requires

Identify the word or phrase that needs to be changed for the sentence to be correct:

- 5- The stories that she makes out for her children ought to be written down and published. No error.
- A
 - B
 - C
 - D
 - E

- 6- We are expecting to go abroad this Summer. No error
- A
 - B
 - C
 - D
 - E

Choose the word most similar in meaning to the capitalized one.

DISRUPTION:

- 7-
- comfort
 - luxury
 - trouble
 - freedom
 - calm

INEVITABLE:

- 8-
- doubtful
 - deny
 - unexpected
 - certain
 - unusual

Choose the lettered word or phrase that is most nearly opposite in meaning to the word in capital letter.

ERADICATION:

- 9-
- suppression
 - termination
 - control
 - establish
 - extinction

INTERRUPTION:

- 10-
- break
 - continuity
 - injury
 - difference
 - crack

PHYSICS

11-

Due to the magnetic force, a positively charged particle executes uniform circular motion within a uniform magnetic field 'B'. If the charge is 'q' and the radius of path 'r', which of the following expression gives the magnitude of the particle's momentum?

- A. $\frac{qBr}{r}$
 B. $\frac{qB}{r}$
 C. $\frac{q}{(Br)}$
 D. $\frac{B}{(qr)}$
 E. $\frac{r}{(qB)}$

12-

A transverse wave on a long horizontal rope with a wavelength of 8m travels 2m/s. At $t = 0$, a particular point on the rope has a vertical displacement of A where A is the amplitude of the wave. At what time the vertical displacement of same point on the rope be -A?

- A. $t = \frac{1}{8}$ s
 B. $t = \frac{1}{4}$ s
 C. $t = \frac{1}{2}$ s
 D. $t = 2$ s
 E. $t = 4$ s

13-

An object is placed 60 cm in front of a concave spherical mirror whose focal length is 40cm. which of the following best described the image?

	NATURE OF IMAGE	DISTANCE FROM MIRROR
A.	Virtual	24 cm
B.	Real	24 cm
C.	Virtual	120 cm
D.	Real	120 cm
E.	Real	240 cm

14-

An object is placed 60cm from a spherical convex mirror. If the mirror forms a virtual image 20 cm from the mirror, what is the magnitude of the mirror's radius of curvature?

- A. 7.5cm
 B. 15cm
 C. 30cm
 D. 60cm
 E. 120cm

15-

Find the unit vector parallel to the vector: $\vec{B} = 6\mathbf{i} + 12\mathbf{j} - 4\mathbf{k}$.

- A. $\frac{1}{14} \mathbf{i} + \frac{12}{14} \mathbf{j} - \frac{4}{14} \mathbf{k}$
 B. $\frac{1}{14} \mathbf{i} + \frac{12}{14} \mathbf{j} + \frac{4}{14} \mathbf{k}$
 C. $\frac{1}{14} \mathbf{i} + \frac{17}{14} \mathbf{j} - \frac{4}{14} \mathbf{k}$
 D. $\frac{1}{14} \mathbf{i} + \frac{12}{14} \mathbf{j} + \frac{9}{14} \mathbf{k}$
 E. $\frac{1}{14} \mathbf{i} - \frac{12}{14} \mathbf{j} - \frac{1}{14} \mathbf{k}$

16-

The quantity of heat required to raise the temperature of one mole of a substance through 1K, and its units are J - mole⁻¹K⁻¹ is called:

- A. Carnot Engine
 B. Molar specific heat
 C. Kinetic specific heat
 D. General Gas Law
 E. Boyle's Law

17-

A shot leaves a gun at the rate of 160m/s. Calculate the greatest distance to which it could be projected and the height to which it would rise? ($g = 10\text{m/s}^2$).

- A. 1560m, 540m
 B. 2560m, 640m
 C. 3560m, 740m
 D. 4560m, 840m
 E. 9595m, 348m

18-

A car covers the first half of a certain distance with a speed V_1 and the second half with a speed V_2 . Find the average speed during the whole journey?

- A. $\frac{V_1 V_2}{2(V_1 + V_2)}$
 B. $\frac{2V_1 V_2}{(V_1 + V_2)}$
 C. $\frac{2V_1 3V_2}{V_1 + 5V_2}$
 D. $\frac{V_1 V_2}{5V_1 V_2}$
 E. $\frac{V_1 - V_2}{4V_1 + 9V_2}$

19. A rescue helicopter drops a package of emergency ration to a stranded party on the ground. If the helicopter is traveling horizontally at 40m/s at a height 100m above the ground, where does the package strike the ground relative to the point at which it was release.

A. 120m
B. 130m
C. 140m
D. 180.7m
E. 200.3m

20. The radius of the moon is 27% of the earth's radius and it's mass is 1.2% of the earth's mass. Find the acceleration due to gravity on the surface of the moon.

A. 0.431 m/s²
B. 1.615 m/s²
C. 2.431 m/s²
D. 3.615 m/s²
E. 4.431 m/s²

21. Two capacitors $C_1 = 2\mu$ and $C_2 = 4\mu$ F are connected in series across a 100V supply. Find the effective capacitance.

A. $\frac{1}{2}\mu$ F
B. $\frac{3}{2}\mu$ F
C. $\frac{5}{2}\mu$ F
D. $\frac{4}{3}\mu$ F
E. 2μ F

22. A sample of an Ideal gas occupies a volume 'V' at pressure 'p' and absolute temperature 'T'. The mass of each molecule is 'm'. If 'k' is the Boltzmann constant then the density of the gas is:

A. mkT

B. $\frac{P}{KT}$

C. $\frac{P}{kTV}$

D. $\frac{mP}{kT}$

E. $\frac{2mPT}{k}$

23. A car is waiting at a traffic signal and when the signal turns green, the car starts ahead with a constant acceleration of 2 m/s². At the same time as a bus traveling with a constant speed of 10 m/s overtakes and passes the car. How far beyond its starting point will the car overtake the bus?

A. 40M
B. 30M
C. 90M
D. 120M
E. 100M

24. A battery has an e.m.f. of 6.0 volts and an internal resistance of 0.4 ohm. It is connected to a 2.6 ohm resistor through a SPST (single pole, single throw) switch.

A. 0
B. 0.8
C. 2.6
D. 5.2
E. 6.0

25. When the switch is closed, the potential difference between the terminals of the battery is, in volts:

A. 0
B. 0.8
C. 2.6
D. 5.2
E. 6.0

26- Assume that you two balls of identical volume, one weighing 2 newtons and other 10 newtons. Both are falling freely after being released from the same point simultaneously. Which of the following will then be true?

- I- The 10-N ball falling freely from rest will be accelerated at a greater rate than the 2-N ball.
 II- At the end of 4s of free fall, the 10-N ball will have 5 times the momentum of the 2-N ball.
 III- At the end of 4s free fall, the 10-N ball will have the same kinetic energy as the 2-N ball.
 IV- The 10-N ball possesses greater inertia than the 2-N ball.

- A. I, II and III only
 B. I and III only
 C. II and IV only
 D. IV only
 E. None of the above

27- A ball moving horizontally with speed 'v' strikes the bob of a simple pendulum at rest. The mass of the bob is equal to that of the ball. If the collision is elastic, the bob will rise to a height:

- A. $\frac{v^2}{g}$
 B. $\frac{v^2}{2g}$
 C. $\frac{v^2}{4g}$
 D. $\frac{v^2}{8g}$
 E. $\frac{v^2}{7g}$

28- A point source of light is placed at the principal focus of a concave lens. Which of the following will be true of the refracted light?

- I- It will diverge.
 II- It will be parallel to the principal axis.
 III- It will seem to come from a point 1/2 of the radius of curvature from the lens.
 IV- It will converge.
 A. I, II and III only
 B. I and III only
 C. II and IV only
 D. IV only
 E. None of the above

29- The dimensions of volume and acceleration (respectively) are:
 A. LT^1 and LT^{-2}
 B. LT^2 and LT^{-1}
 C. L^3 and LT^{-2}
 D. L^4T^{-1} and LT
 E. LT^3 and T^2

30- A vector such as the velocity of a body undergoing uniform translational motion, which can be displaced parallel to itself and applied to any point is known as:
 A. Unit vector
 B. Free vector
 C. Null vector
 D. Position vector
 E. Resultant vector

31- What is the flux density at a point 3 cm from the long straight wire, when there is a current of 25A in the wire? ($\mu_0 = 4\pi \times 10^{-7}$)
 A. $0.23 \times 10^{-1} T$
 B. $1.67 \times 10^{-4} T$
 C. $2.99 \times 10^{-6} T$
 D. $3.63 \times 10^{-8} T$
 E. $9.99 \times 10^{-7} T$

32- If an object is placed 30 cm from a convex lens whose focal length is 15 cm, the size of the image compared to the size of the object will be approximately:
 A. twice as large
 B. more than twice as large
 C. 1.5 times as large
 D. smaller
 E. the same size

33- When a conductor of cross-sectional area $5.0 \times 10^{-4} m^2$ carries a current of 6.0A, the drift velocity of the conduction electrons is $1.2 \times 10^{-4} ms^{-1}$. What is the number density (number per unit volume) of the conduction electrons?
 A. $4.0 \times 10^{-28} m^{-3}$
 B. $1.6 \times 10^{-27} m^{-3}$
 C. $2.5 \times 10^{-27} m^{-3}$
 D. $6.3 \times 10^{-28} m^{-3}$
 E. $1.3 \times 10^{-34} m^{-3}$

34- A thermocouple is connected across a galvanometer of resistance 30Ω . One junction is immersed in water at 373 K and the other in ice at 273K. The e.m.f of the thermocouple is $90\mu V$ for each 1K difference in temperature between the junctions, and the thermocouple resistance is 6Ω . What current will flow in the galvanometer?
 A. $1.8\mu A$
 B. $250\mu A$
 C. $300\mu A$
 D. $1.5mA$
 E. $1.8mA$

35. The first law of thermodynamics may be written as $\Delta U = Q + W$, Where ' ΔU ' is the increase in internal energy of the system, ' Q ' is the heat transfer to the system and ' W ' is the external work done on the system. Which of the following is correct for the case of an isothermal expansion of an ideal gas?

A $W > 0$
 B $W = 0$
 C $\Delta U = 0$
 D $\Delta U > 0$
 E $Q = 0$

36. A body of mass 5kg, initially at rest, is moved by a horizontal force of 2N on a smooth horizontal surface. Find the work done by the force in 10s?

A 40J
 B 30J
 C 50J
 D 20J
 E 10J

37. A particle moves from position $\vec{r}_1 = 3\hat{i} + 2\hat{j} - 6\hat{k}$ to position $\vec{r}_2 = 14\hat{i} + 13\hat{j} + 9\hat{k}$ under the action of a force $\vec{F} = 8\hat{i} + 2\hat{j} + 6\hat{k}$. find the work done by the force.

A 50 units
 B 75 units
 C 125 units
 D 155 units
 E 200 units

38. A body start sliding on a rough horizontal surface with a speed of 10 m/s. If the coefficient of friction is 0.2, find the distance traveled by the body before coming to rest. ($g = 10 \text{ m/s}^2$)

A 15m
 B 25m
 C 35m
 D 40m
 E 55m

39. A battery whose e.m.f. is 40V has an internal resistance of 5Ω . If this battery is connected to a 15Ω resistor 'R', what will be the voltage drop across 'R'?

A 10V
 B 30V
 C 40V
 D 50V
 E 70V

40. A particle of charge -0.04C is projected with speed $2 \times 10^4 \text{ m/s}$ into a uniform magnetic field 'B' of strength 0.5T. If the particle's velocity as it enters the field is perpendicular to 'B', what is the magnitude of the magnetic force on this particle?

A 4N
 B 8N
 C 40N
 D 80N
 E 400N

CHEMISTRY

41. Catenation is a process in which carbon shows the properties of:

A Making single bond
 B Hybridization
 C Making long chains or rings of carbon atoms
 D Isomerism
 E Breaking of bonds

42. The maximum number of electrons that an orbital can accommodate is / are:

A 0
 B 1
 C 2
 D 3
 E 4

43. Which of the following is not a nucleophile?

A HO^-
 B NH_3
 C BF_3
 D CN^-
 E NH_2^-

44. Chlorine is manufactured commercially by the electrolysis of aqueous sodium chloride (brine). Which other products are made in this process?

A Hydrochloric acid and hydrogen
 B Hydrogen and sodium
 C Hydrogen and sodium hydroxide
 D Sodium and sodium hydroxide
 E Hydrochloric acid and sodium

45. Sodium reacts with water more vigorously than lithium because it:

A Has higher atomic weight
 B Is more electronegative
 C Is more electropositive
 D Is a metal
 E Has high melting point

The crystals formed as a result of Vander waal's interactions are:

- 46-
A. Molecular crystals
B. Covalent crystals
C. Metallic crystals
D. Ionic crystals
E. None of the above

47- Which of the following statements about H_2S is false?

- A. It is a covalent compound
B. It is a gas with bad smell
C. It is a strong reducing agent then H_2O
D. Its molecule is non-linear
E. It is a weak base in water

48- The rain drop acquires spherical shape and ink spreads over blotting paper due to:

- A. Surface tension
B. Adhesive forces
C. Viscosity
D. Polarity
E. Latent heat of vaporization

49- 950 torr corresponds to:

- A. 3.5 atm
B. 1 atm
C. 3 atm
D. 1.25 atm
E. 2.25 atm

50- The enthalpy change accompanying the gain of an electron by a neutral gaseous atom to form negative ion is called:

- A. Ionization potential
B. Electro negativity
C. Electron affinity
D. Lattice energy.
E. Potential energy

51- Sigma bond is formed by:

- A. Transferring the electrons
B. head on overlapping of atomic orbital
C. Mutual but unequal sharing of electrons
D. Parallel overlapping of atomic orbital
E. All of the above

52- The heat of a reaction can be calculated by using:

- A. Joule's law
B. Ohm's law
C. Hess's law
D. Faraday's law
E. Boyle's law

53- According to law of mass action, "The rate of chemical reaction is proportional to:

- A. Products
B. Product of molar concentration of reactants
C. Initial concentrations of reactants
D. Catalyst
E. Pressure

54- For, the reaction $N_2 + 3H_2 \rightleftharpoons 2NH_3$.
The production of NH_3 will be favoured at

- A. High pressure and catalyst
B. Low pressure only
C. Low pressure and catalyst
D. High pressure only
E. Catalyst only

55- The sum of exponents of the molar concentration of the reactants is equal to:

- A. Molecular
B. Polarity
C. Activation energy
D. Rate of reaction
E. Order of reaction

56- Nacent hydrogen used in the formation of methane, is obtained from the reaction of:

- A. $NaHCO_3$ with Zn
B. HCl with Zn
C. KOH with Zn
D. H_2O with Zn
E. CH_3I with Zn

57- "Zymase" a group of 14 enzymes, used in the fermentation of starch, is present in:

- A. Bacteria
B. Yeast
C. Fungi
D. Algae
E. Virus

58- The general formula for aldehydes is:

- A. R-OH
B. R-COOH
C. R-CO-R
D. R-X
E. R-CHO

59- When an element exists in more than one crystalline form, the phenomenon is termed as:

- A. Isomorphism
B. Allotropy
C. Isomerism
D. Anisotropy
E. Enthalpy

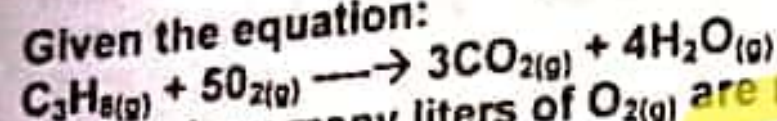
60- The formula for "Plaster of Paris" is:

- A. $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$
- B. $2\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$
- C. $(\text{CaSO}_4)_2 \cdot \text{H}_2\text{O}$
- D. $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$
- E. $\text{Na}_2\text{SO}_4 \cdot \text{H}_2\text{O}$

61- Linear combination of atomic orbitals (LCAO) results in the formation of:

- A. Sigma bond
- B. Pi bond
- C. Bonding molecular orbitals only
- D. Bonding and anti-binding molecular orbitals.
- E. All of the above

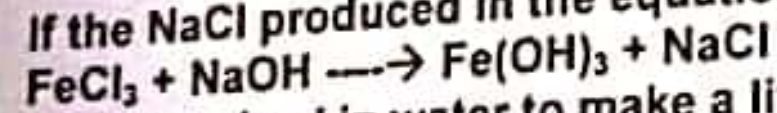
62- Given the equation:



At STP, how many liters of $\text{O}_{2(g)}$ are needed to completely burn 5.0 liters of $\text{C}_3\text{H}_{8(g)}$?

- A. 5.0
- B. 10
- C. 10.5
- D. 15
- E. 25

63- If the NaCl produced in the equation:



was dissolved in water to make a litre of solution, the molarity would be:

- A. 0.1M
- B. 3M
- C. 8M
- D. 0.5M
- E. 1.5M

64- When 18×10^{-3} moles/dm³ of acetic acid react with 22×10^{-3} moles/dm³ of ethyl alcohol to form 40×10^{-3} moles/dm³ of ethyl acetate and 40×10^{-3} moles/dm³ of water. Find the value of equilibrium constant (K_c)

- A. 4.04
- B. 3.14
- C. 3.04
- D. 2.02
- E. 1.04

65- Harmful and undesirable reaction of metal when exposed to atmosphere or chemical agent is known as:

- A. Allotropy
- B. Electroplating
- C. Collision
- D. Cracking
- E. Corrosion

66- What is the product of both fermentation reactions and fractional distillation?

- A. an ester
- B. an acid
- C. an alcohol
- D. a soap
- E. as base

67- During condensation of polymerization, two monomers may be joined by the removal of a molecule of:

- A. carbon dioxide
- B. hydrogen
- C. oxygen
- D. water
- E. none of the above

68- Which gas is likely to deviate most from ideal gas behavior?

- A. HCl
- B. He
- C. CH_4
- D. N_2
- E. O_2

69- Purification of bauxite whose major impurity is Silica (SiO_2) is carried out through:

- A. Baeyer's method
- B. Hall's method
- C. Sjerpek's method
- D. Contact method
- E. Electrolytic method

70- The range of pH below _____ and above _____ of soil represents its sterility.

- A. 5...10
- B. 10...5
- C. 3...10
- D. 10...3
- E. 5...3

BIOLOGY

71- The muscles attached to the bones are:

- A. voluntary and smooth
- B. Involuntary and smooth
- C. Voluntary and striated
- D. Involuntary and striated
- E. Smooth and striated

The law of Dominance is illustrate in the garden pea by:

- heterozygous tall x heterozygous tall
- heterozygous tall x heterozygous tall
- heterozygous tall x Homozygous tall
- pure short x pure short
- heterozygous tall x pure short

In the Hardy-Weinberg principle, $p^2 + 2pq + q^2 = 1$, q^2 represents the frequency of the

- A Homozygous dominant
- B heterozygous dominant
- C heterozygous recessive
- D Homozygous recessive
- E Blended genes

Fungi do not contain:

- A Cell wall
- B Hyphae
- C Chlorophyll
- D Mycelium
- E Spores

75- Identify the incorrect statement about Charles Darwin's Theory:

- A The individuals of species have variations among them
- B There is always a tendency of over reproduction in a species
- C Vast gradual changes result in the origin of a new species
- D Favourable variations survive and unfavorable will be exterminated
- E Intra specific competition occurs between different species and inter-specific competition occurs among the individuals in a species.

76- Identify the incorrect statement from the following:

- A Apical growth increases the length of stems and roots
- B Xylem is situated on the outer side of the cambium ring and the phloem on the inner side
- C Secondary growth increases the diameter of stems and root.
- D The cells in elongation phase don't divide.
- E The cells in formative regions are closely packed together.

77- According to _____, protein layers are not confined to the surface of the membrane but embedded in lipid layers.

- A Lock and key mode
- B Induce fit model
- C Fluid mosaic model
- D Lotka Volterra model
- E All of the above

Down Grads

355

11 Years Past Papers

78- In a typical nucleotide the nitrogenous base is attached to _____ carbon of pentose sugar.

- A 6th
- B 5th
- C 4th
- D 3rd
- E 1st

79- Binomial nomenclature was first time proposed by _____

- A Charles Darwin (1859)
- B Rodolph Virchow (1855)
- C Louis Pasteur (1862)
- D Carolus Linnaeus (1707)
- E Robert Brown (1773)

80- _____ causes amoebic dysentery in humans:

- A *Pelomyxa palustris*
- B *Entamoeba histolytica*
- C *Trichonympha*
- D *Trypanosoma*
- E *Radiolaria* ooze

81- The main process that occurs in the dark reaction in photosynthesis is:

- A That water is split
- B Light energy is converted into chemical energy
- C That glucose is oxidized
- D That carbon dioxide is fixed
- E None of the above

82- _____ is commonly known as Hook worm:

- A *Ancylostoma duodenale*
- B *Ascaris lumbricoides*
- C *Enterobius vermicularis*
- D *Hirudinaria*
- E *Wuchereria*

83- An organism appears to be a segmented worm. Upon observation it is determined that the organism has a closed circulation, a mouth and an anus, and does not have an exoskeleton. The organism most likely belongs to the phylum:

- A Mollusa
- B Annelida
- C Echinodermata
- D Arthropoda
- E Chordate

84- Ileum is about _____ long.

- A 3.6 millimeters
- B 3.6 centimeters
- C 3.6 inches
- D 3.6 meters
- E 3.6 kilometers

85- The Latin words of the name given to a human being, *Homo Sapiens*, include the:

- A Genus and family
- B Family and order
- C Order and class
- D Genus and class
- E Genus and species

86- In paper chromatography xanthophylls will give ____ color:

- A Orange
- B Grey
- C Yellow
- D Blue-green
- E Yellow-green

87- The propulsive movement of the gastro-intestinal tract (GI tract) is:

- A Peristalsis
- B Epiptosis
- C Antiperistalsis
- D Anus
- E None of the above

88- The Calvin Cycle consist of ____ main reactions:

- A 3
- B 6
- C 9
- D 13
- E 16

89- Identify the incorrect statement about the Bathyal zone:

- A It ranges from surface to depth of about 2000 meters
- B It consist of pelagic and benthic zone
- C It is aphotic
- D It contains producers that prepare food for consumers
- E None of the above

90- Deamination in the liver initially produces:

- A ammonia
- B arginine
- C ornithine
- D urea
- E uric acid

91- The causes of cyanosis include:

- A deficiency of vitamin C
- B varicella zoster virus
- C degeneration of the cartilage of joints
- D ventricular septum defect
- E None of the above

92- The Prolactin hormone responsible for activation of mammary glands to start producing milk is a hormone of:

- A Pituitary gland
- B Pancreas
- C Thyroid gland
- D Thymus gland
- E Adrenal gland

93- Which of the following bones are present in the palm of hand?

- A Carpals
- B Metcarpals
- C Phalanges
- D Tarsal
- E Radius

Questions 94-96

Hemophilia is a disorder in which blood fails to clot. Saad, a male hemophiliac, marries Sara, a normal woman and together they have four children, two boys (Ahmed and Ali) and two girls (Alia and Ayesha). None of the children display the symptoms of hemophilia. Ahmed, Ali, Alia, and Ayesha all marry normal individuals and have children. None of Ahmed's or Ali's children, male or female, display symptoms of hemophilia, but the son of Alia and Ayesha display symptoms of hemophilia while the daughters of Alia and Ayesha do not.

94- Which of the following best explains the reasons that Ahmed, Ali, Alia, and Ayesha do not display symptoms of hemophilia, even though their father, Saad, is a hemophiliac?

- A Hemophilia is an X-linked disorder and Saad can only pass on his Y chromosome
- B Hemophilia is an X-linked disorder and even though Alia and Ayesha received a hemophiliac X chromosome from Saad, Sara gave them a normal X chromosome
- C Hemophilia is a Y-linked disorder and therefore cannot be displayed in females
- D Hemophilia is a Y-linked disorder and Ahmed and Ali must have received an X chromosome from Saad
- E Hemophilia is an X-linked disorder and even though Ahmed and Ali received a hemophiliac X chromosome from Saad, gave them a normal X chromosome

95- If one of Ali's daughters marries a normal man, what is the probability that one of their children will display symptoms of hemophilia?

- A 0%
- B 25%
- C 50%
- D 5%
- E 100%

96- Which of the following individuals are heterozygous for hemophilia?

- A Saad, Ahmed and Ali
- B Ahmed, Ali, Alia, and Ayesha
- C Saad and Sara
- D Alia and Ayesha
- E Ahmed and Ali

- 97- Which of the following would be most likely to occur in an ecosystem?
- A As the number of prey decreases, the number of predators increases
 B As the number of predators increases, the number of prey increases
 C As the number of prey decreases, the number of predators decreases
 D As the number of prey increases, the number of predators decreases
 E As the number of predators decreases, the number of prey decreases

- 98- Which biome contains maples, Oaks, and bears?
- A Tundra
 B Tropical rain forest
 C Temperate grasslands
 D Taiga
 E Deciduous forest

- 99- The major sign and symptom of Microcephaly is;
- A Sexual defects
 B Excessive number of toes
 C Mental retardation
 D Small skull in proportion to the normal body size
 E Split in upper lip and gap in the roof mouth

- 100- The reaction involved in chemotropic nutrition is:
- A $6\text{CO}_2 + 12\text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{CO}_2 + 6\text{H}_2\text{O}$
 B $2\text{H}_2\text{S} + \text{O}_2 \rightarrow (\text{CH}_2\text{O}) + \text{H}_2\text{O} + 2\text{S}$
 C $\text{NH}_4 + 3\text{O}_2 \rightarrow 2\text{NO}_2 + 2\text{H}_2\text{O} + 4\text{H}^+ + \text{energy}$
 D $\text{CH}_3\text{COCOCH} \rightarrow \text{CH}_3\text{CHO} + \text{CO}_2$
 E $5\text{GAP} + 3\text{ATP} \rightarrow 3\text{RuBP} + 3\text{ADP} + 2\text{P}_i$

NTS PAST PAPER 2009

ANSWERS KEY

Q.NO.	ANSWER	Q.NO.	ANSWER	Q.NO.	ANSWER
01.	A	34	B	67	D
02.	C	35	C	68	A
03.	C	36	A	69	C
04.	E	37	E	70	C
05.	B	38	B	71	C
06.	E	39	B	72	C
07.	C	40	E	73	E
08.	D	41	C	74	D
09.	D	42	C	75	C
10.	B	43	C	76	E
11.	A	44	C	77	B
12.	D	45	C	78	C
13.	D	46	A	79	E
14.	B	47	E	80	D
15.	B	48	A	81	B
16.	B	49	D	82	D
17.	B	50	C	83	A
18.	B	51	B	84	B
19.	D	52	C	85	D
20.	B	53	B	86	E
21.	D	54	A	87	C
22.	D	55	E	88	A
23.	E	56	B	89	D
24.	E	57	B	90	D
25.	D	58	E	91	A
26.	C	59	B	92	D
27.	B	60	C	93	A
28.	E	61	D	94	B
29.	C	62	E	95	B
30.	B	63	E	96	A
31.	B	64	B	97	D
32.	E	65	A	98	C
33.	D	66	E	99	C
			C	100	D
					C

12 YRS PAST PAPERS

Durr
N.T.S

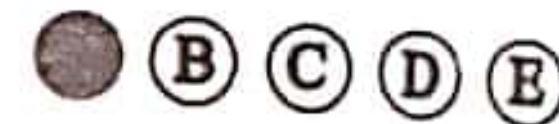
PAST PAPER – 2010

HOW TO FILL ANSWER SHEET
(IN CASE OF CONFUSION ASK NTS STAFF FOR HELP)

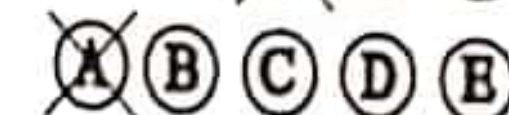
- Please Fill the appropriate circle in the column of Color of Question Book
- Please Sign in the Column of Candidate Signature
- Please Write your Question Book No. in the Column of Question Book No.
- Fill the appropriate Circle completely LIKE THIS •

Color of your Question Book (Fill only one circle)		Name MOHSIN	NTS Signature
Yellow	☐	Father's Name Ali Hassan	Candidate Signature
Green	☐	Roll No. 11424	Question Book No. 30792
White	☒	Do not write or mark anything in this Box	PASTE PHOTOGRAPH
Blue	☐		
Pink	☐		

CORRECT FILLING



INCORRECT FILLING



NOTE

INCORRECT AND IMPROPER FILLED CIRCLES WILL BE MARKED AS "WRONG ANSWER" BY THE MACHINE

QUESTION BOOK NO.
PINK BOOK

NAME

Roll No.

Signature

INSTRUCTIONS

CANDIDATE IS ADVISED TO CHECK HIS / HER QUESTION BOOK (PAGE NO. & QUESTION NO.) BEFORE THE START OF TEST. REPORT ANY MISPRINTING PAGE MISSING QUESTION MISSING TO NTS INVIGILATOR SO THAT YOUR BOOK CAN BE CHANGED. WILL NOT BE RESPONSIBLE FOR ANY SUCH COMPLAINT REPORTED AFTER THE START OF TEST.

PLEASE DO REMEMBER YOUR ROLL NO FOR FUTURE CORRESPONDENCE

No Correspondence / query will be entertained without Roll No.

There is **NEGATIVE** marking @ 1/4 (0.25) for each wrong answer.

Duration of test is 120 MINUTES (100 Questions).

Fill only one circle for each question. All questions carry equal marks.

Question Books are in five colours and Shuffles.

Do not write or mark any thing on Question Book except Name, Roll No and Signature on Title page of Question Book. Do not tear off or remove any paper from Question book.

No mobile phones, calculators and wrist watches with calculator are allowed in the Examination Hall.

Put all your belongings under your seats.

Two plain papers are given inside the Question Book for rough work. No extra sheet will be provided.

You are not allowed to leave your seat during test.

You will not leave your seat after test until advised by the Chief Supervisor.

See back of your Question Book for instructions regarding solving & filling the Answers Sheet.

In the Case of any confusion get help from invigilation staff.

Before the submission of the NTS Copy of Answer Sheet to the invigilator at the end of the test, please tear off Candidate copy and keep it with you for your record.

Answer Key will be displayed on NTS website (www.nts.org.pk) on November 01, 2009 at 7:00pm.

ENGLISH

Choose the most similar in meaning to the capitalized one.

1. DISRUPTION:

- A. Comfort
- B. Luxury
- C. Trouble
- D. Freedom
- E. Calm

2. INEVITABLE:

- A. Doubtful
- B. Deny
- C. Unexpected
- D. Certain
- E. Unusual

Choose the lettered word or phrase that is most nearly opposite in meaning to the word in capital letters.

3. ERADICATION:

- A. Suppression
- B. Termination
- C. Control
- D. Establish
- E. Extinction

4. INTERRUPTION:

- A. Break
- B. Continuity
- C. Injury
- D. Difference
- E. Crack

Identify the word or phrase that needs to be changed for the sentence to be correct

5. The stories that she makes out for her children ought to be written down and published.

No error

- A
- B
- C
- D
- E

6. We are expecting to go abroad this summer. No error

- A
- B
- C
- D
- E

Read the passage to answer questions 7-8

It is of course a trifle absurd to speak of Asia as a unity, and only opposition to Western Imperialism has caused people to think in these terms. Asia contains half the population of the world and at least three very distinct civilizations: that of Islam, that of India, and that of China. These differ from each other just as much as they differ from the civilization of Christendom, and there is not the faintest reason to expect them all to act in unison. What is to be hoped is, an endeavor after cultural or political unity, but a determination to uphold independence at home and to respect it elsewhere- and when I speak of independence, I am not thinking only of politics, but also of culture. There is a great danger of too much cultural uniformity. No great civilization has ever been cosmopolitan.

7. In the view of the author, what is the harm of cultural uniformity?

- A. It will not lead to the formation of a great civilization
- B. It will lead to the formation of a great civilization
- C. It will lead to the formation of rich culture
- D. It will lead to the change of environment
- E. It will lead to the destruction of values

8. The author is in favor of:

- A. Cultural uniformity
- B. Cosmopolitan
- C. Political and cultural independence
- D. Unity of Asia
- E. None of the above

Complete the sentences by choosing the most appropriate word, from the given lettered choices (A to E) below each.

9. Nearly everyone dreams of building _____ ideal house:

- A. its
- B. their
- C. his
- D. him
- E. them

10. Life _____ water, light, a moderate temperature and a variety of chemical elements:

- A. finds
- B. reveals
- C. designs
- D. calculates
- E. requires

PHYSICS

11. A particle moves from position $\vec{r}_1 = 3\hat{i} + 2\hat{j} - 6\hat{k}$ to position $\vec{r}_2 = 14\hat{i} + 13\hat{j} + 9\hat{k}$ under the action of a force $\vec{F} = 8\hat{i} + 2\hat{j} + 5\hat{k}$. Find the work done by the force:

- A. 50 units
- B. 75 units
- C. 125 units
- D. 155 units
- E. 200 units

12. A body starts sliding on a rough horizontal surface with a speed of 10 m/s. If the coefficient of friction is 0.2, find the distance traveled by the body before coming to rest. ($g = 10 \text{ m/s}^2$)

- A. 15 m
- B. 25 m
- C. 35 m
- D. 40 m
- E. 55 m

13. A battery whose e.m.f. is 40 V has an internal resistance of 5 Ω . If this battery is connected to a 15 Ω resistor 'R' What will be the voltage drop across 'R'?

- A. 10 V
- B. 30 V
- C. 40 V
- D. 50 V
- E. 70 V

14. A particle of charge 0.04 C is projected with speed $2 \times 10^6 \text{ m/s}$ into a uniform magnetic field 'B', of strength 0.5 T. If the particle's velocity as it enters the field is perpendicular to 'B', What is the magnitude of the magnetic force on this particle?

- A. 4 N
- B. 8 N
- C. 40 N
- D. 80 N
- E. 800 N

15. Due to the magnetic force, a positively charged particle executes uniform circular motion within a uniform magnetic field 'B'. If the charge is 'q' and the radius of its path is 'r', which of the following expressions gives the magnitude of the particle's linear momentum:

- A. qBr
- B. $\frac{qB}{r}$
- C. $\frac{q}{(Br)}$
- D. $\frac{B}{qr}$
- E. $\frac{r}{qB}$

16. A transverse wave on a long horizontal rope with a wavelength of 8 m travels at 2 m/s. At $t=0$, a particular point on the rope has a vertical displacement of $+A$, where A is the amplitude of the wave. At what time will the vertical displacement of this same point on the rope be $-A$:

A. $t=1/8$ s
B. $t=1/4$ s
C. $t=1/2$ s
D. $t=2$ s
E. $t=4$ s

17. The dimensions of volume and acceleration (respectively) are:

A. LT^3
B. LT^2 and LT^{-1}
C. L^3 and LT^{-2}
D. L^4 , T^{-1} and LT
E. LT^3 and T^2

18. A vector such as the velocity of a body undergoing uniform translational motion, which can be displaced parallel to itself and applied to any point is known as:

A. Unit vector
B. Free vector
C. Null vector
D. Position vector
E. Resultant vector

19. What is the flux density at a point 3 cm from the long straight wire, when there is a current of 25 A in a wire: ($\mu_0 = 4\pi \times 10^{-7}$):

A. 0.23×10^{-1} T
B. 1.67×10^{-4} T
C. 2.99×10^{-5} T
D. 3.63×10^{-4} T
E. 9.99×10^{-7} T

20. If an object is placed 30 cm from a convex lens whose focal length is 15 cm, the size of the image compared to the size of the object will be approximately:

A. Twice as large
B. More than twice as large
C. 1.5 times as large
D. Smaller
E. The same size

21. When a conductor of cross-sectional area $5 \times 10^{-8} \text{ m}^2$ carries a current of 6 A, the drift velocity of the conduction electrons is $1.2 \times 10^{-1} \text{ ms}^{-1}$. What is the number density (number per unit volume) of the conduction electrons:

A. $4 \times 10^{23} \text{ m}^{-3}$
B. $1.6 \times 10^{27} \text{ m}^{-3}$
C. $2.5 \times 10^{27} \text{ m}^{-3}$
D. $6.3 \times 10^{28} \text{ m}^{-3}$
E. $1.3 \times 10^{24} \text{ m}^{-3}$

22. A thermocouple is immersed in water at 373 K and the other in ice at 273 K. The e.m.f. of the thermocouple is $90 \mu\text{V}$ for each 1 K difference in temperature between junctions, and the thermocouple resistance is 6Ω . What current will flow in the galvanometer:

A. $1.8 \mu\text{A}$
B. $250 \mu\text{A}$
C. $300 \mu\text{A}$
D. 1.5 mA
E. 1.8 mA

23. The first law of thermodynamics may be written as $\Delta U = Q + W$, where ΔU is the increase in internal energy of the system, Q is the heat transferred to the system and W is the external work done on the system. Which of the following is correct for the case of an isothermal expansion of an ideal gas:

A. $W > 0$
B. $W = 0$
C. $\Delta U = 0$
D. $\Delta U > 0$
E. $Q = 0$

24. A body of mass 5 kg, initially at rest, is moved by a horizontal force of 2 N on a smooth horizontal surface. Find the work done by the force in 10 sec:

A. 40 J
B. 30 J
C. 50 J
D. 20 J
E. 10 J

25. An object is placed 60 cm in front of a concave spherical mirror whose focal length is 40 cm. Which of the following best describes the image:

	Nature of Image	Distance from Mirror
A.	Virtual	24 cm
B.	Real	24 cm
C.	Virtual	120 cm
D.	Real	120 cm
E.	Real	240 cm

26. An object is placed 60 cm from a spherical convex mirror. If the mirror forms a virtual image of 20 cm from the mirror, what is the magnitude of the mirror's radius of curvature:

A. 7.5 cm
B. 15 cm
C. 30 cm
D. 60 cm
E. 120 cm

27. Find the unit vector parallel to the vector: $\vec{B} = 6\hat{i} + 12\hat{j} - 4\hat{k}$

A. $\hat{b} = 4/14\hat{i} + 12/14\hat{j} - 4/14\hat{k}$
 B. $\hat{b} = 6/14\hat{i} + 12/14\hat{j} - 4/14\hat{k}$
 C. $\hat{b} = 6/14\hat{i} + 17/14\hat{j} - 4/14\hat{k}$
 D. $\hat{b} = 6/14\hat{i} + 17/14\hat{j} - 4/14\hat{k}$
 E. $\hat{b} = 9/14\hat{i} + 12/14\hat{j} - 1/14\hat{k}$

28. Two capacitors $C_1 = 2\mu\text{F}$ and $C_2 = 4\mu\text{F}$ are connected in series across a 100 V supply. Find the effective capacitance:

A. $1/2\mu\text{F}$
 B. $3/2\mu\text{F}$
 C. $5/2\mu\text{F}$
 D. $4/3\mu\text{F}$
 E. $2\mu\text{F}$

29. A rescue helicopter drops a package of emergency ration to a stranded party on ground. If the helicopter is traveling horizontally at 40 m/s at a height of 100 m above the ground where does the package strike the ground relative to the point at which it was released ($g = 9.8\text{ m/s}^2$):

A. 120 m
 B. 130 m
 C. 140 m
 D. 160.7 m
 E. 200.3 m

30. The radius of the moon is 27% of the earth's radius and its mass is 1.2% of the earth's mass. Find the acceleration due to gravity on the surface of the moon:

A. 0.431 m/s^2
 B. 1.615 m/s^2
 C. 2.431 m/s^2
 D. 3.615 m/s^2
 E. 4.431 m/s^2

Questions 31-32:

A battery has an e.m.f of 6.0 volts and an internal resistance of 0.4 ohm. It is connected to a 1 ohm resistor through a SPST (single pole, single throw) switch)

31. When the switch is open, the potential difference between the terminal of the battery is in volts:

A. 0
 B. 0.8
 C. 2.6
 D. 5.2
 E. 6.0

32. When the switch is closed, the potential difference between the terminals of the battery is, in volts:

A. 0
 B. 0.8
 C. 2.6
 D. 5.2
 E. 6.0

33. Assume that you have two balls of identical volume, one weighting 2 Newton's and other 10 Newton's. Both are falling freely after being released from the same point simultaneously. Which of the following will then be true:

I. The 10- N ball falling freely from rest will be accelerated at a greater rate than the 2 N ball.
 II. At the end of 4s of free fall, the 10 N ball will have 5 times the momentum of the 2 N ball.
 III. At the end of 4s of free fall, the 10 N ball will have the same kinetic energy as the 2 N ball.
 IV. The 10 N ball possesses greater inertia than the 2 N ball.

A. I, II and III only
 B. I and III only
 C. II and IV only
 D. IV only
 E. None of these

34. A car waiting at a traffic signal and when the signal turns green, the car starts ahead with a constant acceleration of 2 m/s^2 . At the same time a bus travelling with a constant speed of 10 m/s overtakes and passes the car. How far beyond its starting point will the car overtake the bus:

A. 40 m
 B. 30 m
 C. 90 m
 D. 120 m
 E. 100 m

35. A sample of an ideal gas occupies a volume 'V' at pressure 'P' and absolute temperature 'T'. the mass of each molecule is 'm'. If 'k' is the Boltzmann constant then the density of the gas is:

A. mkT
 B. $\frac{P}{KT}$
 C. $\frac{P}{kTV}$
 D. $\frac{mP}{kT}$
 E. $\frac{2mPT}{k}$

36. A ball moving horizontally with speed 'v' strikes the bob of a simple pendulum at rest. The mass of the bob is equal to that of the ball. If the collision is elastic the bob will rise to a height:

A. $\frac{v^2}{g}$
 B. $\frac{v^2}{2g}$
 C. $\frac{v^2}{4g}$
 D. $\frac{v^2}{8g}$
 E. $\frac{v^2}{7g}$

37. A point source of light is placed at the principal focus of a concave lens. Which of the following will be true of the refracted light:

1. It will diverge
 2. It will be parallel to the principal axis
 3. It will seem to come from a point $\frac{1}{2}$ of the radius of curvature from the lens
 4. It will converge

A. 1, 2, and 3 only
 B. 1 and 3 only
 C. 2 and 4 only
 D. 4 only
 E. none of the above

38. The quantity of heat required to raise the temperature of one mole of a substance through 1 K, and its units are $\text{J-mole}^{-1}\text{K}^{-1}$, is called:

A. Carnot engine
 B. Molar specific heat
 C. Kinetic specific heat
 D. General gas law
 E. Boyle's law

39. A shot leaves a gun at the rate of 160m/s. Calculate the greatest distance to which it could be projected and the height to which it would rise: ($g=10\text{m/s}^2$):

A. 1560m, 540m
 B. 2560m, 640m
 C. 3560m, 740m
 D. 4560m, 840m
 E. 9595m, 348m

40. A car covers the first half of a certain distance with a speed V_1 and the second half with a speed V_2 . Find the average speed during the whole journey: :

A. $\frac{V_1 V_2}{2(V_1 + V_2)}$
 B. $\frac{2V_1 V_2}{V_1 + V_2}$
 C. $\frac{2V_1 3V_2}{V_1 + 5V_2}$
 D. $\frac{V_1 + V_2}{5V_1 V_2}$
 E. $\frac{V_1 - V_2}{4V_1 + 9V_2}$

41. What is the product of both fermentation reactions and fractional distillation:

- A. an ester
- B. an acid
- C. an alcohol
- D. a soap
- E. a base

42. During condensation polymerization, two monomers may be joined by the removal of a molecule of:

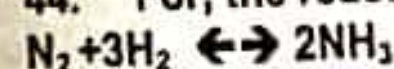
- A. carbon dioxide
- B. hydrogen
- C. oxygen
- D. water
- E. none of the above

43. Given the reaction:
 $C_3H_8(g) + 5O_2(g) \rightarrow 3CO_2 + 4H_2O(g)$

At STP, how many liters of $O_2(g)$ are needed to completely burn 5.0 liters of $C_3H_8(g)$?

- A. 5
- B. 10
- C. 10.5
- D. 15
- E. 25

44. For the reaction:



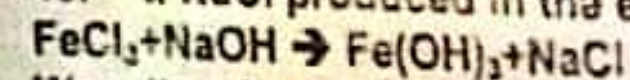
The production of NH_3 will be favored at:

- A. High pressure and catalyst
- B. Low pressure only
- C. Low pressure and catalyst
- D. High pressure only
- E. Catalyst only

45. The range of pH below _____ and above _____ of soil represent its salinity.

- A. 5....10
- B. 10....15
- C. 3....10
- D. 10....3
- E. 5....3

46. If NaCl produced in the equation:



Was dissolved in water to make a liter of solution, the molarity would be

- A. 0.1M
- B. 3M
- C. 8M
- D. 0.5M
- E. 1.5M

47. When 18×10^{-3} moles/dm³ of acetic acid react with 22×10^{-3} moles/dm³ of ethyl alcohol to form 40×10^{-3} moles/dm³ of ethyl acetate and 40×10^{-3} moles/dm³ of water. Find the value of equilibrium constant (K_c):

- A. 4.04
- B. 3.14
- C. 3.04
- D. 2.02
- E. 1.04

48. Harmful and undesirable reaction of metal when exposed to atmosphere or any chemical agent is known as:

- A. allotropy
- B. electroplating
- C. collision
- D. cracking
- E. corrosion

49. Catenation is a process in which carbon shows the properties of:

- A. making single bond
- B. hybridization
- C. making long chains or rings of carbon atoms
- D. isomerism
- E. breaking of bonds

50. Which gas is likely to deviate most from ideal gas behavior:

- A. HCl
- B. He
- C. CH₄
- D. N₂
- E. O₂

51. The maximum number of electrons that an orbital can accommodate is/are:

- A. 0
- B. 1
- C. 2
- D. 3
- E. 4

52. Which of the following is not a nucleophile:

- A. HO⁻
- B. NH₃
- C. BF₃
- D. CN⁻
- E. NH₂

53. Purification of bauxite whose major impurity is silica (SiO₂) is carried out through:

- A. Baeyer's method
- B. Hall's method
- C. Serpek's method
- D. Contact method
- E. Electrolytic method

54. Chlorine is manufactured commercially by the electrolysis of aqueous sodium chloride (brine). Which other products are made in this process:

- A. hydrochloric acid and hydrogen
- B. hydrogen and sodium
- C. hydrogen and sodium hydroxide
- D. sodium and sodium hydroxide
- E. hydrochloric acid and sodium

55. Sodium reacts with water more vigorously than lithium because it:

- A. has higher atomic weight
- B. is more electronegative
- C. is more electropositive
- D. is a metal
- E. has high melting point

56. The crystals formed as a result of vander waal's interactions are:

- A. molecular crystals
- B. covalent crystals
- C. metallic crystals
- D. ionic crystals
- E. none of the above

57. According to law of mass action, "The rate of chemical reaction is proportional to:

- A. products
- B. product of molar concentration of reactants
- C. initial concentration of reactants
- D. catalyst
- E. pressure

58. The sum of exponents of the molar concentration of the reactants, is equal to:

- A. molecularity
- B. polarity
- C. activation energy
- D. rate of reaction
- E. order of reaction

59. Nascent hydrogen used in the formation of methane, is obtained from the reaction of:

- A. NaHCO_3 with Zn
- B. HCl with Zn
- C. KOH with Zn
- D. H_2O with Zn
- E. CH_3I with Zn

60. "Zymase" a group of 14 enzymes, used in the fermentation of starch, is present in:

- A. bacteria
- B. yeast
- C. fungi
- D. algae
- E. virus

61. The general formula for aldehyde is:

- A. R-OH
- B. R-COOH
- C. R-CO-R
- D. R-X
- E. R-CHO

62. When an element exists in more than one crystalline form, the phenomenon is termed as:

- A. isomorphism
- B. allotropy
- C. isomerism
- D. anisotropy
- E. enthalpy

63. When an element exists in more than one crystalline form, the phenomenon is termed as:

- A. $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$
- B. $2\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$
- C. $(\text{CaSO}_4)_2 \cdot \text{H}_2\text{O}$
- D. $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$
- E. $\text{NaSO}_4 \cdot \text{H}_2\text{O}$

64. Linear combination of atomic (LCAO) results in the formation of:

- A. Sigma bond
- B. Pi Bond
- C. bonding molecular orbital's only
- D. bonding and anti-bonding molecular orbital's
- E. all of the above

65. Which of the following statements about H_2S is false::

- A. it is a covalent compound
- B. it is a gas with bad smell
- C. it is a strong reducing agent than H_2O
- D. it's molecule is non-linear
- E. it is a weak base in water

66. The rain drop acquires spherical shape and ink spreads over blotting paper due to:

- A. surface tension
- B. adhesive forces
- C. viscosity
- D. polarity
- E. latent heat of vaporization

67. 950 torr corresponds to:

- A. 3.5 atm
- B. 1 atm
- C. 3 atm
- D. 1.25 atm
- E. 2.25 atm

68. The enthalpy change accompanying the gain of an electron by a neutral gaseous atom to form negative ion is called:

- A. ionization potential
- B. electro-negativity
- C. electron affinity
- D. lattice energy
- E. potential energy

69. Sigma bond is formed by:

- A. transferring the electrons
- B. head on overlapping of atomic orbitals
- C. mutual but unequal sharing of electrons
- D. parallel overlapping of atomic orbitals
- E. all of the above

70. The heat of a reaction can be calculated by using:

- A. Joule's law
- B. Ohm's law
- C. Hess's law
- D. Faraday's law
- E. Boyle's law

BIOLOGY

QUESTION 71-73

Hemophilia is a disorder in which blood fails to clot. Saad, a male hemophiliac, marries Sara, a normal woman and together they have four children, two boys (Ahmed and Ali) and two girls (Alia and Ayesha). None of the children display the symptoms of hemophilia. Ahmed, Ali, Ayesha and Alia all marry normal individuals and have children. None of Ahmed's or Ali's children, male or female, display symptoms of hemophilia, but the sons of Alia and Ayesha display symptoms of hemophilia while the daughters of Alia and Ayesha do not.

71. Which of the following best explains the reason that Ahmed, Ali, Alia and Ayesha do not display symptoms of hemophilia, even though their father, Saad, is a hemophiliac.

- A. Hemophilia is an x-linked disorder and Saad can only pass on his Y chromosome.
- B. Hemophilia is an x-linked disorder and even though Alia and Ayesha received a hemophiliac X-chromosome from Saad, Sara gave them a normal X chromosome.
- C. Hemophilia is a Y-linked disorder and therefore cannot be displayed in females.
- D. Hemophilia is a Y-linked disorder and Ahmed and Ali must have received an X-chromosome from Saad.
- E. Hemophilia is a Y-linked disorder and even though Ahmed and Ali received a hemophiliac X-chromosome from Saad, Sara gave them a normal X-chromosome.

72. If one of Ali's daughter marries a normal man, what is the probability that one of their children will display symptoms of hemophilia.

- A. 0%
- B. 25%
- C. 50%
- D. 75%
- E. 100%

73. Which of the following individuals are heterozygous for hemophilia:

- A. Saad, Ahmed and Ali
- B. Ahmed, Ali, Alia, and Ayesha
- C. Saad and Sara
- D. Alia and Ayesha
- E. Ahmed and Ali

74. The propulsive movement of the gastro-intestinal tract (GI tract) is:

- A. Peristalsis
- B. Epiglottis
- C. antiperistalsis
- D. Anus
- E. None of the above

75. The Calvin Cycle consists of _____ main reactions:

- A. 3
- B. 6
- C. 9
- D. 13
- E. 16

76. Identify the incorrect statement about the Bathal zone:
- It ranges from surface to depth of about 2000 metres
 - It consists of pelagic and benthic zones
 - It is aphotic
 - It contains producers that prepare food for consumers
 - None of the above

77. Excretion in the liver initially produces:
- Ammonia
 - Arginine
 - Ornithine
 - Urea
 - Uric acid

78. The causes of Cyanosis include:
- Deficiency of vitamin C
 - Varicella-zoster virus
 - Degeneration of the cartilage of joints
 - Ventricular septum defect
 - None of the above

79. The prolactin hormone responsible for the activation of mammary glands to start producing milk is a hormone of:
- Pituitary gland
 - Pancreas
 - Thyroid gland
 - Thymus gland
 - Adrenal gland

80. Which of the following would be most likely to occur in an ecosystem:
- As the number of prey decreases, the number of predators increases
 - As the number of predators increases, the number of prey increases
 - As the number of prey decreases, the number of predators decrease
 - As the number of prey increases, the number of predators decreases
 - As the number of predators decreases, the number of prey decreases

81. The law of dominance is illustrated in the garden pea by:
- Homozygous tall x heterozygous tall
 - Heterozygous tall x heterozygous tall
 - Homozygous tall x homozygous tall
 - Pure short x pure short
 - Homozygous tall x pure short

82. Identify the incorrect statement about Charles Darwin's theory:
- The individual of species have variations among them
 - There is always a tendency of over reproduction in a species
 - Vast gradual changes result in the origin of a new species
 - Favorable variations survive and unfavorable will be exterminated
 - Intra specific competition occurs between different species and inter-specific competition occurs among the individuals in a species

83. Identify the incorrect statement from the following:
- Apical growth increases the length of stems and roots
 - Xylem is situated on the outer side of the cambium ring and the phloem on the inner side
 - Secondary growth increases the diameter of stem
 - The cells in elongation phase don't divide
 - The cells in formative regions are closely packed together

84. Proteins float in membrane like ice burg in sea:
- Lock and key model
 - Induce fit model
 - Fluid mosaic model
 - Lokta voltera model
 - All of the above

85. In the Hardy-Weinberg principle, $p^2 + 2pq + q^2 = 1$, q^2 represents the frequency of the:
- Homozygous dominant
 - Heterozygous dominant
 - Heterozygous recessive
 - Homozygous recessive
 - Blended genes

86. Fungi do not contain:
- Cell wall
 - Hyphae
 - Chlorophyll
 - Mycelium
 - Spores

87. In a typical nucleotide the nitrogenous base is attached to _____ carbon of pentos sugar:
- 6th
 - 5th
 - 4th
 - 3rd
 - 1st

88. Binomial nomenclature was first time proposed by:
- Charles Darwin (1859)
 - Rodolph virchow (1855)
 - Louis Pasteur (1862)
 - Carlous Linnaeus (1707)
 - Robert brown (1773)

89. _____ causes amoebic dysentery in humans:
- Pelomyxa palustris
 - Entamoeba histolytica
 - Trichonympha
 - Trypanosoma
 - Radiolarian ooze

90. The main process that occurs in the dark reaction in photosynthesis is:

- A. That water is split
- B. Light energy is converted into chemical energy
- C. That glucose is oxidized
- D. That carbon-dioxide is fixed
- E. None of the above

91. _____ is commonly known as hook worm:

- A. Ancylostoma duodenale
- B. Ascaris lumbricoides
- C. Enterobius vermicularis
- D. Hirudinaria
- E. Wuchereria

92. Ileum is about _____ long:

- A. 3.6 centimeters
- B. 3.6 millimeters
- C. 3.6 inches
- D. 3.6 meters
- E. 3.6 kilometers

93. The Latin words of the name given to a human being, Homo Sapiens, include the:

- A. Genus and family
- B. Family and order
- C. Order and class
- D. Genus and class
- E. Genus and species

94. In paper chromatography xanthophylls will give _____ color:

- A. Orange
- B. Grey
- C. Yellow
- D. Blue-green
- E. Yellow-green

95. Which of the following bones are present in the palm of hand:

- A. Carpals
- B. Metacarpals
- C. Phalanges
- D. Tarsal
- E. Radius

96. Which biome contains maples, Oaks, and bears:

- A. Tundra
- B. Tropical rain forest
- C. Temperate grasslands
- D. Taiga
- E. Deciduous forest

97. The major sign and symptoms of microcephaly is:

- A. Sexual defects
- B. Excessive number of toes
- C. Mental retardation
- D. Small skull in proportion to the normal body size
- E. Split in upper lip and gap in the roof of mouth

98. The reaction involved in chemotropic nutrition is::

- A. $6\text{CO}_2 + 12\text{H}_2\text{O} + \text{light} + \text{chlorophyll} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{H}_2\text{O} + 6\text{CO}_2$
- B. $2\text{H}_2\text{S} + \text{CO}_2 + \text{light} \rightarrow (\text{CH}_2\text{O})_n + \text{H}_2\text{O} + 2\text{S}$
- C. $\text{NH}_4 + 3\text{O}_2 + \text{light} \rightarrow 2\text{NO}_2 + 2\text{H}_2\text{O} + 4\text{H}^+ + \text{energy}$
- D. $\text{CH}_3\text{COOH} + \text{enzyme} \rightarrow \text{CH}_3\text{CHO} + \text{CO}_2$
- E. $5\text{GA3P} + 3\text{ATP} \rightarrow 3\text{RuBP} + 3\text{ADP} + 2\text{P}_i$

99. The muscles attached to the bones are:

- A. Voluntary and smooth
- B. Involuntary and smooth
- C. Voluntary and striated
- D. Involuntary and striated
- E. Smooth and striated

100. An organism appears to be a segmented worm. Upon observation it is determined that the organism has a closed circulation, a mouth and an anus, and does not have an exoskeleton. The organism most likely belongs to the phylum:

- A. Mollusca
- B. Annelida
- C. Echinodermata
- D. Arthropoda
- E. Chordate

NTS

PAST PAPER 2010

ANSWERS KEY

Q.NO.	ANSWER	Q.NO.	ANSWER	Q.NO.	ANSWER
01.	C	34.	E	67.	D
02.	D	35.	D	68.	C
03.	D	36.	B	69.	B
04.	B	37.	E	70.	C
05.	B	38.	B	71.	B
06.	E	39.	B	72.	A
07.	A	40.	C	73.	D
08.	C	41.	D	74.	A
09.	C	42.	E	75.	D
10.	E	43.	A	76.	D
11.	E	44.	C	77.	A
12.	B	45.	B	78.	D
13.	B	46.	A	79.	A
14.	E	47.	E	80.	C
15.	A	48.	C	81.	E
16.	D	49.	A	82.	E
17.	C	50.	C	83.	B
18.	B	51.	C	84.	C
19.	B	52.	C	85.	D
20.	E	53.	C	86.	C
21.	D	54.	C	87.	E
22.	B	55.	C	88.	D
23.	C	56.	A	89.	B
24.	A	57.	B	90.	D
25.	D	58.	E	91.	A
26.	D	59.	B	92.	D
27.	B	60.	B	93.	E
28.	D	61.	B	94.	C
29.	D	62.	E	95.	B
30.	B	63.	B	96.	E
31.	E	64.	C	97.	D
32.	D	65.	D	98.	C
33.	C	66.	E	99.	C
			A	100.	B